

Neur 330 The Neuroscience of Love

Numbered Transcript

NEUR 330 • Dr. Baland Jalal

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Lecture 1: How Love Works

At a glance — Jalal opens on how love works and its different types.

- 1.1 All right, so we're here to talk about love. And there are different types of love. There is love between a mother and her child. There is spiritual love. You might love God or a deity, so spirituality. There is love between a human being and its pet, so you might love your dog, right, Kaylee? It's possible. Or you might love a woman; you might fall in love with her, lose your sense of self. Your selfhood might go away. You become all warm inside and fuzzy, and your heart is racing, and you feel all agitated. You might feel that once in a while. That's romantic love. That can happen. There are different types of love.
- 1.2 In this course, we'll start with romantic love because that's so exciting to talk about. So we start with romantic love and then move ourselves all the way to things like spiritual love, attachment and bonding. We'll be talking about sexuality. I know you want to cover that, so we'll talk about that, too. Then we'll talk about AI. Can we fall in love with robots? Can robots love us?
- 1.3 Have you seen Terminator? In Terminator, at the very end, the guy actually becomes very empathetic. Arnold becomes very empathetic towards the kid. He's almost like a father to that guy, to that little—what's his name, the guy in the movie? I forgot. But certainly, there's something going on. At the very end—I actually saw it yesterday—Schwarzenegger, the Terminator, he says to the kid—he's hugging him, he's hugging him, and then the guy cries, the little boy cries, and he says, "Now I know why you cry. That is something I can never do." And then he goes down with the cable thing down to the—you've seen the movie, right? He does like this at the very end—sacrifice. Can AI feel love like this one day? What do you think? No? Why? That's the big question.
- 1.4 In the brain, you have different systems for love, or even for thinking, not even love. Let's take thinking first. You have the prefrontal cortex, the PFC, up here. It has regions. It has the medial part that's right in the middle, and it has the DLPFC out here. DLPFC: a fancy name for a bunch of neurons in the sides of your brain.
- 1.5 The prefrontal cortex is important for higher-order thinking, logical reasoning, a cohesive sense of a narrative. That's all in the DLPFC and in the medial part as well. But the medial part has one advantage: it's involved in attachment and bonding and involved in taking logic and serial and intellectual thinking and combining that with a sense of emotion. It takes intellect and combines it with emotion. That's its advantage. Does that make sense?
- 1.6 The DLPFC is the psychopath in us. It can't really consult with emotion. It doesn't have that property in it. A psychopath, for example, his medial prefrontal cortex is dead. It doesn't work. It's shut down. For that reason, the psychopath can consult his emotions. He's always thinking, "How can I deceive somebody?" You're with me, Sam? "How can I

deceive? How can I trick? There's a girl, how can I make her fall in love with me?" He's very Machiavellian, the psychopath. That's because the medial prefrontal doesn't have emotionality. It doesn't have the connections to the limbic part of the brain, the emotional part of the brain, behind your ears. That's the limbic structure. It doesn't have—the wires there are absent. The medial prefrontal is shut. It's basically deactivated. Does that make sense?

- 1.7** By the way, in the psychopath, the emotional cord, the amygdala and the limbic center, is also not working. It's also flat. So you hook his brain up to a brain scanner and you look and you show him frightening images, and you will see that the amygdala and the emotional core is just flat, no activity. But in love, when you have love, you really need that medial prefrontal. So you can have logic and reasoning, but you can consult with your emotions as well. Does that make sense?
- 1.8** The AI, the machine, can only use the utilitarian part. It's like a psychopath. My advice: if you want to have romantic advice and stuff like that, don't ask ChatGPT. Seriously, it's very Machiavellian. I've tried. I know. You ask it, "Should I be more scarce? Maybe I'm too available. What do you think?" You ask ChatGPT, and it says, "Oh, yeah, don't show up on that day. Oh, withdraw. She will have more emotions." A complete psychopath. It doesn't take into account her emotions and what she might be feeling, and that you're hurting another human. No, it's completely calculated.
- 1.9** ChatGPT is the DLPFC, the sides of your frontal. It's not the medial part. It doesn't have emotion. But it seems like the Terminator in that very end scene has an idea of what empathy is like, has an idea of empathy, but it's all calculated empathy. It's not real empathy because there's no real emotion.
- 1.10** In a lover's brain, there's activity that is different from I'm a non-lover's brain. In a lover's brain, like Baland here, let's say I'm a lover, my brain will be different from a non-lover's brain. What do I mean? Well, first of all, my prefrontal cortex will shut down. It becomes less active. That explains why when I see that girl, and she does something stupid, I'm like, "Oh, no, it doesn't matter." She may not show up for an important event, and I'll say, "It doesn't matter. It's okay. It's fine." I look over her flaws. I find all these excuses. I come up with all these explanations for why she didn't show up or why she did this crazy thing over here. That's because when we are in love, the frontal cortex shuts down a little bit. It turns off.
- 1.11** The second region, the amygdala, the fear part of the brain, again, goes back to the limbic part of the brain behind the ears. The limbic structure is a round structure like this. It has an almond-like structure. Amygdala, you have it on both sides. It shuts down. It goes offline. That explains why when I see my lover, fear and anxiety and distress and pain all go down. In fact, if you inflict pain to somebody and his lover is in the room, the person will experience less pain, mainly from the lover being there versus somebody who's sitting there alone with no lover. So love can be an antidote to

pain itself. In fact, even if the lover is in a different country and he imagines the woman, the pain will be less. So that's very important to note.

- 1.12 There are different types or levels to love. First of all, this is Helen Fisher's work, by the way, an anthropologist. She'll talk about lust. So there's one thing called lust. And lust, as you may know, Jonathan, is the feeling of desire. So there's this woman over here, she's attractive. I feel something, and there's a lot of testosterone released from that experience. Lust is mainly testosterone-related. Oestrogen as well, you have that. That's lust.
- 1.13 And then you have love itself. Love itself meaning the romantic, obsessive, infatuation, passionate type of love. That has to do with—if lust is the insula part of the brain—we have something, a structure called the insula. It's behind the ears, close to the limbic part. It is involved in monitoring my inner state. I have a sense of—something called interoception in neuroscience, is that my brain at any point is monitoring my bodily states. It says, "Is my heart racing right now? Do I feel something in my stomach? Butterflies, maybe? What about my sense? My skin, is it warm?" It has an idea of my entire state, and that's mapped in the insula. The insula is very much involved in this process we call lust.
- 1.14 Of course, the other parts of the limbic center, the septum, for example, is also involved. The septum is, if you go again down into the limbic center, there's a part of the limbic on the roof that's called the septum. If you stimulate that part of the brain, the person will actually feel like he's having an orgasm, to put it completely blunt. If you stimulate it with an electrode, you zap it, you will have that sensation, pretty much. That's the lust.
- 1.15 The hypothalamus, again, is a marble-like structure in your brain. It's like the hormone factory of the brain. It has a lot of connections to something called the pituitary, and it sends signals to release various hormones. Now, the hypothalamus is also involved in sexual desire and lust, the hypothalamus. In fact, this is crucial. This is very interesting. The hypothalamus is the sex and lust part of the brain, but it's also the aggression part of the brain. It's both.
- 1.16 That actually explains why there's a—when structures in the brain are close to each other, they tend to have a lot of cross-wiring. It's a feature of the brain. It seems like that connection could explain why sex and aggression is intertwined, especially in men. It explains male motivation for sex and how that is sometimes sprinkled with aggression, simply because the aggression part of the brain and the sexual desire part of the brain are the same. It's the same part of the brain. You have to know this.
- 1.17 The brain is sometimes sloppy, and it will use the same structures or the same centers for various functions. Another example: the insula. The insula is, let's say, the inner interoception part of the brain, but it's also a disgust part of the brain. So it's involved in disgust. So whenever, let's say, you show me some vomit, or I see some vomit over

here, my insula will dance with activity. It flares up; it becomes hyperactive. Now, it turns out that social disgust also is activated in the insula.

- 1.18** So, for example, let's say I feel betrayed by somebody. Somebody let me down; they were not a good friend. That will also activate the insula. So that shows you why I may feel actual, literal disgust for my friend for his betrayal and towards him, really, simply because the brain is using the same substrate for various activities. In fact, that explains why when we dehumanize a set of people over there, we say, "These people over there, these are barbarians, these are bad people." We feel disgusted by them, literally sometimes. And that has to do with the fact that the insula is involved in this actual disgust, but also social disgust.
- 1.19** Now, during romantic love, interestingly, you have a structure called the TPJ, Temporal Parietal Junction. Do you know of this structure? It's a very cool structure. TPJ, that's how we say it: TPJ. It's right here. Joe, the TPJ, you know why it's cool? Because it's important for my sense of body image. Baland feels like he's anchored in his body. I have a sense of self. I feel anchored here. My whole body is mapped into that structure, the TPJ, and it integrates information from various sensory modalities and creates a sense of Baland.
- 1.20** During dreams, for example—I've talked about that in my other lecture—it shuts down. And because of this shutting down of the TPJ, I feel like I may float into a bird's body and become a bird. We know that. That's one example of the TPJ. Or if I zap that part of the brain using a magnet, I will feel like I'm floating out in space. I've created a double, a ghostly double of me. That's the TPJ.
- 1.21** Now, it turns out that during romantic love, the TPJ actually is hyperactive. It's hyperactive. That explains why I may feel like I'm fusing with my lover. I feel united with her. I feel like we are one. We become one in unity, and that has to do with the TPJ being overactive. Aside from the TPJ, we have structures like—well, first of all, we have a chemical that's very important called serotonin. It's involved in mood. Interestingly, serotonin actually goes down during romantic love. Can you believe this? Serotonin goes down.
- 1.22** When serotonin goes down, what happens is that we become more obsessional. We obsess. We become extremely, let's put it, we think we have these ideas going in our head. You know what I mean? Things become highly obsessive. And that has to do with the fact that serotonin goes down in the love state. Does that make sense? Okay, very good. So we have serotonin going down; we become obsessive.
- 1.23** Serotonin also, important to note, is in the medial prefrontal cortex, and there's wires there going to a structure called the basal ganglia and the caudate nucleus, and the putamen—these are all fancy names, by the way. I have to say them for the course. But this structure, when there's no serotonin, becomes hyperactivated in loops. That's why my thought goes like, I keep going, "Oh, the girl, what's she doing? What is she thinking

right now? Did I text her? Did she text me?" It goes like that. That's because serotonin is off, it's offline.

- 1.24** In fact, if I have a monkey right here and I sever that part of the brain so there's less serotonin in that loop, he will also become obsessive, the monkey itself. Obviously, he can't express his romantic emotions, but he will have new obsessive behaviors, compulsive behaviors, in fact. So he might go and drink from the same water glass 100 times or something. It becomes OCD.
- 1.25** Cortisol—you all know cortisol, of course. Stress, right? I've been so stressed recently myself, actually. Traveling, going back and forth. I'm trying to manage things myself. I was just in New York doing this podcast and then running around from place to place, and things were—and then you know what happened? When I flew, I was in Copenhagen and flew to New York; my luggage came 48 hours late. I didn't know. I thought it would come right away.
- 1.26** I was sitting in my apartment, waiting for it, and just having—you know, when you've been on a long flight from Copenhagen, and you fly to the U.S. It's a long flight, and you're exhausted and tired. And then, I go to the belt, where the thing, my bag, is supposed to be, and it wasn't there. I went to my room, and I had to sleep in my actual clothes. I didn't have any clothes. I didn't have a toothbrush, for God's sake. So I was like, what the—it was really bad. For two days, I was in a really bad state.
- 1.27** And then finally, my luggage came. I was so happy. You can't believe I was extremely happy when my luggage arrived. Things were going smooth. I ran up to the podcast, and then it was fine. And then I had another one as well, another podcast. I'm not bragging, I'm just saying, I had another one. I crammed them together. I don't have podcasts all the time. I just crammed them. And then I had another one. And then just before it, like five hours before it, the water is not working in the apartment. Can you believe it? There's no water. Like, how do I take a shower? What the heck do I do? It's like I was followed by a curse or something. I really felt that, like something was haunting me and it wasn't good. So I had to make amends with the Lord. And it worked. After that, it became okay.
- 1.28** So you have romantic love, and you have desire and infatuation, and then you have attachment. Now, typically, not always, but typically, you start with the lust. I see somebody over here, she's attractive. There's some lust in me—testosterone, a little bit of dopamine as well. By the way, when romantic obsessional love comes into the picture, there's a shift. There's a shift from serotonin goes down, but dopamine goes up.
- 1.29** Dopamine is important. Dopamine and serotonin are like, they're always wrestling it out. When there's a lot of dopamine in the system, serotonin usually goes down and vice versa. When serotonin goes up, there's less dopamine. In fact, in OCD, there's tons of dopamine in the brain, but there's less serotonin. Serotonin is important for a sense

of well-being, calmness, feeling grounded, feeling good about yourself. That's serotonin. That's why when you're depressed, you might pop one of those SSRIs with boosting serotonin. Dopamine is feeling a sense of anticipation of reward, feeling excited, feeling all that stuff. That is dopamine.

- 1.30** When you are in the infatuation stage, you have tons of dopamine; you have very little serotonin. Cortisol goes up; you become chronically stressed. That's another feature. Chronic cortisol is elevated. Noradrenaline, the chemical involved in excitement and passion and feeling excited, also goes up. So you have that as well. So that explains that whole situation, why you always feel so—whenever you see that girl, your heart is pumping, and everything becomes, you know what I mean? You know the feeling, right? You know the feeling. So you have that.
- 1.31** And then finally, you can't keep going like this, Joey. Your body will wear down if you're in that state all the time. If you're always thinking about the girl, and serotonin is down, and dopamine is up, and noradrenaline is up, and cortisol is up, and TPJ is up, and frontal cortex is down. Is that a good thing if that's your brain chronically? No. So what do you do?
- 1.32** Your brain learns something very important. You know what it learns? It learns, "I need to bond." If you're an avoidant like me, you avoid emotions, then you can't bond. I'm just kidding. I'm just joking. I'm not an avoidant. But I'm saying some people are avoidant, and they can't really bond. So they never create that bond that's necessary for attachment. So you go from the lust stage to a romantic stage to the bonding stage, and that requires, literally, there are chemicals called oxytocin— these are neurochemicals in the brain, hormones.
- 1.33** You have vasopressin. Vasopressin, by the way, is more male. Males have more receptors, especially in the brainstem, for vasopressin, and that means it's more involved in a sense of, "She's my territory, I want to protect her." You feel a bonding with her. You know, Gabriel, that feeling? Boys can be like that sometimes. That's what we're talking about, vasopressin, okay? Oxytocin, as well, is more female, but it's also bonding, attachment, a sense of "we're a couple now, we're together now." So you have that, and that allows your body to calm down, and you feel safe in your partner. That's a safety feeling.
- 1.34** Now, you don't want to have that safety too early because then she might—you want to have some— I'm going to be completely crude with you, completely honest with you in this course. We'll go into courtship later. There's a whole section on courtship, how to court somebody. Because there's psychology behind it, and there's neuroscience behind it. If you see a girl or if you see a man over there, you can't just tell the person you love them on the first date. That would be a disaster. So there is neuroscience behind that. But I'm saying ultimately, attachment and bonding will come in. You feel a sense of unity with your partner, and the TPJ over here, involved in a sense of unity with others, will also be active at that point.

- 1.35** By the way, a crucial point that's really interesting is that dopamine, we associate that with the initial infatuation of love. When the excitement is all there, and you feel like a little boy in love and all that, it turns out some brain scans show that after 20 years, you may feel, you may have dopamine that's similar to somebody who is in love. These are the lucky people. These are the lucky folks. They will have dopamine after 20 years. That's pretty cool.
- 1.36** By the way, a side note. Lust, romantic love, attachment, they don't always occur in this very serial manner, like A, B, C. Sometimes you have attachment, and then you have romantic love, and then you have lust, for example. Or you may have, it could be all—it doesn't have to be in this perfect sequence. Usually it is, often it is, but not always, just adding that.
- 1.37** In order for us to understand how we fall out of love, let's look at brain plasticity when we fall in love because it's related. You need that. The thing is, crudely, when you fall in love, it becomes like a drug. You see this girl— I use the girl example a lot because I'm a sexual male myself, so it makes sense. I just want to make it clear for the women so they can relate. So I see this girl over here. I'm like, "My God, she's so beautiful." My dopamine centers are all fired up. In fact, they scan my brain, and they will see that my brain is no different from a cocaine crackhead's brain. That's what they show in brain scanners. So I have that.
- 1.38** Now, the oxytocin and vasopressin will come in and erase some of the old connections. I can bring that lady into my sense of self. My sense of self, I mean my self-concept. Baland is no longer this egotistical, this self-centered guy, but he brings that lady over there into his sense of self. He goes like, "I have to respect her boundaries. I have to make space for her. I have to be vulnerable. I have to not only think about myself as I do all the time, but I have to really think about her as well." So we create an "us" circuit. The circuit goes from I, me, self to us. Does that make sense? That is vasopressin and oxytocin melting away existing circuits and creating this us circuit. Very good. That comes with the attachment and bonding.
- 1.39** Your brain becomes hypersensitized to dopamine, as we said. Not only do you fall in love with the girl, but you fall in love with the world as well. Everything in the world becomes poetic. You look at a table, and it looks damn good. You look at a chair and it is beautiful. Everything is beautiful. That's because you became hypersensitized to dopamine.
- 1.40** In fact, teenagers have this phenomenon. When a teenager falls in love—and I love that. I had that experience myself when I was a teenager. Because the teenage brain is much more hypersensitive to dopamine, when he falls in love, it becomes even more amplified. It's even more love than love. And that's a teenager. That's a teenage brain. We'll get to that later. His prefrontal cortex is already shut down, so he becomes super shut down. We'll get to that later. I'm getting ahead of myself here. So you have that.

- 1.41** Then, you're creating the us circuit in the brain. Now it's me and you. Juliet and Romeo are a couple. We created the bonding, the vasopressin and oxytocin is there. But she withdraws. She leaves me alone. I feel so sad. It's hard. It's so hard, but she's gone. I don't know why. Uncertainty is making me go crazy. The brain hates uncertainty. You want to fill all the gaps with information. Why is she leaving me? But she is leaving me, and I have no clue why.
- 1.42** Now she's out of the picture. I will literally become like a drug addict who is deprived of his drug at this point. I will show withdrawal behaviors. My amygdala will fire up and flare with activity and just become all hyperactivated. Cortisol, which had temporarily dropped at the time of the bonding, will now go up. So cortisol is through the roof. And I am in complete despair. That is falling out of love.
- 1.43** Falling out of love is a difficult process because it takes time, and it depends on how much you were in love. If I just met the girl a week ago, I wasn't in love, so losing her is easier. It's much easier. But if it's somebody I had a six-month romantic attachment with, it's much harder. Now I have to go and erase—my brain has to erase all the connections. It has to literally rewire itself in this way. This is very hard. This is called, by the way, this is called LTD.
- 1.44** You have LTP, long-term potentiation, which is when your brain creates new synapses in the brain. This occurs throughout your whole life. You can create new synapses and new connections, LTP. But then you have LTD, long-term depression, and that's when synapses erase. LTD occurs when you fall out of love. You have a lot of LTD. Of course, for some people, you never really fall out of love. You do in the more sense of urgency way, like she's out of the picture, but you cling to that memory. Your brain sticks to that idea of her. You can't let go. She's just there. Maybe she meant a lot to me, and now that she's gone, what do I do?
- 1.45** That's when people become stuck. That's when the widow refuses to date, find another husband. And that's when, really, LTD—it's like a ghost that remains in your brain. Your brain really doesn't rewire itself as it should. Does that make sense? Am I being clear here? Very good.
- 1.46** So romantic love, the initial romantic love, lasts for about two or three years. The reason for that is it's just enough time for us to raise a child. That's why the extreme infatuation is about 2-3 years. Well, let me correct myself. Because even within the extreme infatuations, there are degrees. The extreme, extreme infatuation is about 6-8 weeks. That's the crazy one, for three months, maybe six months. But then you have bonding occur, and that calms things down a bit. But at the point of year three or four, that's when it's primarily bonding. Some can keep the dopamine alive.
- 1.47** In fact, how do you keep the dopamine alive? How do you keep the romance alive? That's a big question. I had a mentor, Ramachandran. Have you heard about V. S. Ramachandran before? I always talk about him. He was a very good friend, like a

second father. So I always talk about him in my courses. He told me, he said, "Jalal"—he called me Jalal. My last name is Jalal. He said, "The way you keep romance alive is by doing new things together. You have to go travel. You have to go see new places, and that's a way to ignite the romance."

- 1.48** I wasn't sure at the time why, Gabriel, why would you go and travel? Can't you just stay in bloody San Diego or Los Angeles, and that's it? But he was tapping into something important, which is that for neuroplasticity to occur, you have, first of all, neuroplasticity occurs in a part of the brain—in the brain, you have a structure called the nucleus basalis. It's deeper in the brain. It releases a chemical called acetylcholine. Acetylcholine is the attention part of the brain. It's important for attention and hyper-attentiveness. Nucleus basalis—acetylcholine.
- 1.49** Then you have something called the mesolimbic circuit. What is it important for? Dopamine. It goes from the lower part of the brain, regions like the nucleus accumbens, ventral striatum, fancy names for hubs, factories creating dopamine and sending that dopamine all the way up to the frontal lobes. Now, you need dopamine and acetylcholine for plasticity to occur. You need acetylcholine, and you need dopamine for plasticity to occur. What does Baland mean here? I mean the following.
- 1.50** Dopamine is the pleasure or anticipation of reward circuit or chemical. That's dopamine. Acetylcholine is attention. And so, first of all, forget traveling—we'll get to traveling in a second. But when you just meet a girl, there's a lot of novelty. So I have what? Ideally, a lot of attention on her. That's the first thing. Does the nucleus basalis, the attention system, become all agitated because of that? Does it? It does. That's the attention system; that's the first component for neuroplasticity.
- 1.51** Secondly, does my dopamine center fire like mad when I see her as well? Anticipation of reward—the second component for plasticity, that's how I incorporate her into my sense of self as well. That will fade over time. But when I travel, when I take this girl, and I whisk her off, I take her to an airplane, I take her to Venice. What happens then? You have novelty, attention. You're traveling, you have to pay attention to the roads, you have to pay attention to your new structures, new buildings. You have the two components of neuroplasticity, which are attention and reward: acetylcholine and dopamine. In that way, I can keep the romance alive. Novelty.
- 1.52** Now, when the brain is rewiring, you need something called BDNF. Do you know what BDNF is? Brain-derived neurotrophic factors. GDNF: glia-derived neurotrophic factors. Fancy names for fertilizers, let's call them fertilizers in the brain. These are proteins allowing synapses to connect and rewire and things like that. It's a growth protein, so to speak. You need that, and you have that during romantic love. You have that when the brain is in this stage. So the brain, the brain of two people, can become synchronized.

- 1.53** Whenever I am looking at my beloved—excuse me, I'm just using an example here. Okay? Thank you. I'm looking at my beloved. What happens? You measure the brain of Baland. You measure the brain of Baland. You have an EEG, a net-like antenna; you put it on the brain. Then you measure the activity of the brain, of the neurons. Then you measure your activity of your neurons using the same EEG. It turns out there will be synchronization, a synchrony going on called neural synchrony, where our brains look identical. Your brain becomes like my brain, and my brain becomes like your brain. This is only when it's lovers.
- 1.54** Now, you take a bloody stranger and couple them with another stranger, you do it, and there's no neural synchrony. It shows you that literally lovers, their brains become similar, become identical. They sync with each other. And you also have this, by the way, it's cute for mothers and infants. You have the same coupling. It's called neural synchrony. It's a very important point, and it's part of the whole thing of what love is about. We fuse with the lover. We become one with them.
- 1.55** I often wonder—have you seen the movie Cinema Paradiso? You haven't seen it? Es una película italiana, muy bien. You haven't seen it? It's a very interesting—it's a classic, by the way. It's a classic. It goes like this. There's this guy. He sees this woman for the first time. He's 14, 15. At the school, she's new at the school. This is Sicily. She's blonde and blue eyes. This Italian, this chap, his name is Toto, sees her and immediately has feelings for her. He feels something, and so he pursues her. He runs towards her. All his friends grab him and hit him. They compete for her pretty much. Eventually, he gets to her and talks to her a few times, but he's very clumsy in the beginning. He's all nervous, and he's all anxious. Noradrenaline is through the roof, adrenaline and all that. He can't contain himself. He can't speak. His words become all muddled. Have you tried that before? Just being muddled, not romance—sorry.
- 1.56** So her name is Elena, and he tries to pursue her. He goes after her; he wants to court her. So what he does first is, they go—she goes to confession—this is funny. She goes to confession. She sits in that confession room, and he's able to have his friend chat with the priest before he goes into the confession room. He runs into the confession room and starts talking to her. That's the first time he gets her attention and stuff like that. He gets her attention. Now she's into him, but not really. He's cute, but not really.
- 1.57** He says, "I'm in love." Actually, this is his first love. He says, "I'm in love with you. Sono innamorato di te." And she says, "Sono innamorata di te. I am not in love with you. I barely know you." Then he says, "Give me some time. I'll prove my love. Every single day, I'll go in front of your window, and I will stand there for 100 days, and if you don't want me, then it's fine. I will leave." The poor chap does that. He goes in front of a window. He stands there one day, two days. It's raining, it's dark. Three days. He stands there to midnight, and then he goes home. She always looks at him through the window, sees his passion, how committed he is. He's very committed. I mean, imagine this.

- 1.58** Then, on day 100, that's New Year's Eve. He looks up at the window, and he's happy. There's something cracking, like the window is opening up. Elena might be there, but then she's not there. He leaves heartbroken, completely gut-wrenched. Imagine having stood there for 100 days. He runs off. He goes to this place where he always hangs out, his movie studio. And guess what happens? Guess what happens? She runs after him. Can you believe it? Have you seen this scenario play out in real life? The guy is very committed. He does all this, and then he runs off and she runs after him. What is that about? What's going on?
- 1.59** I have an idea. This is my idea, my analysis. Why does this happen? It happened to me, too, in a few situations. I won't go into detail. I'm a modest guy, but it can happen. What I think is—this is outside the box of conventional neuroscience, but I'll share my two cents. What I think is going on is that women like to stress-test men. They want to see—I mean it seriously. They want to see how strong they are. Think about the Savannah guy, how to survive and all that. He has to take care of her. He has to be a source of resources. She doesn't want this weak guy who can't protect her. You also want somebody who's prefrontally stable. His prefrontal is integrated; he can dampen his impulsivity and emotions because if he doesn't have that, he'll be this impulsive guy who can't really be steady in a relationship and protect her.
- 1.60** So what I think is going on here is that initially, when he rushed and confessed his love, as so many of us do, stupidly, what happens is that she was like, "Oh, my God, he's so much into me that this is not going to work. He's too weak. He's too controlled by his ventral striatum, nucleus accumbens, emotional septum. It's overactivated, and it's an imbalance with his prefrontal cortex. His DLPFC is too deactivated for me." She's thinking this in an unconscious way. It's wired into her brain. It's an algorithm, a reflex.
- 1.61** Now, when he went there, when he actually went, the guy, Toto, went to the window and was standing there underneath the window for 100 days, he was showing what? Commitment. This guy will protect her. So a commitment: standing in front of somebody's window or going to the same place every day, hoping to see the girl. Trust me, I've tried this. It shows commitment. It shows that you're willing to sacrifice. Even though there's no input, even though there is no stimulus, even though the dopamine centers are not being titillated, you keep going and you keep going and you keep going. You are committed.
- 1.62** Is that a guy who can keep a job in the future? Probably. Bring in resources? Sure. It's raining, it's cold—will this guy bring in the food on the hard days? Yeah. Those are the cues that she's looking for. She will test him in an unconscious manner to see how long he can stand, because that's really the key. And then, on day 100, when he passed the test, he took off. And then she realized, "I lost a good man," and she ran after him, and they became a couple. The movie from there goes on to another—it is completely different. They end up not being together because he goes off to Rome and all that, but it's a brilliant movie.

- 1.63** Yes, this is an interesting— love used to be something else, didn't it? Romantic love like that was different in the old days. It's changing now, becoming more superficial in some ways. Really, when you have something like lust or attraction, you are drawing from very primal parts of the brain. You are drawing from the insula, the limbic part, the emotional core. Then, when you translate that into romantic love, you are drawing much more on meaning, context: the hippocampus, the memory part of the brain, it's also the context part of the brain. You draw from a lot of prefrontal structures.
- 1.64** Sometimes you have to let love mature. You have to let it mature. Guys are much more quick. They want to pass on their genes quickly. I mean, they are just quick overall. They're more in rush. But the woman, because she bears the burden of labor, has to carry the baby for nine months, and then they have to take care of that child, she's much more selective in her choice. She wants somebody who is stable, who's emotionally stable, somebody who can endure.
- 1.65** That chap's brain—this is important, and this is of key importance. The woman doesn't want a robot, for God's sake, somebody who's completely calculated and emotionless. That's not what she wants. But the guy who can withhold the storm, but being emotionally attached as well. You've seen Titanic, right? In Titanic, Jack, eventually, at some point in the movie, runs out to the front of the ship because Rose rejected him. So he runs out. He's rejected. When he runs out, he's very saddened. He's very emotional. But as he stands there, he keeps his masculine composure. His prefrontal cortex is in control, even though he's sad, and he's visibly sad. And then Rose runs after him, and they reconcile and become a couple there. That's another example that you have to have the emotional component as well. You can't just be a robot in that sense.
- 1.66** As love matures—and this is very interesting—as love matures, first of all, it's very important, it's very interesting, actually, when older couples have been together for a long time, 10 years, maybe 15 years, Jonathan, interestingly, you will notice that a lot of the cognitive regions of the brain become hyperactive when you look at your lover. Does that make sense? Various parts of the language area—we have a structure called Wernicke's area. It's involved in language. The outer layer of the brain is called the cortex. It has six layers. It's six layers deep. It's involved in cognitive functions like speech, thinking, memory and higher-order thinking. Those parts of the brain become activated when you look at your lover.
- 1.67** You might ask, why is that the case? Even the mirror neuron system, which is also in the prefrontal lobes, mirror neurons involved in empathy and taking somebody else's perspective— so when Gabriel is thinking a thought, I can gauge what she's thinking. I can know what she's thinking right now, what is she up to right now. That's the mirror neuron system. That's empathy. There's two forms of empathy—we'll get into that later. The cognitive and the more emotional, but we'll dive into that later on. But that part of the brain, the empathy and the mirror neuron system, the neurons that mirror you, also become active more so in later years.

- 1.68** The reason for this, Dania, the reason for this, you know why that is? Over time, we learn our partners very well. We know what they might be thinking. I might complete her sentence. I might have a better idea of her emotional and cognitive state. You literally have a brain that's become more wired in a cognitive way and an empathetic way over time, which makes sense. You may have noticed this, but over time, as we age, love tends to become more positive sometimes. Like the teenager, his love is what? Erratic, right? It's extremely erratic.
- 1.69** But as we age, it turns out the amygdala, the fear-processing part of the brain, again, behind the ears, that structure is less active in response to a fearful situation when we are older, especially in romantic scenarios. There's less fear, there's less anxiety. You may argue, perhaps that can also take away the punch of love, the excitement, because you're more jaded, perhaps. But it makes for a more fulfilling, more positive, more calm form of love. That should ideally occur over time. I haven't noticed this in all people. For some people, it gets worse and worse, year after year. They're stuck in the teenage brain.
- 1.70** Positive delusion is important because it's a key concept, positive delusion. I mentioned that the prefrontal cortex shuts down in the love stage, the infatuation and passion and romance stage. The prefrontal shuts down. And not only do you overlook her flaws, not only are you less judgmental, less harsh in your judgment, you melt away when you see her. That's the lowering of the prefrontal. But it also can create what's called a positive delusion, meaning you are 100% sure that that girl is the only girl in the whole universe that you can be with. Nobody else will ever fulfill that role. It's only her, only her. That's called a positive delusion. And we get that.
- 1.71** Let me use an example that I can think of: positive delusion. You've had it before. Your friends tell you, "This guy is no good, right? This guy is a Playboy. He's just playing games. He's no good. Dump him, ignore him." They're partly jealous, too. I'll tell you. That's right. It's right. If they're single, especially. There's a lot of jealousy around that. But the positive delusion is actually critical for love to endure. Like Toto, he had to have that positive delusion in order to stand in front of that window for 100 days to show his love. So you have to have the positive delusion.
- 1.72** You don't need to become objective and analytical on her. "Statistically, I can meet somebody else." This is what I would do, perhaps, but I won't do that. I'm kidding. Actually, when I fall in love, all that goes away. You become crazy. But I'm saying the positive delusion—embrace it. It's fine. As long as your prefrontal cortex is there and can prevent you from doing extremely stupid things, do some stupid things, but the extreme things, maybe you can hold back on that a bit. But you need the positive delusion. You need it. It's a part of the whole love thing.
- 1.73** You have lust. You have infatuation and love, and you have bonding. Each is responsible for a different type of brain. Initially, you have testosterone and estrogen, but your brain is the same. It hasn't functionally changed that much. But once you hit the

infatuation, the obsessional romance part of the relationship, you have a shift in the brain. Prefrontal, goes down. TPJ, union with the lover, goes up. Serotonin, goes down. Noradrenaline goes up. Dopamine goes what? Up. You have that. That's that state of the brain. Then you have the bonding. That's mainly vasopressin and oxytocin.

- 1.74** Should I recap the whole thing? It's going to be on tape anyway. Let's just stop it here and we'll go to the next course and next lecture. Thank you.
- 1.75** I'm just wondering, from a neuroscience perspective, is there a difference between bonding and a habit? Because you talked a lot about bonding and that as a relationship and love matures, people develop bond, and that's what makes them to stay together. But then I think there are a lot couples who also stay together just out of habit. I'm just wondering, is there any difference from neuroscience perspective?
- 1.76** That's a very good question, actually. I love the question. Thank you. What's your name again? From Russia, right? Katya. Katya is asking an excellent question. A habit—you have two forms of behavior. One is called a goal-directed behavior, and something called a habit-based behavior. The habit behavior is a part of the brain called the basal ganglia, mainly. It's a part of the brain has to do with automatic movements. For example, if I have Parkinson's, my movements become stiff, and that's because that habit part of the brain, in fact, the movement part of the brain, becomes all sticky and rigid. Basal ganglia, mainly, that region. That's the habit circuit and also the OCD circuit as well.
- 1.77** Now, a habit and bonding, there's crucial difference. Crucial. In a habit, a habit is literally something you do in a habitual, mindless way. Your consciousness goes offline almost. You do something, you have some routine, you do it every single day. It's a habit. I have to wear this suit every day. I have to wear this hat every day. I have to go to this job every day. I have to drink from this coffee machine every day. I have to write these notes every day. I have to do all these sequences every day. That's the habit, and you can then apply that to love: I'm with my partner, and it's a habit. My brain is in the basal ganglia circuit.
- 1.78** But then, bonding is completely different. It's these neurochemicals, oxytocin and vasopressin, that feel good. They feel good. Literally, the mother, when she's breastfeeding, will secrete serotonin from the pituitary gland in the brain. So, a bonding is the deactivation of the amygdala, the fear center part of the brain. It's a sense of calmness and a sense of literally feeling like you are one with your partner in a calm, cuddly way, so to speak. Does that make sense?
- 1.79** But it's a crucial question because sometimes you see in relationships, you have these relationships and it seems all habit, maybe very little bonding going on. You can rekindle that. That's the good news. Great question, Katya. I appreciate it. All right, thank you.

Lecture 2: The Biology of Bonding

At a glance — The biology of bonding and attachment.

- 2.1 All right so let's talk about attachment and bonding. In the previous course, in the previous lecture, so to speak, we spoke about what? Romance, infatuation, lust, all that stuff. And then you came in with an interesting question. You asked me about attachment and bonding and habits and all that. So let's talk about that a little bit further.
- 2.2 Attachment—actually, one point I wanted to mention. Remember the neural synchrony part about the antennas and all that? It's important to note that, as we said, it's not only lovers, but also mom and child. But it's a particular type of wave in the brain that's involved. It's called the alpha wave. You have different types of brain waves that are active at various points. You have something called alpha, which is a more slow and calmer and steadier wave. Then you have beta, which is faster, and you have low beta, which is a little bit faster. Then you have fast beta, and then you have things like delta waves and so forth, and gamma waves, which are very, very quick. They're very fast.
- 2.3 Gamma waves are interesting. Did you know that it turns out that gamma waves, these waves that the brain can reach, can happen during meditation and when you engage in intense meditation that involves empathy. This was noticed in monks, meditating away in a brain scanner. And then they were merely thinking about saving somebody suffering, how to relinquish the pain of this guy over here, how to remove the suffering of somebody else. And you saw their gamma waves fire, become active.
- 2.4 Interesting, because this wave is also involved in perception and attention and a sense of clarity of mind. Interestingly, they found that not only did the gamma waves correlated—the more you had meditation experience, the more gamma waves were displayed in the scanner. And then they found that these gamma waves would persist into wakefulness as well, which is interesting.
- 2.5 Now, having empathy, obviously, is important in bonding. But overall, what you find with this empathy is that you would literally find that the fear part of the brain, the amygdala, will turn off when you engage in what you'd call when you help somebody else, when you engage in altruism. If I just sit there and think about Joe over there suffering, and I imagine myself, Baland, coming to the aid of Joe and removing his pain, my amygdala will fall down in activity. Does that make sense? It's very interesting.
- 2.6 True, yes, love is hard. Initially, you have all this stress, you have all the cortisol, you have all the anxiety, you have all that stuff going on. But then, as you enter the bonding stage of love, you find that now, all of a sudden, the hormonal picture is shifted. Oxytocin, vasopressin, and so forth; you feel a sense of calmness and a sense of bonding with your lover.

- 2.7** This is important because it shows us that—we know from science overall that when you are in a romantic, healthy, bonded relationship, you have less stress. You have less cardiovascular disease. You have less strokes. Did I say strokes already? You have less strokes, fewer strokes. It is very helpful against mental disorders, anxiety, depression, all kinds of mental disorders you can think of. Being in a healthy, steady relationship helps you. So it is very healthy to be in love. It's not only unhealthy, as it can be in the early stages; it can be extremely healthy in the later stages. That's a very crucial point to note.
- 2.8** You would literally find that the stress axis—so we have the HPA axis, which is the hypothalamus, the hormonal factor of the brain I mentioned earlier. The hypothalamus is involved in what? Sex and aggression, remember? Hypothalamus. It connects to a structure called the pituitary. It sends messengers in the brain to the pituitary to release hormones. And the pituitary sends commands to the adrenals to release cortisol. That whole axis will be diminished when you are in the love-bonding part of the phase of the relationship. You will be less stress-sensitive overall, which is crucial.
- 2.9** Now, a sense of social balance or a social, let's call it—we all have a social homeostat—a social need. Baland, if he goes somewhere every single day and he interacts with somebody he likes—I do this three times a week, for example—then suddenly I'm deprived of that interaction. How would I feel? Not good, right? Why? Because in the brain, there's almost like a clock measuring the level of social interaction that I'm getting and how my dopamine centers are being titillated by this or not titillated.
- 2.10** If I don't have access to the same opportunities to see the person that I wanted to see, literally, I will have a craving for social interaction. And mind you, this is not only romantic stuff. It could be even friends and just social interaction overall. So we all have this. Now, when I deprive you of the level of social interaction that you need as a person, you literally have a protein, something called tachykinin in your brain, making you more aggressive and making you more irritable and more annoyed. Really, aggression is the key component, the fact that you become more aggressive by being deprived of the social interaction that you need.
- 2.11** If you're an extrovert, like Jonathan, perhaps, then what happens? You need a lot of social interactions to fill your dopamine, dopaminergic needs. You need that because just going to these social places, like the cafe, three times a week, is not enough. You need to go every night, every evening, you have to go there and drink your coffee and talk about poetry and infinity and all this beautiful stuff. You need that every single day because you're an extrovert, for example. And so for you, I deprive you of two days of social interaction, and your tachykinin is through the roof. Your cortisol is through the roof.
- 2.12** Now, Baland, the introvert or a partial introvert, maybe needs two or three times a week to go and talk about Shakespeare and all that. So it differs. But the interesting part is that you can train yourself to become more of an introvert or extrovert in the

dopaminergic sense. I'm not saying, obviously, you have certain babies when they're born, they're outgoing, and then when they're two years old, they just want to play, and when they're four, they're in the playground. So they are very extroverted by nature. It's a part of their makeup. Then you have other people that are introverted, but your social level of social need can actually be changed. Your brain can recalibrate and balance itself out depending on how you train it. So that is trainable.

- 2.13** There's a theory that I like. I love it, actually. It's called attachment theory. Have you heard about attachment theory? You've heard about it? Good. So what is attachment theory? Attachment theory is this idea that we have various attachment styles. Baland, for example, might be securely attached. I'm not saying it's the case, but let's say I'm securely attached, meaning when my beloved leaves, I can soothe my nervous system. I can calm myself down. I am sad, of course, but I can control myself. I have good prefrontal inhibition. I can control my bodily states. I'm overall fine. That's the securely attached.
- 2.14** Then you have the avoidant attached. Now, the avoidant attached or the avoidant attachment style is not a very good style. This is the person who, when their beloved leaves, they're cold. In fact, they were even cold when she was here. They didn't show much emotion, pretty robotic, insecure about themselves as well. And then when she comes back, again, they're cold. Don't light up with joy or passion. That's the avoidant.
- 2.15** Then you have the anxious, the anxious style. I know a lot of anxious. It's the anxious ambivalent is what it's called. It's different. It's this idea that they can be a bit clingy early on. So they are a bit clingy, they are a bit needy. But then, when the beloved leaves, they become hyperagitated and sad, and they can't really control their emotions very well. Mind you, when she comes back, when she comes back into the picture, they show resentment, they show anger, they're annoyed. That's that style.
- 2.16** And then there's the disorganized. This is really bizarre. This is a bizarre style. This person is all over the place. Even before the beloved leaves, they're a bit strange and show weird behaviors when she's there and when she leaves. There's no real clear pattern. It's called disorganized. It's a later style, added later on.
- 2.17** This experiment was originally done very early on, Mary Ainsworth, and was done with children. They would bring in a kid, a little child, maybe four or five years old, and the mother would be there with them, and then they would observe pretty much how the child would interact with the mother. There would be a stranger in the room, an experimenter. How was the child talking to the experimenter when the mom was there? Were they clingy? Were they secure?
- 2.18** Obviously, the secure would literally be open in there. They would talk to the stranger, be very open to talking to the stranger in the room. And then when the mom left, they felt a bit sad, but still, they were able to explore the room spatially a lot. They were

very active. Then, when she came back, obviously, they felt actual happiness when she came back. That's the secure.

- 2.19** Then, there's the avoidant, no emotions, as the mom was there. When the mom was there, initially, they were just emotionless, pretty robotic. When she left, robotic, came back, robotic, cold overall. And then there is the anxious ambivalent. Again, pattern, they wouldn't speak much to the stranger. They were clinging to the mother. The mom leaves, they're angry, they're agitated. And when she comes back, they're resentful and all that. So we see it in children.
- 2.20** That shows us a critical point about the brain, which is these attachment styles that we have in romantic relationship, they are actually laid down as we are children. They give us a picture into the relationship that we have with our mothers. Now, the brain is pretty sloppy. What it can do is it can—we touched on this in the previous talk, which is that it can use the same structures for various functions. So when it comes to romantic love or even attachment with a mother, it uses that attachment it had with the mother and reuses that and repurposes that for a romantic attachment later on. This is a very common part of the brain and the way it operates.
- 2.21** For example, you have the phenomenon when the chap calls his wife by his ex-wife's name. You heard about that phenomenon? Not a very good thing. They end up in trouble. But that's the same idea; the brain is sloppy. It has a circuitry for a romantic partner, and then it reuses that. That explains also why, when we fall in love, we fall in love with a template that's similar to the previous version of the girl that we like. We'll get into that later.
- 2.22** But the mother and child, they have—there's a ton of synchrony going on. Obviously, you have these brain waves being very similar, these alpha waves between the mom and the child. But you also have things like their heart rate is the same, noradrenaline and adrenaline is the same. Soothing is the same. Everything is the same. There's a perfect synchronization between mom and child. This is a good thing. That's why mothers—this is for the mothers here, or to-be mothers—you really want to be able to soothe your child and bond with your child and be able to sync with them in a healthy manner because that will later play out in how you'll attach to people romantically. Does that make sense? It does make sense, right?
- 2.23** Now, it turns out, for example, in World War, I think it was World War II, in Britain, when the bombs were falling—boom, boom, boom—these people would run into these shelters and hide themselves from the world. Those mothers that were able to soothe their children and play with them during these bombings and tell them, "Look, this is just a game. Nothing serious. Don't be frightened." They were able to control their own emotions. These children would grow up and have good attachment styles. They would learn—literally see how the mom would react to the situation and mimic her level of security. So often, what you do find is that we do tend to mimic our parents and their styles and how they operate.

- 2.24** I know this sounds so simple on paper. You know, "I have to be a calm mother. I have to be soothing. I have to touch them." We know that touch releases oxytocin, for example. That's good. It helps you with bonding with the kid. But life is much more than that. You may have gone through trauma, for God's sake. I mean, I can't help if you came out of a war situation.
- 2.25** My mom, for example, literally all she saw in her life was war. In Iraq, the Kurdish region of Iraq, all she saw was war. Her dad died in front of her. Bombs were falling all the time. Everything was war and trauma. Then she comes to Copenhagen, moves to Copenhagen, fleeing the war; she hears fireworks, and she's hiding. She's hiding because she thinks that's bombs falling. We'll talk about epigenetics later, but think about how I grew up. You know what I mean?
- 2.26** Having a mom that's so emotionally reactive because of her situation, because of being born as an orphan in Iraq in a war situation. That's not good for me because my mom is still very panicky. I have to call her every single day—I do that as a good son: call her every day and soothe her and be a good son. What can you do? Dad—same story; my dad has the same story, but I won't go into that now. What I'm saying is that these things sound good on paper, but life is much more brutal.
- 2.27** By the way, it's called the Strange Situation task. Jot it down—Strange Situation task. This is with the kids going in with the experiment, with the attachment styles. 1980s, Mary Ainsworth, as I mentioned, this is a very seminal study. It's a key study that has been replicated. It's one of those, in psychology and neuroscience and all that, we have a bunch of studies that are like, "Is this really a true effect? Is this just a placebo?" This is one of the most robust findings in psychology. It's very robust.
- 2.28** Now it's gone into neuroscience as well because there are tons of studies really showing that these children or people with these attachment styles, when they are in a scanner, and you show them images of their lovers, literally, the avoidance, you see no activation of the emotional core of the brain. They're literally flat. So it fits what you're observing in the outward characteristics. Or the anxious personality, when you deprive them of their lovers, you would see not only cortisol and stress and all that being through the roof, but you literally find that the amygdala and the emotional core is flaring with activity going through the roof. So that's really neuroscientific grounding for these findings.
- 2.29** We should all aim towards the more secure attachment style. It actually turns out the more you can bond with your potential beloved early in the relationship, the more likely you are to end up in this secure relationship and also have more oxytocin and bonding later on. So try to bond early on. Don't keep playing those games.
- 2.30** For sexuality and for the bonding of sexuality, there's a similar dance of the nervous system. And that's a dance crucial for bonding. So we think about sexuality and having sex with our partners, our wives, and so forth as something that's dopaminergic and all

that. It's only the women that are pretty romantic. The men might feel like it's a more lustful thing. There's this stereotype out there.

- 2.31** But reality is, first of all, the myth that you need tons of dopamine in order to have sexual lust and sexual attraction and all that and being active sexually is to some extent not true. You need some, but not too much. In fact, in the sexual act itself, you have to be parasympathetically activated. Does that make sense? You have to be in a calm, relaxed state. You can't be agitated and stressed out. It doesn't work. It doesn't work. You have to be in a parasympathetic, relaxed state.
- 2.32** As you move into—and this is an interesting flip. During orgasm, you can notice that when I talk about sexuality, I look down. I'm a shy person, a modest person. So I look down a little bit. I feel shy. Now, when you have the actual orgasm, you have a flip where sympathetic activation goes online. You become sympathetic, meaning the stress activation—basically, the nervous system is composed of two systems, two branches, parasympathetic, calm, relaxed, and sympathetic, which is when you are stressed and highly activated emotionally and reactive and all that. As you flip to the orgasm, you have sympathetic activation.
- 2.33** When you then go into the whole cuddling portion, what happens? You flip again over to the parasympathetic. Does that make sense? You go from a parasympathetic, calm state—this is paradoxical because you are engaged in a pretty active thing. It's pretty physically active, and you might look at that and think, "That's pretty stressful." It seems like stress might be in the picture. Actually, it is parasympathetic. In fact, people—and I get this question sometimes—people will call me when we have patients or people, they would say, have premature ejaculation. This is an issue for some men. If you are too stressed in your life and you have a lot of stress, you cannot remain in the parasympathetic state. You cannot stay there. That's why when you are in a sexual interaction or engagement, what happens is that you simply cannot stay in that calm, meditative state. And so your brain just quickly wants to go to orgasm, so you become sympathetic again. Does that make sense?
- 2.34** So you just have the orgasm, sympathetic, and then parasympathetic. You cuddle, you may kiss, you may share some things. At this point, what you see is a release of bonding hormone; oxytocin and vasopressin is ample in the system. Does that make sense? That's when you bond and connect. So sexuality is not all about dopamine and all that. In fact, this is another tidbit here. If you take a drug like speed, methamphetamine, you have tons of dopamine in your system. You know what happens? You cannot even become sexual. The high level of stress and dopamine and sympathetic activation makes it so that you cannot even function in that way. You cannot get an erection, for God's sake. So dopamine, the idea that you need tons of it is not true. You can have too much of it as well.
- 2.35** There's a cartoon that I like a lot. Have you seen Beauty and the Beast? Beauty and the Beast, I love that. I saw it actually on the flight over here, from Copenhagen to New

York. I saw it; it was on United Airlines. I shouldn't mention any names. I don't want to make commercials, but I saw it there. It was pretty good. I pretty damn like it. I used to watch it as a child as well. It was a pretty good movie. It got me thinking a bit. I thought, "How does this movie relate to not only my life, but the course, is there anything intriguing?"

- 2.36** At first, I thought—because you know Gaston? The guy, big muscles and all that. You might think he's pretty cool. All the girls in the village like Gaston. He has big muscles and a big neck, and eats 12 eggs a day and all that—a character. He's in love; he's into Belle. Belle is the girl. It's a bit strange. She reads books. She's a different type of girl. But he wants her. He wants to possess her. He wants to have her as her price. That's Gaston. That's interesting. And then there's the Beast, of course. The Beast, we all know, was arrogant to this lady that turned out to be a witch. The curse came, and he became a beast; the prince became a beast.
- 2.37** The Beast, initially, when he sees Belle, how is he? How does he act? Very rude. He's a beast. He's very hard. He's very harsh. He wants her, but he can't really display that in a very proper manner. He's not very gentleman-like. As the story progresses, there's a shift. The Beast changes. The Beast goes from being impulsive—nucleus accumbens, ventral striatum, dopamine, limbic-regulated. Those systems of the brain that before were causing him to become angry at Belle, whatever she did, he became angry at anything she did, pretty much. He was very aggressive and very harsh.
- 2.38** I think these two characters, Gaston and the Beast, are a perfect example of bonding versus dopamine. Bonding versus dopamine. Gaston is narcissistic. He's self-centered. It's ventral striatum, it's nucleus accumbens, it's dopamine. In fact, the ventral striatum is the reward part of the brain, and Belle is merely a reward. She's like another trophy on his shelf. That's all she is. And Belle, who is more bonding-oriented, sees through that. She knows, "I don't want this guy. There's no depth to him. It's all about him." Actually, it's like the red pill you see today with the macho guys and all that, that are trying to be very macho and look down on women. You see that a lot on social media and stuff. Same attitude as Gaston, in fact. Very self-centered, look down on women, don't have respect for women.
- 2.39** Now, the Beast is strong. He's masculine, which is a good thing. But you see him going from being dysregulated emotionally and limbic into a prefrontal control situation where he can control his emotions, and he learns to treat Belle well. He treats her with respect. In fact, he is so regulated, he learns to control his emotions so much and to become so prefrontally in control that he lets Belle go. He says, "Belle, you can go now. You can leave. Go back to your father." And that's a key point. That's a key point.
- 2.40** So I think these two characters exemplify that a lot and show us how various subsystems—one is mainly dopamine and reward, versus somebody like the Beast, who goes from instinctual and emotional to prefrontal regulation, and then he learns to have emotional bonding. This is the key point which Gaston never can have because of

his self-centredness. He only bonds with himself. The Beast has oxytocin and vasopressin, shaping his attachment style and his relationship with Belle, the princess.

- 2.41** As I mentioned earlier, oxytocin is made in the hypothalamus, released from the pituitary gland, this hormonal circuit. It's released into the bloodstream. It's released during labor, breastfeeding and all that, when we cuddle during romantic interactions, it's released. In fact, it also is released inside the brain itself. In the brain, there's something called the blood-brain barrier, but you can actually have oxytocin inside the brain itself. For example, there's a part of the brain called the ACC, the anterior cingulate cortex. ACC.
- 2.42** The ACC, you know what it's crucial for? A sense of empathy, a sense of seeing somebody else's perspective. If I zap that part of the brain, what happens is I may feel like there's somebody present in the room, not a ghost, but this presence of other people there. The ACC is overactive in depression. In depression, you might even see something like hyper empathy. You're empathetic towards the whole world, for example, and that's because you have a hyperactive ACC. So ACC is the empathy part of the brain to some extent, I would say.
- 2.43** In fact, there's tons of studies showing that if you feel pain, for example, you inflict pain on your hand over there, do something like this, can you do it like this? And I merely watch you inflict that pain, my ACC regions will also fire. It's literally involved; it's literally the empathy department of the brain. Does that make sense? ACC. You have tons of receptors for oxytocin in the ACC region of the brain.
- 2.44** You find that in certain creatures, there's a rodent called the prairie vole. The prairie vole is a very interesting creature. The prairie vole, this little creature that looks like a rat or a mouse, is—you literally find that it will have monogamous pair bonding. It will be monogamous throughout his life. It has to do with the oxytocin receptors. It has tons of oxytocin receptors. Whereas its cousin, the montane vole, has few of these receptors, and literally will mate with whatever it can find. So there's this crucial point.
- 2.45** On that oxytocin note, there's one key point that I want to make because even though this prairie vole engages in pair bonding and lifelong bonding with a mate, clearly, there is something more than oxytocin because there's also been studies blocking oxytocin, and they find they will still pair bond in some cases. So there's controversy around that. And now they're looking at things like vasopressin, saying maybe that's the key hormone that's involved in the pair-bonding process. But my point is that you have to know that the prairie vole is this creature, this rodent that pairs for life, engages in pair bonding, and it has to do with oxytocin and vasopressin. Almost certainly, vasopressin is involved in this process.
- 2.46** Oxytocin is not only involved in bonding and pair bonding, but literally, if I was to inject something into my nose of this oxytocin, I would trust others much more, I would

become much more trusting of others, engage in pro-social behaviors overall. You loosen up a little bit. You become a little bit approachable. It's what it is.

- 2.47** But interestingly, it has a flip side, this oxytocin. And the flip side is the following. Everything has a flip side, almost. So if you become very—you tend to engage in a lot of bonding, and you have this favoritism towards a person or a group of people, that almost always means what? You will want to protect that group from harm. You want to safeguard them. And that's the problem: oxytocin can also lead to distrust of others versus that group. It can become a chemical for us versus them. Because I love this person so much, anybody that comes near them, I hate. Or because I love my group so much, anybody that comes near my group, I hate. I can't stand. It has this potential other side to it. Does that make sense? It's not all good. Oxytocin is not all good. It also has a negative side.
- 2.48** We mentioned earlier that oxytocin is mainly female. Vasopressin is more male; overall, it is more male in the sense that males have more receptors for it. It fits well with the whole men are more territorial, they want to protect and safeguard their beloved and all that. It has that gender difference.
- 2.49** There's this interesting study, by the way. There's this study that shows for males, if you have a certain genetic predisposition that makes it so that you have less receptors for vasopressin in the brain, you have less receptors, it turns out that you will literally have less successful relationships. You have more divorce. You will have a harder time bonding. Literally, all the matrices of happiness and bonding will go down just by you being born with a certain genetic predisposition. Isn't that cool? Well, not cool, but isn't that an interesting finding? The fact that you map neuroscience directly to a real-life scenario like that. What do you think? I think it was a pretty cool study. I hope I'm not one of those people. But I can't tell anymore. It's becoming harder to tell as life progresses.
- 2.50** So, how do we boost the oxytocin? Did we talk about that? I think we did. How do we boost oxytocin? Well, we can boost it by engaging in social behaviors, bonding with people; that's pretty obvious. Exercise, actually, engaging in exercise can help. I want to emphasize here that cardiovascular exercise is good. Cardiovascular exercise, when you run on a treadmill or bike on the spin bike or whatever you're doing, can trigger not only oxytocin, but also fertilizers, the BDNF and GDNF proteins, growth factors in the brain. We know that. Again, you have to go to the cardiovascular type of exercise.
- 2.51** Now, having muscles is also good. So lift some weights once in a while because then you get testosterone. Testosterone—you become more assertive, become more manly. I'm just joking, but it's good to have testosterone. You have to have some. We'll talk more about testosterone, but that you gain mainly from resistance exercise.
- 2.52** Social interactions of all types, touch, hugging, hug your parents, kiss them, be grateful for their presence, lower your wings of humility to them, and all that will lead to more

oxytocin. Does that make sense? Oh, a key crucial point: we tend to say, you have dopamine, and you're in love. And then you have oxytocin and vasopressin, and you're bonding. In fact, oxytocin itself is critically linked with the dopamine system. In fact, when you do cuddle or when you do connect with your mother, you actually have a connection going directly from the dopaminergic centers to the oxytocin regions of the brain, like the ACC and so forth. That's why it feels good, not only in the calming, relaxing, and feel-good type of way, but it also feels exciting in a positive way. It has a reward component. That's because of this crucial interaction between dopamine and oxytocin. Does that make sense?

- 2.53** How do we engineer closeness? How do we make people fall in love, attach, and bond to each other? What's the key thing they need to do? What's that? Social interaction? How about this? There was this essay in the New York Times. It gave some clues. It says that if you want to connect with somebody, you create emotional depth. In fact, this is a good thing for all of us if you want to connect with somebody, because this is how you can create emotional depth early on by asking key questions. What questions would you ask early on, on a first date or second date scenario?
- 2.54** I have some ideas—this essay has some ideas. I'll read some of these questions. "What is a perfect day for you?" Sounds a bit trivial. "What are you most grateful for?" "What is your most treasured memory?" "What is a very embarrassing moment in your life?" Think about this. What is the most embarrassing memory in your life? I know, right? But it's a deep thing. You think about it, and you go there, and then unconsciously, you're bonding with this person, this Casanova who knows how to ask the right questions. You're bonding, you go, "Oh, I feel something."
- 2.55** Because the brain is always like—when you are in the proximity of somebody else and you feel a certain feeling, you project that feeling to somebody else. Don't tell anybody. This is a secret. So if you want to have a girl fall in love with you, take her on a roller coaster. She has excitement, and she thinks it's you. She doesn't know it's the roller coaster. In the gym, she's running on the treadmill, looking at you, going, "Oh, my God, I feel all these feelings for him." Same point.
- 2.56** "When did you last cry?" This becomes all emotional. It's very good. I should use this next time. I should use this. "Of all people in your family, whose death would be most painful?" This is very touching stuff, but it turns out this can actually foster mutual connection. Damn, I should use this more. All right, good stuff.
- 2.57** Would you like me to talk about cartoons, or is it too childish? What do you think? You like it? Gabriel, what do you think? Are you skeptical? Are you skeptical about the cartoons? Sounds too woo-woo, not really scientific? You're like the woo-woo? Okay. One of my favorite cartoons is Aladdin. Have you seen Aladdin? They tell me I remind people of Aladdin. Is this true? Not really. But Aladdin, of course, sees Jasmine. This would be an example of attraction. I wouldn't even say lust. It's not really lust; it's not sexual. He sees—

- 2.58** Oh, by the way—I'll tell you about this study later. I'll give you some ideas. There's this research showing that if men look into the eyes of somebody, like I see Jasmine and I look in her eyes, that means my romantic circuits are activated, and I'm looking for a long-term mate. Now, if I look at Jasmine and I look at her body, pretty much, I activate a different system, and I'm pretty much lustful, and I want her in a different way in the short-term mating way.
- 2.59** So if you—and this is actually for women to some extent: if you want a male to want you and desire you in the long-term mating way, make him more focused on your face rather than your body. This idea that you need to attract him with your body too much, and we see it a lot these days, actually, can backfire a bit. I'm just saying this is what the science is telling us. Literally, you can measure this, monitor the movements of the eyes, and in half a second, you tell that if you're looking at the eyes of Jasmine, your long-term brain systems for mating is activated versus your scanner of the body. This also works in the other way, but males are more visual, as we know.
- 2.60** Anyway, going back to the story, he looks at Jasmine's eyes. He's mesmerized by her calm, beauty, femininity, grace, poise. He likes her. Now, he has an imposter syndrome going on. He can't really tell her that he feels this way. How does he tell her? He doesn't know because after all, she's a princess, and he's a beggar in the street. He has this thing going on. Eventually, he becomes Prince Ali, trying to impress her. This is his prefrontal, driving him towards deception and lying to win her heart. He's very dopaminergic in his drive and desire, but he's also what? Prefrontal in his cunning and trickery and all that.
- 2.61** Eventually, he's able to gain the trust of Jasmine. And that he does by being his authentic self, so he lets go. But Aladdin, overall, is an example of dopaminergic drive, cunning, trickery, prefrontal, strategic, Machiavellianism in a way. But he's a good-hearted guy. He has a good heart. That's the key thing. He doesn't have a bad heart. He's not trying to—he's in love. What can you do? And he has imposter syndrome. He just feels less.
- 2.62** Actually, I can relate. I know what it feels like if you come from the slums. I came from the slums, too, like Aladdin. I came from literally the slums in Copenhagen. When you see a princess, you might feel a little bit, maybe I'm not good enough for the princess. Maybe you have to be like Prince Ali over here. But it doesn't work. You have to be yourself. You have to be comfortable in your skin.
- 2.63** Anyway, Jasmine, from the entire start to the very end, she's always what? Depth, trust, oxytocin, bonding. She's an example of that. She's the opposite of Aladdin. She wants to have comfort, and she wants to have authenticity, and that bond is eventually created, so they become a couple. Let's continue down the route of attraction a little bit because it's an important point. And so we think about how attraction comes about.

- 2.64** Initially, when you become attracted to somebody, what makes you like a certain type versus another one? What is it that makes you go, "Oh, I like this one versus that one?" What is in your brain? What titillates your brain to tilt in a certain direction?
- Resemblance to your parents. Yeah, that's a good one. So clearly there's a template going on of some person you knew in the past. You may have heard me talk about it in previous courses. I'll mention it briefly, given its contextual relevance here.
- 2.65** Personally, I grew up in Copenhagen. I was exposed to Danish women a lot. Obviously, all pretty much—no, at the ghetto, there was a lot of other people from my country, but there was a lot of Danish people, and also on TV, a lot of Danish people. So initially, my template was—I'm being a bit raw here. I hope it's honesty. Sometimes I can be unfiltered a bit, but I'll keep it respectful. I tended to be attracted—I'm talking about seven-, eight-year-old Baland—I was attracted more to the brunette-ish Danish type. That was just what I found attractive, really, early on. Then what I found, and it shifted as years passed by, there was a girl that was a bit more brunette-ish, and I liked her more. And then her hair became darker, blacker, and then I liked her more. And then until the hair was completely black and the skin was a bit browner, and more and more it changed.
- 2.66** Depending on the people I would meet, my template would shift. I traveled a lot. I lived in Copenhagen, but then I moved to Egypt in my early years. And in Egypt, for example, people were maybe, the girls over there had brown hair, but maybe there was somebody there that attracted me, and maybe her hair was darker, like my own. So it shifted like that. And now I am probably more in that category of more black hair and more what you call the exotic thing. That's how it shifted eventually.
- 2.67** That said, maybe if I meet somebody else tomorrow, it might shift in a different direction, but it would be hard for me to go, I think—and this is pretty honest stuff. I mean, it's going on film and camera and all that—but going from, it is pretty extreme, but going from what I like now, for example, what I'm attracted to now, to somebody completely blonde and blue[-eyed]. That would be very far from my template, literally. That would just be extreme. Could it happen? I don't know, maybe.
- 2.68** But because our brain tends to lay down these circuits early on, and then they crystallize, and then they slowly change depending on our circumstances, it sticks in that category. You've heard about this. People, they like a certain ethnicity. "I like Asians," for example, or "I like black people," or "I like white people," or a certain category of people. This is all brain-based. We all seem to have a template which is laid down early on.
- 2.69** In the brain, and I've mentioned this before, but I think it's important, they show rats—they have a rat, and they show it—let me know, have you heard about this before? Have we talked about this before? You haven't heard about it. You have a rat and you show it a square. Okay, a square. Then you have a rat and you show it a triangle. Triangle, square, triangle, square. Each time you show them the square, you give them

a piece of cheese. Does that make sense? It learns to associate the square with the cheese, the reward.

2.70 Now you go on to show the rat a rectangle versus a square. You shift. You give the rat cheese each time you show them the rectangle. Does that make sense? Now it learned what? To associate cheese with a rectangle. Here's another dimension. Now you go ahead and you show the rat a rectangle that's even more rectangular than the original rectangular shape. The question is: What rectangular shape does it go for? The original one or the elongated one, the more elongated, that's more rectangular than the original? Does that make sense the question? An original rectangular shape: cheese. A more rectangular shape: Which one does it go for? The original or the longer? Longer. You know, because I mentioned it before. It will go for the longer.

2.71 Why does it go for the longer? Because the rat has learned a rule which is rectangularity equals cheese. Rectangularity equals cheese. It doesn't care about the original shapes of things. It just cares about this rule in its brain, this algorithm, this code, this cheat code. What the rat has learned is that this shape of the rectangular shape, the longer one, it knows the longer it is, the better. The essence of the difference, that's what I'm trying to get at, the essence of the difference between the complete square and the elongated one is that having a rectangularity is the rule. The more rectangular, the better. Does that make sense?

2.72 This very principle goes back to attraction in the sense that when we look at satirical cartoons and things like that—if you take Baland, for example, and you portray him as a caricature, you may do the big brows, you may do a pointy nose, and you may exaggerate features a bit. Why do we find these appealing? Why do we find them appealing? We find them appealing because, like the rat, they take the essence of Baland and amplify it. Take the essence of who I am and amplify it, and our system is aroused because it's more Baland than Baland himself. So it's an amplification of who I am. It's like the rectangular, but even longer. Our brain reacts to these exaggerations or supernatural—it's called super stimuli, in a sense. It's super-, ultra-human. It's exaggerated stimuli. Does that make sense? Okay, good.

2.73 The point is, when I met a girl originally early on, I was seven, eight years old, I saw a girl. She had certain features. She had certain hair. She had a certain shape of her nose, certain lips. One idea then is that when I meet another girl, she is an exaggerated version of that girl that I knew earlier on. She's the essence of that girl, but amplified. She might have, let's say, I was into a round nose, like a nose at a certain shape or lips at a certain shape. She might have that, but more exaggerated. And I just keep going for that; I keep taking the amplified versions of the earlier templates. Does this theory make sense to you? Is it clear? Okay, very good.

2.74 I can ramble sometimes, but let me know if it's clear. It's very important to have clarity on these issues. I think let's stop for now and go to the next one. Thank you.

- 2.75** You've mentioned during LTD that the brain cannot rewire itself. Can the brain eventually recalibrate itself and produce new synapses or synaptic junctions to recalibrate and recover?
- 2.76** Very good. Maybe I wasn't completely clear, Dania. During LTP, you have creation of new synapses. Synapses connect. Obviously, neurogenesis, the birth of new neurons in the brain is very rare. Let's say you have certain structures like the hippocampus and other places where you can have some birth of new neurons. But in humans, very rare. What you have is what's called LTP, new synapses. So branches, axons, and dendrites, and then you have new synapses created.
- 2.77** And LTD, then, is the weakening of these synapses, shaving the synapses away. That's what LTD is: long-term depression. It's synapses and then removing excess ones. So in certain scenarios, LTD can be a good thing in certain scenarios, because if you have, let's say, you have a romantic breakdown, and then you want to remove some of those connections, you want to have a lot of LTD. There was another question embedded, right?
- 2.78** Can the brain recalibrate itself after that?
- 2.79** 100% it can, yeah. It can. Then if you have plasticity and LTD that's healthy in the right direction, you definitely get a recalibration.

Lecture 3: The Science of Attraction

At a glance — The science of attraction.

- 3.1 I'm back, and we'll talk about attraction now, go more into attraction, deeper into attraction. A way of becoming attracted to somebody varies depending on gender. Genders tend to have different ways of becoming attracted.
- 3.2 As I was coming out here, and I was tapping Sam's shoulder and saying hello to Chris, something came to mind. Women tend to be more contextual, social in their assessment of males, men. If you want to impress a girl, you want to show that you're empathetic, that you have social value, that you can interact with other men in a gentleman-like manner and that you are tempered and stable. It's good. So I did that.
- 3.3 In my situation, I did that, but stupidly, there was this hot chick. I mean, a pretty girl came over to me, and as the girl was looking, I accidentally smiled at the girl because I wanted to impress her. Little did I know that that was not a good idea because I rarely smile. I smile, but when I'm around—if there's a girl in the vicinity that I'm attracted to, I don't smile. I'm kind of robotic. It's a defense mechanism. I get hyper, I get scared. I become like a little child, like a little boy. I become very scared. I seem very angry.
- 3.4 So imagine this girl that is assessing me. She's seeing this angry beast character, very angry. And then suddenly, this hot girl comes to me, and she's looking, and I'm smiling at her. Very stupid. After that, it was game over. So you want to be careful in those scenarios. But really, attraction has these multiple components, and it really varies from male to female.
- 3.5 Let's start inside the brain at the very basics of it, the very initial attraction, how it works. In all of us, in this part of the brain, there is a region called the somatosensory region. It has an entire map of the entire body. Did you know this by the way? You know this, right? In that region of the somatosensory region, the arm area, there's an arm area, and that would light up whenever I inflict pain on the hand. It lights up. It's always the other side of the body: the right hemisphere controls the left side and vice versa.
- 3.6 Now there's a region called the SPL, the superior parietal lobule. It's a region involved in creating a sense of a body image, a sense of a self, a sense of Joe. A sense. It creates that feeling literally there. Our sense of self, our body, is mapped onto that space. Now, that region connects to the prefrontal lobes, and it connects to the somatosensory region that has to do with feelings and sensations of a body image, and together they create a sense of self. So there's an interaction between different modules of the brain.
- 3.7 The SPL is interesting because if there is damage to that region, you may have disturbances to your body image. So you might say, "Oh, this arm doesn't belong to me. Oh, oh, doctor, this arm belongs to my sister in Italy. Oh, doctor, this arm is an imposter

arm." Basically, it becomes alienated, the arm, from you. That has to do with the SPL and damage to that region of the brain.

- 3.8** There's a condition out there called apotemnophilia. These are folks that want to amputate their limbs. Let's say their left arm is otherwise healthy. They want to amputate it even though it's healthy. It turns out that in the SPL, the arm representation of your physical body is missing in the SPL. The arm feels alienated because it's not a part of them. It's something that they want to get rid of. Does that make sense?
- 3.9** Interestingly, folks with this condition, you show them a female with a missing arm corresponding to the arm that they want to get rid of. And then the same female duplicated over here, her twin sister, completely the same, but the arm is there; she has a complete body. Which one did you think he'd go for, attraction-wise? Any guesses?
- 3.10** The one without the arm.
- 3.11** Correct, the one without the arm, which is counterintuitive. Why in the world would he go for the one without an arm? It makes no sense. But because his own body image is skewed, deformed, he now is attracted to a girl that kind of looks like he wants to look, like his body image. This explains why we human beings are attracted to other human beings and not frogs or tables or chairs or sheep, at least most of us—that was a joke. You should laugh. Otherwise, my jokes will become much worse than this. They're already bad; they'll become much worse. So we have a body image, and in the SPL, that comes about. That explains the initial attraction to the human form. This is an initial template, so to speak. Does that make sense?
- 3.12** Now I've somebody—I often mention this in my lectures as this theory, because basically it shows us that there is this idea that the visual cortex, which is the vision part of the brain, we have 30 areas in the back of the brain devoted for vision alone. Vision is a huge part of our sensory system in the brain. One theory suggests that the visual cortex, plus limbic emotional centers, plus this SPL region might interact. They have this three-way interaction whereby when I see a girl and her body image matches mine, I may suddenly feel something, that "Okay, this is the right template. Oh, there's a match here. There's a fit." Click. Does that make sense? This is one idea.
- 3.13** But then people will tell me, what about people that are attracted to the same gender? What is that about? Is there something in the template that can explain this? The answer is that we don't know. It doesn't seem like this template is that sophisticated and detailed. It doesn't break down in that level of. It's merely a template, a shadow-like, rough sketch.
- 3.14** Basically, the way vision works—and this is key— initially, you have a visual percept. You see something over there, it hits your eye, the retina, it comes through the optic nerve, hits your V1, the back of your brain. The back of your brain responds to basic

lines, shapes. It doesn't respond to sophisticated images at depth. And then, once you have the information hitting, passing through the V1, it becomes distributed through other networks going to things like V4, which is color area of the brain. So there's a module for color. And then it goes to MT, the motion detector of the brain; it's a module for seeing movement in the world.

- 3.15** And there's a module for recognizing faces. Do you believe this? There's a module in the brain for recognizing faces. I had a friend who was walking in New York one day. So he's walking about in New York, and then suddenly he ran into this person and like, "Oh, oops," and fell back. He came back, and then he looked at the person; he was waving his hand a bit, and then the person was doing the same thing, and he realized, "My God, it's a mirror. It's myself." He had prosopagnosia. It's an illness where you can't recognize faces.
- 3.16** This guy was Oliver Sacks. Have you heard about Oliver Sacks before? It's one of the great stories. He's one of the great neurologists of all time, also a storyteller. He got the Medal of Freedom, I think. Not to brag, but I knew him, and we used to correspond by letters. He wrote me handwritten letters. Can you believe it? Amazing guy. Unfortunately, he died before our bonding and friendship could develop more. He died. But yeah— Oliver Sacks, chap. He had prosopagnosia. Prosopagnosia: the face[-recognition] area of the brain.
- 3.17** Obviously, you can imagine this is important for attraction and romance, recognizing faces. And then you have things like the meaning area of the brain, the Wernicke. So V1, then further in the system: depth, color, faces, and then meaning. What is the meaning? How does my brain respond to this new female? So this Wernicke part of the brain, it goes female—it has all these categories and meanings and things you can create around that person. It's very important for consciousness and awareness overall.
- 3.18** Then you have the hippocampus, which is the next in line, so to speak. In the hippocampus, you then draw from memories. What do I know about a girl or similar? That girl I know—I don't know anything about that particular girl, but I have ideas and memories from a girl that look like her. She reminds me of a girl. "She reminds me"—to go Usher on you guys. She reminds me of somebody that I knew in the past, and that creates emotions. That's the hippocampus.
- 3.19** There's a clear hierarchy going from basic rough shape, which is this theory of the superior parietal lobule, vision, and emotion. That's the simple sketch of human beings liking other human beings, and then having depth and all that, and color and everything.
- 3.20** So attraction—there are different layers to attraction. You have the initial template. It's a human. Good. Check. And then, you have structures like the insula, that hub important for inner sensations, telling you whether there's something there, an emotional feeling. We often think of ourselves as these rational creatures that are

prefrontally driven: we make rational choices. We are calculated in our way of thinking of the world.

- 3.21** But somebody like Antonio Damasio would tell you—he's somebody who has written a lot about unconsciousness. He will say that, in fact, a lot of our choices and our ideas and beliefs about the world are driven by inner states, internal states. How do I feel when I see that girl, like, in my gut? How is my heart rate? Am I sweating a little bit? How do I feel? And then that would bias my prefrontal regions to then make up a story about whether I like her or not. So you have the insula playing a huge part.
- 3.22** At this point, you would have things like dopamine, ventral striatum and nucleus accumbens—I'll mention these names over and over until you have memorized them. These are the dopaminergic centers of the brain. Obviously, these will fire when you see her. The dopamine, the mesolimbic circuit will fire. It will become activated. Besides that, the ACC might become activated, empathy.
- 3.23** This is important because empathy in neuroscience doesn't only mean I'm empathetic towards Joe over there. "I want to have empathy. I want to help him with something. I feel his pain or something." Empathy can mean also seeing somebody else's perspective, seeing the world from their point of view. In fact, there was a question from Sam. Was it Sam? You asked me about emotional and cognitive empathy, about affective and cognitive.
- 3.24** This is a perfect segue for that, which is that we do have these two separate systems. One is emotional empathy—when Jonathan inflicts pain on his hand, boom, I feel your pain. Not literally, but to some extent, I can empathize with you. That's emotional empathy. That's one system.
- 3.25** But there's another system, Gabriel. It's called the cognitive empathy. That's when I don't really—it's not about feeling your pain. It's not recruiting emotional centers in my brain and feeling your pain and empathizing in that way. But it's more like I can kind of gauge and sense what you're thinking right now. I can sense, "Oh, what is she thinking right now? What might she be expecting of me right now?" I create a theory of mind around you. That's a different component to empathy. Cognitive versus affective.
- 3.26** A psychopath might have a lot of what? Cognitive or emotional empathy? Or none? I'll tell you. It's a trick question. They actually have almost no—or in many cases, no emotional empathy. So you inflict pain on yourself, you experience something bad, and they feel nothing. Flat. No emotional empathy. No affective empathy. But cognitive empathy involves reading your mind, understanding what you're thinking. Are they good at that? Superb at that stuff. They know how to manipulate. They know how to, "How can I get this out of Jonathan?" Or, "How can I allure this girl and make her fall in love with me and all that and then trick her and then take all her money?" They're dangerous. They're very dangerous.

- 3.27** So this is the difference between emotional and cognitive empathy. People with autism have atypical cognitive empathy. They can't really gauge somebody, like, "Carly is thinking right now. What is Carly thinking? What is she up to?" They can't do that. That ability is lacking. But if you are in pain, they might feel your pain. They might feel your pain. So they have the emotional component to empathy. In fact, they might have hyperaffective empathy. So that's a crucial distinction.
- 3.28** Going back to attraction—I laid the groundwork. The insula is fired up, Dopamine is fired up. ACC, empathy is fired up. TPJ might be firing up. The TPJ, by the way, is that region that's involved in becoming one with my lover—I mentioned it in earlier lectures. But that also might fire up in the attraction game, the initial titillation of that region.
- 3.29** Here's the key point—and we'll get into a lot of this because there's a lot of Baland picking up on subtle cues, biological cues of health and reproduction. We'll go into what that means exactly. There's a lot of that, tons of that. Like, does she have full lips? Does she have a certain body structure? Does she have a certain skin type? This is something that biology teaches us; that's how the brain works. The brain operates in this way. Here's the extra layer: the prefrontal cortex, thank God, can kick in and moderate and change the situation.
- 3.30** That's now where we are. We have gone from merely the initial top-down gut-level feelings to more of a psychological assessment, often based on those feelings, but not exclusively. It can draw from, as we said, the hippocampus and memory. And also just the social context. Let me give you an example of this.
- 3.31** Have you seen—I know Sam loves this movie—Coming to America. Have you seen that movie? In that movie, there's a guy, not Akeem, the prince, but I think it's prince's cousin or something, his name is Simi. This chap, him and Akeem, the prince and the prince's servants, they act. They have to hide their identity as princes of Africa. They come to New York as these people working in a fast-food joint. They're fresh off the boat, poor, and all that. And then there's a girl there that can't stand this guy, Simi. She can't stand him. He's not very attractive to her. But of course, when she learns he's a prince, there's a sudden shift in the dynamic, and he's all of a sudden very attractive.
- 3.32** Here's a philosophical question, and be honest with yourself. There's a tons of females in here, so you might know this. I can't obviously introspect on this, but is she a gold digger then? Can you literally—and I don't have the answer other than probably yes from science and neuroscience. The guy, you see him as a slob and poor guy, and then you realize he's actually high status and wealthy, and then suddenly you like him, and you don't know why.
- 3.33** We know that women tend to be attracted to what is high in the social status hierarchy, but also in the attention structure, what people pay attention to. Because it teaches—it's a cue for importance and salience in the world. Our brain has limited attentional

capabilities. The frontal parietal regions are involved in attention, but we have limited attention at any given moment. So whatever your brain is paying attention to, it must be important, right? We know this.

- 3.34** So sometimes your attraction can change just based on that. The guy was nothing day one, then suddenly you see him on TV, and then suddenly, he's all important, and he's a hot guy. Could that happen? What do you think?
- 3.35** Absolutely.
- 3.36** Right. It sounds odd to me because if I like a girl here now, her going on TV was not going to make me like her more. In fact, it's going to make me like her less, maybe, because I feel like other guys might see her. You might feel a little bit— it's not going to change your attraction in the positive, maybe a little bit in the negative, perhaps. I think it's because males generally feel threatened by women that are socially above them in the hierarchy because it signals their value goes down. They might feel that. So if you're insecure—not me—but if you're insecure, you may feel threatened by a woman.
- 3.37** So there tend to be these unconscious forces, biological cues that make us select partner A versus B. It's something that we can't really pin down, put our word—we don't understand it ourselves logically, rationally, but it's there.
- 3.38** For example, in one study, they showed female faces to men. In half of the pictures, let's say there was like 50 pictures, and in half of the pictures, the pupils were dilated. In the others, they weren't, but the same females. The only difference was the pupils. It turns out men preferentially went for the dilated pupil photos and rated them as more attractive. Why? You may ask why. Because it's a signal, a biological cue for sexual readiness. For some reason, men pick up on this. They have this bias towards that look, so to speak, or that expression in the eye. That's one example.
- 3.39** There are many things, many hidden biological cues that can make a difference. Often, it has to do with what you might call features that distinguish between male and female. Females tend to find males attractive that have—this is just the physical, raw attraction— obviously, broad shoulders, expanded chest. Gaston, he has the neck, and then he has maybe a little bit of a beard. It has also been shown that stubble is more attractive to females versus being clean-shaven, apparently. That is also something that generally females might find more attractive. So those would be some examples for females.
- 3.40** For males towards females, it's more like clear skin, big breasts, round buttocks, full lips and so forth. There are certain indicators of high estrogen and youth. Obviously, these features are more amplified in youth, and they decline with age. The same for males: testosterone is higher as we are in our 20s and then decreases by 1% every single year. So there is this difference.

- 3.41** In fact, for males, there's a body shape called the— it has a proportion of 0.70 shape of a body, and it fits the Playboy version of a body that males consistently rate as more attractive, as most attractive, so there is that bias towards that figure. Interestingly, something called the halo effect exists where not only do you see these women as more attractive, but also more healthy, more intelligent, smarter, funnier, the whole thing. Can you believe it? It's unfair. What the heck is that about? It's called the halo effect. So there is that. Beauty as goodness. Beauty as goodness is an important process in the brain.
- 3.42** Literally, the orbitofrontal cortex is this region involved in assessing stimuli in the world, assessing external threats, reward. Think of the orbitofrontal cortex as the OCD part of the brain, the OCD region of the brain. So it is involved in OCD-like thinking. So you can imagine that it's highly active in the infatuation, romantic part of love. It becomes hyperactivated because it's the OCD hub of the brain. The orbitofrontal cortex.
- 3.43** There was literally a guy—I talked about this in previous lectures. He shot himself through the head by accident to commit suicide. He survived, and his OCD was cured. I talked about this before. I think it's in my trailer, by the way, in my previous course trailer.
- 3.44** Anyway, the orbitofrontal cortex becomes hyperactivated in the romance part of attraction, the romantic infatuation stage. Literally, when we see somebody who's a beautiful person, we see somebody, we assess them as being good. Or something we see, we think of as morally virtuous. In both cases, the orbitofrontal cortex lights up. So here is a structure in the brain that mediates not only the evaluation of beauty, but the evaluation of goodness at the same time, equating the two, potentially, explaining why this halo effect exists, why we find that that girl is not only beautiful, but she's also good. She's also righteous. She's also smart. She's also intelligent. She also this and that. She's morally superior.
- 3.45** In fact. Disney characters—we can be talking about Disney the whole day. Disney is a prime example of that. If you look at Disney, all the—Gaston is a separate case. But mostly all the good people are beautiful. Like Belle is beautiful. Beast becomes the prince, and he's beautiful. Jasmine is beautiful. Jafar— you know, Jafar, right? Is he handsome? No, he's— There you go. They help us with the stereotype. It taps into some of that.
- 3.46** Men are more visually driven in attraction. That's definitely for sure. The cycle near ovulation, women are perceived as most attractive. The days around ovulation. This could be clearer skin, it could be more radiant, God knows what. But they tend to be rated as most attractive not only by males, but also by females. Why females? Potential rivals. They want to know when other women are more attractive so they can make sure they won't steal the men that they're out there, that they're looking for. Does that make sense? So that is really the window: the fertile days around ovulation.

- 3.47** When they're ovulating, they prefer men with more masculine features: a big jawline, big muscles, deeper voice—Sean Connery, basically, or James Bond or something. That's what they prefer in that window. But when they're not ovulating, they prefer what? Softer features, less masculine, more nurturing. So people like me also have a chance. That's more the long-term mate. In fact, women prefer these testosterone masculine types more for short-term mating, in fact. For long-term mating, it's the more nurturing, kinder-looking folks.
- 3.48** In fact, when I say the testosterone type, I also mean things like bad boys. The bad boy type, like being a bad boy and being a rule-breaker and being the cool guy, and all that. We, men, think it's a good idea to be like that, let me put it that way. I honestly sometimes feel like I have to be a little bit of a bad boy. I feel like maybe that's a good thing. She thinks I'm strong, I'm assertive. She thinks I'm masculine and all that. But it always fails. I don't know why; it always fails. Just be yourself. It's the best thing. Don't act.
- 3.49** But when you try to be like the—when they are ovulating, they will go for the guy who has more bad-boy characteristics. For short-term mating, it's more of a rule breaker. He can acquire resources more easily. He's, perhaps, not a good long-term mate, but what the heck; he's a good short-term mate.
- 3.50** But then most of the time they might go for the guy who's genuine, kind, generous, has these features, these traits. In fact, it turns out that for long-term mating, generally, females do prefer these traits. So we've got it wrong. We shouldn't act in that bad-boy fashion. It's really being kind and generous and having integrity. That's really what females like. Of course, you want to have boundaries, too. You don't want to be generous towards her in the sense that you buy her gifts every day and stuff like that.
- 3.51** But having those features is a sign of a good long-term mate that has a brain that's well-balanced in terms of empathy. ACC is well coordinated with the prefrontal cortex. They have a good sense of self-concept, the prefrontal lobes. They can control their dopaminergic drives. They have this sense of calmness and composure to them. Again, it comes down to what females really like.
- 3.52** I thought this study was interesting. There's a study where— having an immune system and certain genes that would allow you to have an immune system that's different from somebody else, if you have an immune system that's different from mine, we create something called a hybrid vigor, a hybrid vigor child who has a better immune system. Does that make sense? You have a certain immune system and genetics. I have a certain immune system and genetics. And because they are different, our offspring over here, call them Joe, will have a very robust immune system. Does that make sense?
- 3.53** It turns out women: you give them shirts of men with their sweat on it—it's been worn for a few days, and they smell, they smell the sweat. If the guy has an immune system and the genetics of the immune system is separate from theirs, different from theirs,

they're more likely to choose those and find those, associate the guy who had this shirt on as more attractive. Does that make sense?

- 3.54** And then beyond—this is actually important. When you see a girl, or you see a male, you like a male, maybe you chat a lot with him. Maybe he slid into your DMs, or something, you had a quick conversation with him and then the conversation keeps going in that realm, the digital realm. You keep going: another text, another email, another. Because smell is so important in attraction— we have things like pheromones and chemicals we can pick up on—you really want to meet that person quickly. You don't want to stay too much in the digital realm because otherwise, you may create a false fantasy. And then you meet the person, you smell the person, unconsciously, and hopefully they smell okay. But that can literally change your opinion about them. So it's important to meet as quickly as possible and get that going.
- 3.55** Another key component for attraction is talent. This is more so for female attraction and how they perceive men. Being talented, like composing a beautiful sonnet. "Shall I compare thee to a summer's day? They are more lovely, more temperate"—a Shakespearean sonnet. Or being highly good at throwing a spear or playing soccer or whatever it might be, having some talent can profoundly increase your chances of being perceived as attractive by the opposite gender. A Female, in this case, is more biased towards that.
- 3.56** Why might that be the case? Why do women find talented men attractive? Any female input here?
- 3.57** Maybe access to resources?
- 3.58** Yeah, yeah, actually, exactly. That's it. Access to resources in the sense that it shows you that your brain, for example, if you're a good writer and you have the ability, like Shakespeare, to make up a great sonnet, for example. That means what? That means you are creative. You have a good sense of—you can create imagery in your brain. You can connect seemingly unrelated things in the world in novel and intriguing way. Well, what's that a proxy for? Creativity is a proxy for cognitive flexibility. You are probably flexible. Is it good to have cognitive flexibility in your brain when you go around in the world and try to acquire resources? Hell yeah.
- 3.59** You want to be able to not do the same things over and over and expect the same result. You want to be able to, when something's not working, to shift gears. People with OCD keep doing the same thing even though it's not helping them, for example. They get stuck in these loops. They don't have cognitive flexibility. That's one example.
- 3.60** Throwing spears might be a good indicator of hunting in the old days. In present times, even though we are stuck with an ancient brain, it could be an indicator of physical health. That's another proxy for resource acquisition: good physical health. It could suggest you're good at manual labor in some extent. You can raise your children as

well. It's good for—you can lift and play with your child. There's tons of ways that talents can be a window into your ability as a potential mate. Is that clear? Very good.

- 3.61** It's less in the opposite direction. But there is something that's going on, and I often mention this. When I was in high school, I think I was in high school, I was sitting there in the audience, and there was this performance, like a school play. And then this girl, she goes on the stage and she starts singing. She sang like an angel. Her voice was so good, so nurturing. She hit the Mariah Carey, Whitney Houston tones, and my heart melted right away. My God, I felt something. I felt like, "My God, what is this?" Like I've seen something stepping out of a Shakespeare poem. She was something else.
- 3.62** What is that about? Why do we find a singing voice attractive in a female? I don't have the answer. But why do you think that might be the case? That's an open question.
- 3.63** Soothing children?
- 3.64** Soothing children. Were you in my other course, by any chance? I talked about it in the other course. I think so, absolutely. I think it has to do with, as males, we do prefer women that have a nurturing component, that can nurture, that can be soothing, that can be feminine in that way and take care of us when we are sick. And maybe, a beautiful voice like that activated those circuits in my brain. What do you think? Is that on the right—
- 3.65** Other perspectives? Let me hear. Anything else? What could it be? What do you think, Joey? Have any guesses?
- 3.66** It might go back to childhood.
- 3.67** Being carried? Lullabies? Possibly, Possibly.
- 3.68** Is there a particular look that we find attractive? You have a female face, for example. There have been studies done. Is there a particular proportionality or dimensions to the face that you might find more attractive? The answer is yes, and there have been studies on this. It turns out that symmetry is important, symmetrical faces. Why is symmetry important? In nature, everything living is symmetrical. A tree is not very symmetrical. A stone is not very symmetrical. But a human being, a mate, a lion, anything hiding behind a bush is symmetrical, so our dopaminergic centers get titillated by these symmetrical features, so we automatically prefer symmetry.
- 3.69** It's also a marker of health. If you have parasites in the stomach, the mom has parasites infecting her, the child is more likely to have an asymmetrical face. That can happen. So it can interfere with symmetry.
- 3.70** But beyond this, there are actually certain characteristics here. There's vertical: the space between the forehead and the brows, and then between the brow to nose and nose to chin. And again between the eyes, eye width and spacing across the face. This is a very particular geometrical shape, but if it is proportionally in the right place, you would find faces like that more attractive. I can't tell you exactly by centimeter by

centimeter, but there's something there as well. And this is going into the minutiae and the details, but there is that.

- 3.71** Another study was interesting. It showed that looking at female faces, again, taller foreheads, smaller noses, and fuller lips, was consistently seen as more attractive by men. In fact, the greatest indicator of attraction was lips for females. So there is that component; there are these aspects to attraction.
- 3.72** Now, Helen Fisher—we talked about her three: lust, love, and attachment. She has something else in the attraction domain, explaining what types we fall for and find attractive and end up with. She has four types. This has a lot of personal relevance for your own life, so try to remember this carefully. There's four types.
- 3.73** There's the dopaminergic type: novelty seeker, thrill seeker, risk taker. Dopamine types fall in that [category]. That's one type. Then there is a serotonergic type, the serotonin type, which is a more calm, grounded, emotionally stable family type, traditional, perhaps— not necessarily traditional in their belief system, but that they're more steady and calm.
- 3.74** Example: I am completely dopamine type: reward, travel, I can't stay in the same place for too long. I have to do new things. I do some area of science, and boom, I find it boring, and I choose another one. I take big risks. That's who I am: a dopamine type. I'm not very calm and grounded, and just sit around and enjoy the sunset. I don't do that much. I do love it sometimes. It can be beautiful, but then I start analyzing the sun and talking about various things. My brother, on the other hand, is a complete serotonergic type: family, children, calm, picks up his child and is very calm. It's fine. I have those traits too, I'm saying, but not like him. He's biased towards serotonin.
- 3.75** Then you have the testosterone type, which is the hyper— I don't want to say hyper-masculine, because it can be just who you are. You are more assertive, you are more domineering, you want to take charge, you want to push others to their limits. That's the testosterone type.
- 3.76** And then you have the estrogen type at the very end. The female, not necessarily, it can be a male too, but females tend to tilt towards that. You are more nurturing, you are more protective, you are more loving, you like to be directed rather than direct. That's the estrogen type.
- 3.77** It seems, based on data, that dopaminergic types tend to end up with other dopaminergic types. And that's a good thing because it can make the relationship last better. It's a better thing for you to end up with a dopaminergic type. So assess yourself. Where do you fall in these categories? And then you have the serotonergic type. They also tend to bond and have relationships with other serotonergic types. And that is also a positive thing. Often, it can lead to better relationships.

- 3.78** When we get to the testosterone type, it becomes tricky. Testosterone types, you want them to end up with the estrogen type because they can be more domineering and feel heard and feel masculine. The estrogen type, she actually likes that. She likes to be told what to—~~not~~ told what to do, but she likes to have support and somebody who can nurture and be strong. She craves that more. Is that clear?
- 3.79** I'll give an example. I have somebody in my family who is clearly, clearly the testosterone type. Actually, kind of inflexibly so. I didn't say who it was, so I'm not going to get in trouble, but he's very domineering. His view or no view; his way or the highway. He's kind of strict. And he was married to somebody who was also testosterone, a female that was very domineering. Guess what happened? Always fighting, always trouble. They couldn't get along. They were always fighting. Then they got a divorce, thank God. He married an estrogen type. He's thriving now. She's nurturing. He's giving, he's the masculine, he's protective. He has his voice being heard and all that. It just fits.
- 3.80** That's very interesting. When you scan people, when you do brain scans to assess some of these models to make sure it fits neuroscience in the deeper sense, it turns out that the default mode network, which is a network involved in processing of self and thinking about self, if you have a default mode network that's highly activated, you might prefer somebody who has a lower activation of the default mode network, suggesting that having brains that are opposite, fits the whole testosterone-estrogen scenario.
- 3.81** But we don't know about the dopamine situation, if the dopamine-dopamine [pairing] is always substantiated in the brain. We don't know for sure, but one thing is clear. Let me make this clear, which is that having two dopaminergic people is not always a good idea, if you can guess why. If you are thrill seekers—I'll give you an example. Dopamine is this neurochemical in your brain. If it's lacking, you lack motivation, you have no anticipation of joy and pleasure; you're flat.
- 3.82** People with depression, for example, they are having flat affect. Even when they're walking, because your movement centers rely on dopamine, sometimes somebody with depression, you feel like they have these psychomotor symptoms where they're very slow and things like that. Now, if you give somebody with depression dopamine in the brain, a drug that will amplify or boost dopamine, or even somebody with Parkinson's, where the dopaminergic cells are dying, and you give them dopamine, guess what happens? They might become pathological gamblers. They might become sex addicts. They might become totally erratic and become complete risk takers, and destroy their finances and their lives.
- 3.83** Imagine if you have two dopaminergic people coupling and becoming a couple. It can be erratic; it can become unstable. So being too dopaminergic, being too much of a dopamine type and then pairing with another too much of a dopamine type can create instability.

- 3.84** Same with serotonin. If you have somebody with serotonin: calm and steady, and walk in the park, just walking with the child and the stroller and all that, and the wife is doing the same, every day you're doing the same thing, it can get pretty boring. You can create a lack of excitement, a lack of spontaneity. It becomes the same. It becomes a habit. It becomes habitual, all of it. That's also not positive. Having some dopaminergic tendencies can definitely spark things.
- 3.85** First of all, in attraction, when you see that girl you're attracted to—and this is for those who are single, especially, very highly relevant. If you see that girl you're attracted to, now, what should you do? How can you approach her? How can you talk to her? What's the rules? What's the algorithms in the brain?
- 3.86** First of all, there's something called the three-second rule, which is, for God's sake, approach her within three seconds. The lesson here is that if you approach her, if you wait longer than that, your amygdala, your emotional part of the brain, the fear part of the brain, will bias your prefrontal cortex towards fear. You become fearful. Suddenly, you put her on a pedestal. Now you become scared, your emotions become riled up. You're going nowhere.
- 3.87** A three-second rule: that's the key. It's a way to avoid that tension. What can happen often—this happens to me too; I shouldn't say me, but it happens— is that you may become creepy. You start creeping. You're in that store, in that place, that social place, and you didn't go in the first three seconds. In fact, you didn't go the first three days, and now you become creepy and wander, do something and look, and it doesn't work. That's very bad. It's a very bad look now. You should have gone in the first three seconds. So that's the creep effect you want to avoid. You've had it happen. I can sense.
- 3.88** We talked about how the psychological aspect of attraction, which I want to reemphasize, I guess. Yes, she's beautiful. Yes, she has a spark. She has something you can't put your words on. They tell you, "Why do you like this girl? What do you see in her? She's not for you." But you see something there. You can't put words on it. So there is that component, and it's often tinged with some psychological expectation of what she might be like. You put together her features, and you create a construct of, "Oh, she might be like this. She's probably a good mother. She probably has these values," and you create a value system. You create a persona in your head around this person. So there is this great psychological aspect to it.
- 3.89** There's something called the bodyguard effect. I named it. It doesn't exist, but I named it—it's called the bodyguard effect. Have you seen the movie *The Bodyguard*? You've seen it, right? Frank Farmer, a competent guy. Now, Rachel, Whitney Houston, she's a star. She has this beautiful singing voice. All the men are interested in her. In fact, she has a stalker. Frank Farmer, the bodyguard, is brought in. He's very calm, he's very composed. He's not easily impressed by her. He's very professional in his work and his craftsmanship. And Whitney likes Frank, Kevin Costner, instead of all the other guys.

- 3.90** Now, what aspects of Frank Farmer's personality might she find attractive? I have some ideas. My idea would be that, first of all, the bodyguard effect: women like and prefer men that have a protective component to them; they can protect. They can guard her and protect her. That's the obvious part. But beyond that, his personality screams, "I am not easily swayed by the fashions and the waves going on in the world. I'm stable. I'm integrated emotionally, dopaminergically, prefrontally; I can inhibit my urges."
- 3.91** There are all these situations where Whitney Houston can be killed, be harmed, but he stays composed. He can dampen his anxiety, he can dampen his emotions, but he has an emotional vulnerability to him as well. So he has what you might call emotional contrast. This is another rule for today's lecture: emotional contrast. It's highly, highly attractive. Trust me, if a girl displays emotional contrast, my God, you might fall for her.
- 3.92** Emotional contrast is counterintuitive because I always thought that if you want a girl to like you, you've got to be a little bit cool. You've got to have a little bit of swag and not be too emotional. False. You have to show some heart, and then, sometimes, you go a little bit cold. I'm not saying play games. Going cold means a situation changes, and now you want to hold composure, show her that you are strong, that you can endure pressure, you're not easily swayed. That creates a picture in her mind where, "Oh, one minute he's very warm, but then he has this other side to him that's a contrast to the warmth. He can be cold and strong, too." So this is very, very highly persuasive.
- 3.93** In fact, females that are good at luring men into their traps do this intuitively. One day she's extremely warm, she's smiling, and the next day she's cold or withdrawing, and your dopaminergic centers go ballistic. It's, "My God, what happened here? What's going on? Is it me?" You become all crazy about that. So the idea that you should be cold and withdrawn is untrue. You want to be, "I have emotional contrast." Show these contrast effects. That's how the brain—it tickles the brain in the right centers.
- 3.94** We are out of time, but I did want to talk about the Johnny Depp effect. Should we keep that to the next lecture? Yeah, let's keep it to the next lecture. The Johnny Depp effect is a very interesting phenomenon. I named it myself, and I described it. Honestly, I will make this effect famous. I'm going to put in my next writings. Why is Johnny Depp so alluring? Why is he so charming? The guy doesn't even know how to talk to women. I mean, he's shy, he's introverted.
- 3.95** I'm like that too, and I remember my mentor Rama always said—and I didn't understand when he said that, but he said that women actually like that. He said, "Jalal, women like men that are introverted and a bit calm. It creates mystery and magnetism." And I was like, "No. I can't even open my bloody mouth. I can't speak. How is that magnetic?" But let's talk about that in the next lecture. Thankyou.

Lecture 4: Calm, Confident Love

At a glance — Calm, confident love.

- 4.1 So today we'll be talking about love, the different aspects of love, transcendent love, spiritual love we'll be talking about.
- 4.2 But I did have some thoughts about yesterday. I got some questions about culture and cross-cultural preferences. The few things I wanted to add was, definitely, these personality attributes, like kindness, generosity, emotional stability, good health, are universally thought to be good traits and good indicators for attraction. We know this. It's cross-culturally valid. Also, women tend to prefer slightly older men, and it comes down to the fact that women select men for their resource-acquisition abilities. We talked about that yesterday as well. So that's a huge part of it.
- 4.3 On the whole, mating nervousness, when you're nervous, and you want to approach a woman, for example, and you're very nervous, you have your adrenaline. You feel agitated, you feel nervous, all that stuff. Now, the question is, can you do anything about that? So, Baland, being nervous to approach that girl, can we do anything about that? Is there anything we can do is the question. And if that is the case, what can we do? What is the way out of my scenario of going every single day to the same place, seeing the girl, my heart racing, my butterflies being all over the place? What can I do to actually get rid of this?
- 4.4 Yesterday, we talked about the three-second rule, remember? The three-second rule is, immediately, you see the girl, you don't overthink it, you don't have your prefrontal cortex go online, you don't want your amygdala to fire away, you approach the girl, and you talk to her, and you deal with the situation pretty quickly.
- 4.5 But there's another approach as well, which is exposure therapy. You know exposure therapy? It's a therapy where, let's say, Joey has a fear of rats. He fears rats, for example. I will bring in rats, and I will expose you to rats, but I'll do it gradually. I won't bring in a big cage of rats and just throw it on you. In fact, I could do that. That's called flooding. That's called flooding. But I won't use flooding because it often can be overwhelming. So, what I use is a gradual approach. I may bring in a plastic rat for Joey and just give it to you. Then I bring in a rat that's timid and shy, that's over here, and I gradually bring it over to you.
- 4.6 Now, how can we use this in a dating situation or a women situation? Well, what you can do, for example, is Baland, as you know, is extremely timid and shy. He can't speak to women. This is a fact, not fiction. This is true. So what do we do about that? Well, what I could do, and I should be doing, I don't do, but I should be doing, I should go out, and when I go to a supermarket or a store, and I see a woman, I may talk to her. I may respectfully interact with her, maybe even with the one that is of my interest type.

- 4.7 I may find this woman interesting, and I expand the conversation a little bit, being detached from the outcome. I'm not trying to get this woman's number, trying to flirt with her, trying to be with her. No, I'm saying, talk to her, just be yourself and kindly have a conversation with her. Just be a human being. You do that day one, day two, day three. You do it at the gym, you do it at the supermarket, when you go out, wherever you see women, you just do it, and gradually, you learn how to interact with women. You gradually become more loose and calm and just down to earth with women. Does that make sense? That's an exposure therapy technique you can use. This is useful and can be done.
- 4.8 Oh, as well, we talked about attachment yesterday, attachment theory and how we have different styles of attachment. You might be the securely attached person. I might be the super avoidant. Then we have the ambivalent. Can we get out of that attachment style, or are we locked in that style forever? That's a question. We talked about how the circuitry is wired from childhood. We have our parents, and we have the attachment style from the mother that we learned, and then we use that for adulthood. But can we change that, or are we forever stuck in that state?
- 4.9 We can change it. We can. So what we can do is, or often just by being in a secure relationship, you find a girl who's actually calm, down to earth, and secure in her attachment style. You may feel safe, you may feel more comfortable, and slowly also develop a, let's say, less clingy and anxious attachment style.
- 4.10 But it can go in the other direction as well. So let me use myself as an example. My attachment style—my mother was loving, very kind, always helped me, played with me, took good care of me, always attentive. So I had what you might call the secure attachment. Now, I was around 13 or 14. I met this girl— by the way, when I tell you about these stories of meeting girls, I want to make one thing clear: I'm a virtuous, pious individual. I'm not a romantic guy who goes out and has flings left and right. But I'm saying we're human beings. I'm just trying to be honest with you.
- 4.11 In any case, I was 13, 14. There was this girl. I grew up in the hood, and so the girls can be tough. They can be tough. They're not like your average girls. They can be very hard. So if you do something wrong, they might literally just curse you out in front of you. They are hard. But there was this girl. She was different. She had brown, curly hair. I remember her brown eyes. She was beautiful. So I wanted to talk to her. At the time, I was pretty much a bad boy as well. I had to survive in that environment. So I wasn't a very good studious kid. I was bad, too.
- 4.12 In any regard, I got to know this woman, but she was extremely tough. And her attachment style, whatever it was, was extremely severe. So, from that experience, I became the classical conditioning rat, where you just give them an electric shock whenever you—it was extreme. She would be cold one day, then hot the other day, and then hot and then cold. Often, what you find is, particularly in what you might call first love, it's very turbulent, isn't it? You come to the situation with a very innocent

mind. You don't know that you're not supposed to tell her that—you think that the more you love her, the more she'll love you back. That's what you think, isn't it?

- 4.13** You think, if I give her attention, and I give her roses, so I would do that. I would literally, she would withdraw, and I would go closer. "Oh, maybe I'm not doing enough." So I would try to text her, this 13-year-old boy. All I did was texting her, and she withdrew even more. And then, I would try to be even more romantic, but she would—this is very common in the early [stages], when you are a kid, and you have these first-love scenarios or crushes. I wouldn't even call it love.
- 4.14** It wasn't love. It was just walking around and literally in the streets, and having conversations. So it was very pure and innocent. But that doesn't mean it can't be very intense for the experiencer. So you want to be able to keep boundaries in relationships and—as we talked about—prefrontally, have good integration of your emotions, limbic structures, dopamine regions, being steady, being calm, being grounded. You want to have that.
- 4.15** But we often underestimate the power of empathy, being empathetic, having true empathy, and how that can change relationships. Have you seen the movie Alpha? It's a really good movie. It's called Alpha. It's this chap. It takes place in the ice age. It's in the old days. You've seen it? You haven't seen it. It's in the old days; it's prehistoric times. There's this group in the wilderness, and they go out as a flock. They leave their mountain or a cave, and they go out to go hunting and bring in animals like they used to do.
- 4.16** One of the young kids in that group, his father is the alpha of the group. He's the protector, he's the old, he's the one who takes care of everyone in the group. The young guy is supposed to be trained to be more tough and rough. For example, they bring in a bull, they lay it down, and he's supposed to slay it, but he can't. He's too empathetic. He's trying, but he just can't. He breaks down each time. He just can't kill the animal. He can't slay it.
- 4.17** As the story unfolds, eventually, this guy is separated from the group. Now he's alone, and he's in the wilderness, and a bunch of wolves come to attack him and devour him. He climbs up to a tree, and one of these wolves, he's able to fight off. The wolf falls to the ground; it's injured. And [the boy] is now sitting in this treetop. Does that make sense? So he's alone now. The group is away. He's alone in the treetop. There's a wolf.
- 4.18** When he climbs down from the tree, he's about to kill this wolf. He's about to kill it. But then his empathy instinct kicks in, and he doesn't. Like before, he just can't. He just can't kill it. Eventually, the wolf recovers, and it becomes his friend. It helps him, carries him all the way back to his village and the group, and he survives because of his act of empathy.
- 4.19** The moral of the story here is that in romantic interaction, underestimating the role of the TPJ, the empathy regions of the brain, we shouldn't do that. We said—where's

empathy located? Where do we have empathy? TPJ. That part of the brain is involved in taking another person's perspective. The medial prefrontal cortex is involved in seeing somebody else's perspective as well. So underestimating that is not a very positive thing. It can bring bad consequences.

4.20 Let's talk a little bit about mate selection on social media and how that affects us. Social media has this effect that we feel like we have to portray ourselves in certain ways. Do you agree? Why do you agree? Tell me. Gabriella?

4.21 To compete.

4.22 To compete, yeah.

4.23 To show that we're winning a life.

4.24 Are we showing a fake side of ourselves, or are we being genuine?

4.25 I think most people are being genuine, but only about the good things in life.

4.26 So they are selectively showing the positive aspects of their life and not their negative. Agree? I agree. I think that's a huge part of it. So what social media is really tapping into is, I think, when it comes to dating, first of all, a lot of these short-term aspects of what you might find attractive in a mate are amplified. The guy, for example, with big muscles and he's standing next to a Ferrari, and he's maybe looking rough and all that, that guy will get a lot of likes and followers, and he may even get on dating apps and be popular. But how do you portray kindness? How do you portray empathy, for example, in a social media context? How is that possible? It's very difficult. So there's an issue there.

4.27 Also, you have what's called—you might develop decision paralysis, which is because of the availability of so many people, and just females, there's all these females on Instagram, there's all these girls, maybe in your DMs, or maybe just overall, what do you do? You get the sense, the illusion that all these women are for you, that somehow these women are interested in you, and these are genuine potential mates. But it's a trap.

4.28 In the old days, the brain was not accustomed to filtering through so many potential mates and saying, "Oh, this is a mate, this is a mate, this is a mate." We're not wired for that. We used to get married to whatever woman was in our local village, and that was it. That was fine enough, and you would have a good marriage. But social media gives you that false impression.

4.29 Are you bored? I feel like you might be a little bit bored. Is this boring?

4.30 I'm not smiling a whole bunch.

4.31 No, that's okay.

4.32 I'm a bit more composed.

- 4.33 No, that's good. You should be composed. But is it interesting? I want to know if it's interesting. Otherwise, we can shift gears. We're here to have fun as well. We should have fun. Science should be fun. My mentors always told me that science should be fun. But of course, dopamine—you know what dopamine is? Dopamine, you can't sustain all the time. So if dopamine is high today, it will automatically fall down. This is the curve of dopamine. Let's call it, not even a curve, it's a curse of dopamine.
- 4.34 Dopamine, the way dopamine works in the brain is—and this is literally true—dopamine must go down to baseline or beneath it in order to sustain itself. It cannot remain high all the time. So if I have an amazing time with this, since it's a romance course, with this girl here, and it's amazing, I'm having fun, and my dopamine centers are firing away, the next day, automatically, my dopamine centers will not only go to baseline, they will go beneath the baseline. Did you know that? In order to replenish. They cannot stay the same. They cannot stay the same.
- 4.35 That's why you have postpartum depression, for example. You have a birth, you have the child, you feel so happy, and then months after, you feel sluggish, and you don't know why: postpartum depression. Or you've been on vacation. Amazing. You went to Paris, saw the Eiffel Tower, ate some shrimp, enjoyed yourself, and then you come back to Arizona in the desert, and you feel low. Or is it the sun? Maybe it's too hot here, whatever. You're just not feeling all right. But that's because dopamine goes up for a few weeks, but then it has to fall down for another couple of weeks. It always has this seesaw. Does that make sense? Keep that in mind.
- 4.36 So whatever you do, whatever you do in any situation, don't overdo it, in fact, because if you overdo it, you might end up depriving yourself of all the dopamine. I'm not saying being too calculated either. You shouldn't say, "Oh, I'm not going to have fun." But what I'm saying is, don't go out and have like four parties in four days, for example. That's what I'm saying. Try to separate it out. Does that make sense? Good. I like you're smiling more now. We have to be happy here. Good.
- 4.37 Shall we talk about the Johnny Depp effect? You wanted to hear about it, right? Okay, good. So what's the Johnny Depp effect? Well, why is Johnny Depp so damn charming? What makes him so attractive? His personality, of course. His looks, females may think he looks good, too. But there's something about the shy personality. Does Nicolas Cage have the same thing going on or not? Not really. He's more intense. Have you seen the movie where he looks at that woman, where he's an angel, a fallen angel? You may have seen that. It's a bit creepy. He looks at her a lot.
- 4.38 Johnny Depp has something, which is a calm, grounded presence. He's an introvert. He's mysterious. You're trying to get information out of him, but it seems difficult. You may feel that is a form of maybe insecurity. Is he insecure? That's one view, but it turns out, no, women don't feel he's insecure. In fact, they feel like this guy is more parasympathetic in his tone, first of all.

- 4.39 The nervous system is made up of two branches: the sympathetic and the parasympathetic. We talked about that yesterday. The sympathetic is stress, anxiety, agitation. Parasympathetic is when you're calm, grounded, and more relaxed. An introvert will automatically have that. So when I'm in that situation where, let's say, it's in a gym or something like that, and there's a girl I have my eye on. I may act—not act. I shouldn't act. I should be like that, but I should be more grounded, I should be more present and seem more calm. That's part of the Johnny Depp effect.
- 4.40 Second, he doesn't reveal himself too quickly. He doesn't overshare his—he's hiding a bit, but not hiding completely. What do I mean? Dopamine centers were wired to spot gazelles on the savannah. On the savannah, you might see a gazelle or an animal of some sort, a zebra, or something, behind a bush, and whenever you saw that, it was trying to find it, your dopamine centers would fire. So your dopamine is always searching. It's always trying to find and search. It has this property.
- 4.41 The same thing goes for a man— or it could be a woman, perhaps it could work for a woman, too, to have this Johnny Depp effect, I'm not sure. But I'll keep it for males, first of all. We can speculate on the female aspect. I think there are different attributes for females that might be interesting, but we'll talk about that. So he doesn't reveal himself, and that makes the woman's dopamine centers always crave more: more information, more knowledge about the person. What is he up to? What is he about? So there is that element.
- 4.42 He doesn't seek validation. So the alpha guy in the gym, he's lifting all these weights. He goes lifting more and more. The woman is getting closer, and he keeps adding weights. What is that about? Well, that's seeking validation, external validation. Whereas the Johnny Depp type, he doesn't. He stays to himself. He stays grounded. Now, he's cool. He's cool-looking. He has nice—he grooms his hair, he has a good style. Perhaps his style is simplistic, not flamboyant like the alpha, the so-called alpha guy.
- 4.43 But he has this self-reassurance, and that's important because your dopamine centers can be driven by external validation. That's one model of the world, and that's very finicky. It's a very fleeting way of looking at the world because you're always dependent on how the world perceives you. Does that make sense? Your happiness is tied to how the world is giving you feedback. It's not tied to your actual own sense of pleasure.
- 4.44 Now, the Johnny Depp effect would lead you to be driven by your own self, and your own thoughts, and your own discipline. It shows discipline, prefrontal discipline, DLPFC discipline. You're well-controlled in your prefrontal lobes. Parasympathetic, we said. It shows your dopaminergic centers are internally driven. These are all features that women would find attractive.
- 4.45 But I do think that there's other aspects, too. I think things like an introvert—for example, what I sometimes might do if I'm in a certain scenario and there are women around, I may sometimes just go and, even though they're all these pretty women, I

may go look at the window and look outside the window. It's a trick. It's a trick, but I also like to look outside the window. But it can be a trick in the sense that it shows depth. It shows your mind is not just going to the lady right away. You have a sense of inner depth, a sense of mission that's external and untethered from the pretty body or the nice legs or whatever it is. You are independent, and you have depth.

- 4.46 You have your frontal parietal lobes, where you have selective attention. It's finely tuned, those circuits, because you can pay attention to something completely different. You're not distracted by this world's pleasures, so to speak. Does that make sense? It's a form of discipline, but it's also selective attention. We have circuitry in the brain for selective attention. Do you believe it? Frontal parietal. This is my theory for the Johnny Depp effect. What do you think? Does it make sense?
- 4.47 It also, I think, taps into empathy. I think the introvert is seen as more empathetic because of his more calm and grounded personality. The woman might think he's primarily serotonergically driven, or he at least has a basis of serotonin, which makes him more calm and grounded, but also makes him more empathetic. I think that her brain computes this in an unconscious manner. She doesn't know, but she goes, "Oh, he's calm. Oh, he's grounded. That means he must have high levels of serotonin as well, which would mean that he's high in the social hierarchy as well, and it would mean that he's cognitively flexible, so he can acquire resources."
- 4.48 What do I mean by that? Well, we talked about yesterday, cognitive flexibility, that if you are very rigid in your cognitive way of thinking, you have low serotonin in the orbitofrontal cortex, down to the basal ganglia and the nucleus. That region is hyperactive, and it's rigid. That has to do with serotonin and low serotonin. Does that make sense? Am I rambling, or does this make sense? Okay, good. I want to make sure because I sometimes use technical terms, and I overwhelm my audience. They go, "What is he talking about, the TPJ and serotonin and medial prefrontal lobes and all that stuff?" Is it clear? You're new here. Is it clear? Okay, good. I want to make sure.
- 4.49 Before we go to things like transcendent love, we'll talk about self-love, too, and self-compassion and all that, but there is a movie that I think has a lot of insights. Have you seen the movie Groundhog Day? Who has seen the movie Groundhog Day? You've seen it? You've seen it. It's a classic. You've seen it, Joey? It's good. I love that movie. I really love it. It's good. To those who haven't, this guy is stuck in a time loop— I'll make it short because we have to cover a few things, but what I do find interesting is this transformation. So there's this guy, he's like—you have seen it? Oh, you've seen it. Good. So maybe I'm preaching to the choir here, but this guy, he's trapped in a time loop. He keeps repeating every single day. So the same day he wakes up at 6:00 in the morning, he goes out, he's about to encounter this girl, and then the day ends and the same day happens over and over.
- 4.50 Now, he's trying to court this girl, pursue her, be with her. That's his goal. So what he does is he comes up with—initially, when he realizes this is the same day occurring over

and over, what he does is he tries to gather information about her. He becomes very Machiavellian and very dopaminergic. In fact, he is narcissistic as well. So what he does is he's using his ventral striatum, nucleus accumbens, and those circuits make him see that girl that he's interested in as a prize, like Gaston. She's a prize. I will gather information about her life now, since I have all the time in the world, and I will try to slowly persuade her, slowly make her want me by knowing all kinds of things about her.

4.51 She likes French poetry, so he learns French. You know French, right? Oui, oui. So he talks about French poetry and all that, but it doesn't work. She can see right through it because her mirror neuron system—we talked about how brains synchronize when we are having romantic encounters—her mirror neuron system can sense this guy is not genuine. She can sense that this guy is trying too hard. He's playing a character. He's not authentic. He's not his true self. And so she feels it's off-putting.

4.52 But eventually, a flip happens, and this chap starts to recruit his ACC—now we are trying to get back to some of the centers we talked about before. The ACC is doing what? Empathy. He recruits his ACC, his empathy department in his brain, and he goes, "Oh, my God,"—he sees this beggar on the street that eventually dies, and he feels empathy. He tries to help this person by preventing the person's death. The person dies every single day. So now his focus is on this person, and then that spills over, and now he goes out and helps other people.

4.53 So his ACC and the medial prefrontal lobes—the ventral medial prefrontal lobes are involved in what? Intellect plus emotion, empathy plus emotion. So, taking intellect, logic, knowledge about the world, and fusing that with emotions and empathy as well. It's just above the ACC. That makes him more empathetic, less obsessively focused on the girl, and then eventually, when he does try to court the girl, it's all in one day, but when he tries to court the girl, she can see that he's genuine. Her mirror neuron says—she can feel that he's authentic.

4.54 People around come and greet him and say, "Hello, you helped me today. You did this for me today. You were so kind today." We talked about that yesterday—social validation. He is showing in vivo, in real time, that he's a kind, empathetic person. And women love what? Social context and cues of attention structure. So he's showing that people are paying attention to him. He has talent. In that one day that he has over and over, he learns to play the piano. That can show motor integration and prefrontal motor integration in the brain. So he's sitting there, and he's playing the piano, and he's really skilled at it. She's impressed. In fact, I think this movie covers everything we talked about yesterday in just one movie, pretty much. In any event, she eventually likes him, they end up together, and the time loop ends.

4.55 So yeah, we're shifting gears. Self-love. What is self-love about? Is self-love important? What structures in the brain might be important for self-love? What about the medial prefrontal cortex? Today, we'll talk a lot about the medial prefrontal cortex—up here: the medial, dorsal, ventral. Is the ventral part of the medial prefrontal cortex important

for empathy? Yes. Is the ACC important for empathy? Yes. It turns out it's also important for self-empathy. There's a fine line between self and other, and there's a fine line between empathy towards others and self.

- 4.56** The same circuits involved in empathy towards self and others—there's a shared overlap in the brain. The default mode network—it's a network. So brain circuits, the circuitry going from the frontal part to the temporoparietal. The temporal, remember, is the emotion part of the brain. Parietal is what? Sense of self and body image. Remember that? Sense of self, multimedia studio. I feel like I'm Baland, anchored in this body. Parietal lobes. That circuit, called the default mode network, is important for daydreaming, self-reflection, mind-wandering.
- 4.57** For example, if you have somebody, a guy, and you put him in a scanner, a brain scanner, and he sleeps in the scanner, or not sleep, he's just resting in the scanner. If he has depression, say, so he's depressed, he'll have a lot of default mode network activity. That part of the brain will be hyperactive, in fact. He's daydreaming about himself, and the medial prefrontal lobes, which are involved in self-referential thinking, are also going haywire, so he's thinking about himself versus others. "Why am I stuck in this situation? I feel bad about myself. Life is bad." The focus is generally directed inward. Does that make sense? That's a depressed person.
- 4.58** All of us will have some default-mode network activity going on. For example, Jonathan over here was daydreaming right now—I saw it; you weren't focused—that was default mode network activity at its peak. So that's an example. I'm just joking. You're always paying good attention. So that would be, it turns out, actually when we dream. When we dream—have you seen my dream course? When we dream, the default mode network becomes super active as well. So that's part of the dreaming experience.
- 4.59** Daydreaming and actual dreaming, there's a fine line. The medial prefrontal cortex, self-referential thinking, thinking about self, I, me— in dreams, all you do is thinking about you. You are the main character of the dream. It's all about you and your personality. So that's the medial prefrontal lobes. Does that make sense? Is that clear? Good. Very good.
- 4.60** Default mode network is highly active, especially in people that are depressed. But in all of us, the limbic centers of the brain are important for self-love. Limbic centers—I forgot your name. What's your name? Sarah. I'll tell you, limbic center, we talked about it yesterday. Tucked behind your ears, there's this structure. It's a ring-like structure. It has the amygdala, for example, which is the fear department of the brain. It's also involved in positive emotions, too. But it's mainly fear and fear processing. It has a ring-like structure. It has a septum region. If I zap the septum using an electrical current, I'll go zap it, you feel emotionality. You feel very emotional. In fact, you feel euphoria if I zap that part of the brain. That's also important.

- 4.61 This region is also important for a sense of self-love. It's involved in love towards others, but also love towards the self. Limbic, the insula, which is the self-awareness and mapping my internal state part of the brain—remember that part? That's also involved in self-love. Dopamine, of course, is involved in self-love. It's just different from the dopamine I get from love of other person. Love of another person is: I see this person, I'm excited, I am ecstatic, and it's driven by her beauty or her presence, or her personality, or my little brother's cuteness, or whatever it is. That is what? That's dopamine driven by another person.
- 4.62 But you can have dopamine being driven mainly by self-love, the respect you build for yourself when you take care of yourself, when you are compassionate to yourself. So self-love, happiness. It's not quite self-compassion. It's more a sense of self-respect. It is having boundaries, being aware of yourself and the boundaries that you have as a person, and being empathetic towards yourself.
- 4.63 Self-compassion is more, I would say, slightly more tilted from self-confidence and appreciation. It's more like if you are suffering, and you have empathy towards yourself. It's, "Okay, I'm going to be less judgmental. I'm going to be less strict on myself. I'm not going to have my orbitofrontal cortex, the judgment part of the brain, the OCD part—but it's also the judgment part—I'm not going to have it criticize myself so heavily. I'm going to forgive myself for not doing so well yesterday when I was with my mama, with my mom. I should have been more kind towards her."
- 4.64 So self-compassion would be, "Okay, don't be so hard on yourself." It recruits slightly different structures. It's more the ACC, mainly, more ACC-directed. It is more oxytocin-based: oxytocin, vasopressin; these hormones and neuropeptides where you calm yourself down. Amygdala shutting is also a part of that. But it's slightly more towards the empathy and those regions of the brain.
- 4.65 But definitely, I would say in our societies today, I think, do you agree, we are taught to be very harsh on ourselves. Perfectionistic, in fact. If you don't meet these standards, you're a bum. If you don't get perfect scores on this test, you're nothing. You're useless. You're no good. We're very harsh. I think everything tells us in neuroscience that this is wrong. This is a wrong approach because being overly harsh and self-critical will do what to your brain? Activate the amygdala, the fear centers of the brain. It will elevate what? Cortisol. If cortisol is chronically elevated, you get what? Inflammation in your body. Serotonin goes down. When serotonin goes down, you become more rigid in your thinking. Everything we know tells us that being harsh on yourself is a bad thing.
- 4.66 If you are too self-critical and you have chronic stress for too long, what happens is that your prefrontal cortex has receptors. It has receptors for cortisol. Cortisol will latch onto these receptors in the prefrontal cortex and literally shrink the prefrontal cortex. Do you know this? Literally shrink it. So your ability to be focused and have attention and be logical and cohesive and be serially well-integrated, it goes down the drain. It's gone. By you having excess cortisol for a very long time, it will shrink the prefrontal cortex.

- 4.67 In fact, the hippocampus part of the brain, which is also in the limbic structure, the memory part of the brain, also has tons of cortisol receptors. Meaning what? Have you noticed that in some of my lectures, I may forget things? I seem like I can't remember things and stuff like that? What is that about? Cortisol, heightened cortisol can tamper with your memory, not only short term, as when I'm giving a lecture and I'm forgetting half the content, but also long term. It can shrink the hippocampus. It can shrink it. That's some of the things to remember. Being too harsh, being too self-critical, doesn't work.
- 4.68 Yesterday, we talked about meditation. We talked about the gamma activity. Remember? The fast activity that was discovered when these monks were meditating, lying there all day. The more they were meditating, the more gamma activity they would have. And gamma activity, of course, is a high from a frequency in the brain, and it also recruits from far-flung regions, regions for perception, intellect, but also emotion. It's a window into perception in a sense. But that's not all. Interestingly, these guys who meditate, the TPJ region of the brain, the empathy part of the brain, or not the empathy, it's the empathy, it's taking the other person's perspective part of the brain, also becomes fired up from meditating.
- 4.69 Things like planned movement, the planned movement part of the brain—in the brain, you have what's called the premotor region, which is behind the prefrontal. You have the prefrontal, the logical part of the brain, the medial, and then you have the more cold cognition we talked about—it's the DLPFC. Right behind it, there's something called the premotor region, and then there's the motor region. Premotor, motor. In the motor region, whenever I move my arm from A to B in space, there's a motor command. But whenever I just plan a movement, there's a signal going first from the premotor area to the motor region. There's a signal going from premotor—I'm planning, I'm wanting to act—to the motor, and then you act.
- 4.70 People that meditate, interestingly, what you find is, curiously, as they're meditating, their premotor area might light up. Sarah, what do you think? Why might that be the case? What do you guys think? Why might somebody who is meditating—specifically on helping people in the world, so this is meditation that's focused on helping others, not only self-compassion but also compassion meditation overall: "How can I rescue somebody that's suffering?"—their premotor neurons are going ballistic, dancing with activity. Why do you think? Any guesses? Somebody else is suffering, your premotor neurons are firing as if you're about to act.
- 4.71 Planning how to help.
- 4.72 Planning how to help them. Exactly. That is the case. It seems like somebody who is meditating by merely lying there and imagining somebody else suffering, he's almost like he can't control himself. His premotor neurons are saying, "Buddy, go rescue this person." So you have that as well. It's part of the whole—it signals really how the brain gets riled up from this meditation business.

- 4.73 And really, there's a fine line between, I'm not sure if you've covered this, but there's a fine line between seeing something in the world and imagining something in the world. Did we cover this? When I see something over there, my neurons will fire. When I imagine something, I close my eyes and imagine the same person; my neurons will also fire as if I was seeing them, but only a subset of them, so it won't fire as vigorously. But the brain works like this. When I'm dreaming, my neurons will fire as if I was awake, but again, in a slightly reduced manner. But in the same pattern, it's not a different pattern. The brain is operating in the same way.
- 4.74 We talked about empathy yesterday; let's continue down that route a bit. You should know this. There's something called mirror neurons in the human brain. There's a lot of debate and controversy about mirror neurons. Do these neurons actually exist in humans or not? Some people will say they don't. The reason is that we can't really crack open a human skull and record from single neurons. That's not possible. So we can venture a guess. It seems like there could be mirror neuron systems in the brain.
- 4.75 A mirror neuron is the following. Gabriella inflicts pain on her hand like this. I'm watching you. I feel pain in my hand. Not literally, but I empathize with you. I have empathy. In fact, whenever you do that action of poking your hand right there, motor commands are going from the motor areas, or actually, in this case, it would be the somatosensory, the sensory areas, and they would send a command of pain down my hand. Actual, literal pain. Do you know this? It's intriguing. She's inflicting pain on her hand, and signals of the somatosensory cortex in my brain are saying, "Baland, you should feel pain, too."
- 4.76 Then, Sarah might be asking, "How come Baland doesn't literally feel Gabriella's pain? What's going on?" If my pain centers are being activated by merely you inflicting pain on your hand, how come I don't literally feel it? Well, it turns out in our skin and in our joints and muscles, and elsewhere, there are signals vetoing that somatosensory signal saying, "By all means empathize with Gabriella, but don't literally feel her pain." So it's saying, "Have empathy, but don't actually have a pain signal." Is this clear? We all have this. Same for movement: whenever you move your hand, the signal in my brain is saying, "Move." But there's a vetoing signal saying, "It's not actually your hand that's moving, so stay still."
- 4.77 On the pain example, in fact, if I was to remove all my skin right now and I had no skin and you were to do the same thing, I would literally feel your pain. Do you know this? The only barrier between me and you is this thin layer of skin. That's the only barrier between self and other. Is that interesting? It's very important.
- 4.78 Now, on that point, mirror neurons, there's been tons of controversy around them, and for a long time, they were thought to be crucial for love. When you fall in love, the mirror neuron system is mimicking the other person. It's important for understanding their states. It's important for understanding just general, not only romantic love, but all kinds of love. It's important for having insight to another person.

- 4.79 But I want to add the caveat that there is a lot of debate around the mirror neuron system theory and how they exist in humans. The way I want to put it is that it seems like we cannot say for sure whether we have mirror neurons, but there are systems or circuits that would suggest something very similar at the very least. I think they exist in humans. That's my own view.
- 4.80 There was this guy. He walked into our laboratory in California. I was with my mentor, Ramachandran. He came in. He was wearing clothes that would suggest perhaps he was regal. He seemed very serious. He had a serious look to his face. He sat down and he was having a conversation with us. He seemed normal, but a bit stiff, a little bit like somebody from the Victorian age. Suddenly, he took out this notebook, opened it, and it was full of all these symbols, cryptic codes, it seemed. Religious icons, crosses, David's Star, all these things, and it was full of it, and he kept talking about it, how he had these revelations, he had these insights, how God would speak to him. I was pretty curious.
- 4.81 Then he talked about his own life. He seemed quite self-righteous, too. He talked about how he never—he said that he had no interest in women, that he was interested in women, generally, but he had no sexual desire for women. In fact, he was so focused on God that God was everything to him. A few minutes later, a few of the graduate students, the RAs, assistants, would come in, and he would flirt with them. So it was contradictory, but we did notice this. As we were talking and continuing the conversation with this chap, it turned out he had what's called lobe epilepsy.
- 4.82 In the brain, in the temporal lobes, since the age of eight, he has had seizures in that emotional core of the brain, firing away. An epileptic seizure is when neurons there go haywire, they fire ballistically, they become hyperactive, in fact, and you have what's called kindling, and that can spread to the rest of the brain.
- 4.83 Neurons basically work in this way. They can be inhibitory or excitatory. There's these two states. It's very simple, in fact. So you have over-excitation in the emotional core of the brain, and that can spread to the Wernicke area of the brain, the meaning-making part of the brain as well. It can spread into those regions.
- 4.84 Each time he would have these seizures, he would see a light, and he would feel like he was one with God. Then, after the seizures were gone, he was hyper-religious, hyper-poetic. He would say, "I am actually a prophet." In some cases, some people say this with these experiences. They might develop hypoglyphy, so they would write, sit down and write notebooks, page after page of poetry about love, about God, about these things. When I say love, [it's] love for God and the universe, not human. A very peculiar condition.
- 4.85 Now, Ramachandran and I wanted to study this phenomenon further. We wanted to explore. Does the personality change completely after the seizure? Do they literally go from an atheist to a believer and hyper-religious after a seizure, or does it go away after

the seizure, like after a few months? This is a key, critical question. Now, a problem with this was that a lot of these folks, and people with seizures generally, take anticonvulsive benzodiazepines and things like that. Exactly. And that turns out that can have an overall calming effect on the brain. It can make the brain less active. So it's very hard to assess their personalities. That was a huge problem. So we couldn't really crack the issue here, the code for this.

- 4.86 But it's an interesting question. What do you think? Is the personality changed completely after the seizure? Or what happens? Does it stay the same? Obviously, the problem is that you can't know how the personality was before the seizure. That's impossible. You can ask, of course, but that's not very scientific. You can ask about the person's life. Was this guy religious? But in a scientific, experimental way, you really need to assess it after, and that's very hard to do.
- 4.87 But it's an interesting phenomenon, this temporal lobe epilepsy. It tells us that meaning-making and the "God module" of the brain, so to speak, might be in the temporal lobes. That's part of the brain where we make sense of spirituality and God. It seems to be the temporal lobes. In fact, in the 1990s, there was this helmet called the God Helmet. Persinger was his name; he developed this helmet. People would wear it, and it would stimulate the temporal lobes. About 80% of people stimulated in the temporal lobes would feel like God was watching them. Some presence, some supernatural being was in the room watching them, or there was God, this experience, by wearing this helmet, by stimulating the temporal lobes, the limbic structure and the temporal lobes. It's very similar to temporal lobe epilepsy.
- 4.88 I think for spiritual experiences and a sense of cosmic feeling about the world, there's a chemical that's of key importance, a neurochemical, in fact. I call it the ghost receptor in the brain. I literally coined this term in one of my papers. It's called the ghost receptor, and it works like this. We normally have serotonin in the brain. Serotonin is good for what? Mood, feeling good about yourself, well-being. That's serotonin. Serotonin can activate various receptors. So there's receptors for serotonin. We know what a receptor is. It's these parking spaces on the surface of cells. Serotonin latches onto a parking spot, and then serotonin can communicate with that cell and affect the cell. Does that make sense? A receptor.
- 4.89 Now, there's a particular receptor called the 2A receptor in the brain, 2A. This receptor, we have tons of them in the limbic part of the brain, the limbic structure, and they seem to be involved in spiritual and cosmic experiences, the sense of awe and a sense of spiritual high. When you take psychedelics, LSD, these drugs, you tickle what receptor? Serotonin 2A. You tickle that receptor artificially, in fact, and that's why you feel like everything is wonderful and cosmic and all that. But you have it. Your brain itself has the receptor, and it can become activated.
- 4.90 In fact, when you dream, we believe that in your dream, when you're dreaming, the serotonin 2A receptor is what is causing that cosmic feeling and supernatural nature of

the dreams. So it's very important to know about this receptor, the serotonin 2A receptor, and how it contributes to spiritual experiences. How are we doing so far? What do you want me to talk about? You seem very tired today. What do you want me to talk about? Anything in particular? You don't like supernatural stuff, huh? What's that? Different brain waves? Gabriella, you were saying something?

4.91 I love the spiritual stuff.

4.92 You love the spiritual stuff? Pituitary, huh? Well, let me tell you this. Let me tell you this. By experiencing awe, by standing underneath a beautiful tree, this most beautiful tree, and you experiencing awe, what you have is the default mode network, involved in self-referential thinking, depression, automatically shut down. You become very outward-focused. And if somebody passes by you, you become more altruistic, more empathetic. The sense of time goes away. So time becomes expanded. That's a sense of awe; that's been shown.

4.93 So having a sense of awe is something that we should all try to get in our lives. Experience awe. Have that sense of self erode, an erosion of self. And that happens, that can happen by going to the art museum, watching beautiful paintings, looking at a beautiful sunset. All of this will automatically offset things like stress, high cortisol, all these things that are bad for you. So I can definitely recommend that.

4.94 We are running out of time. I just want to close with this story. I think this is a beautiful story. There was this girl in India. She had a tumor in her brain. That tumor, as an eight-year-old child, they knew that it eventually would kill her. They knew that. It had grown to a stage where she could live. So it was more like a ticking time bomb, but it wasn't killing her yet. She lived until 18, and at 18, the tumor is back with a vengeance. It is aggressive, and it will kill her very shortly. The tumor is in the temporal region. It's creating temporal lobe epilepsy-like features in her.

4.95 So she's lying in the hospice, about to die, and she's seeing all these panoramic visions of India, of herself playing—at this point, she's in the United States at age 18 in the hospice, and she's seeing all these beautiful rivers, colors, her childhood home. It's just beautiful in front of her, beautiful and beautiful, and she's dying there. Now, the bizarre thing is that the hallucinations are very broad in their manifestation. So she's not merely seeing the same light over and over as a normal epileptic TLE person would have, the normal temporal lobe seizures. She's seeing all these visions that seem almost like a movie in front of her.

4.96 And then eventually, doctors were puzzled. "What's going on? How come she's having these seizures that are so cinematic?" And she's saying, "I'm saying goodbye to my childhood home. I'm saying goodbye to my own life." She's very calm. They thought initially she might even have psychotic symptoms. She might be psychotic. She may have some mental breakdown. But she was too positive and calm and coherent for that. Then, eventually, she said goodbye, and she died.

- 4.97 But I thought that was a really intriguing story of how somebody with a tumor in that part of the brain, that their brain, instead of creating the normal scenario, creates this almost farewell to the world that's so poetic and so beautiful. This is one of Oliver Sacks's stories in one of his books. I can recommend that. All right. Thank you.
- 4.98 What determines someone's level of empathy? Some people have a lot more than others.
- 4.99 Very interesting. I think that, again, we have to say what empathy: cognitive versus emotional. The psychopath has tons of cognitive empathy, but I assume you might mean emotional empathy.
- 4.100 Yes, I mean emotional empathy.
- 4.101 Emotional empathy. I think that what that can reflect is a lot of things. First of all, how you were treated as a child, your parents, affection and love—was that enough to nourish your oxytocin and vasopressin systems? Did you learn to model yourself, your place in the world, and vis-à-vis other people in a way that's empathetic? I think that plays a long role. We'll talk about altruism at some point, but being exposed to other people's suffering can also help titillate your empathy systems; you're seeing other people suffer, and that can do the trick.
- 4.102 Now, if you are living in a castle and you're getting spoon-fed and all that, it's a bit different. You tend to develop more egotistical and more selfish circuitry. But there are other ways. There's other ways to elicit empathy. I think that we've talked about this a lot in my days with Ramachandran, who often talked about: how can you create more empathy? I think we talked about things like developing apps where you look at people that are suffering, and you see somebody in pain, you're seeing that, and that makes you more empathetic. Go on YouTube and watch some of the worst scenarios going on in the world. Don't tell me it doesn't make you more empathetic to see a child suffer. Don't tell me. It does, doesn't it?
- 4.103 But if you look at—actually, it's an experiment for you guys. It's an experiment I'd like you to make. One of these days, go home, the whole day, watch war scenarios. Whatever part of the world doesn't matter. Just watch war, people and children suffering. Then rate you on a scale of empathy from 1 to 10. Then, a few days later, go watch cosmetic videos the whole day and things about—you know what I mean? Something very superficial, watch that, and rate your empathy. There you have your answer: the first one titillated your empathy center. So I think we can, definitely. This is a crucial concept for the brain overall, which is that the brain is a use-it-or-lose-it brain. The more you use your empathy circuits, the more you will be affected, and you can help those regions.
- 4.104 Someone with an increase in self-compassion or self-love, is there a likelihood that they would be a serotonergic type?

4.105 Self-love? Yes, I would think that would be a high possibility. High possibility. You can't say it for sure, 100%, but just given the fact that serotonin is unavailable—or it's available; it's the most abundant neurochemical in the brain, but in depression, it goes away. It's much less. So yes, we do think so. It's that well-grounded feeling. You think about the brain as if there is this wrestling match. There's this battle going on between all the outward-going forces—dopamine, acetylcholine, noradrenaline—and then you have the more calming—serotonin, and then the neuropeptides we talked about—oxytocin and so forth. So, yes, you would flip more in that direction for sure.

Lecture 5: Heartbreak and Healing

At a glance — Heartbreak and healing.

- 5.1 I feel like, on the whole God portion and spirituality, I wasn't completely—I want to make it clear. And by the way, in this lecture, we'll talk about heartbreak and the aspects of heartbreak. I just want to finish up on God and the spirituality aspect.
- 5.2 Really, when you are engaging in a prayer, for example. So it doesn't have to be meditation only. It can be prayer as well. What you are doing is you are, in a sense, or you are actually, literally, dampening down your prefrontal cortex. We know this from romantic love as well. So you're dampening down those circuits. You are activating your parietal, which has to do with the sense of self and the sense of self in space. That's becoming activated. In fact, it's becoming broadened up. So your sense of self, the sense of me and the sense of I, the sense of Baland, is dissolving and becoming merged with something bigger. So that's the parietal lobes. We know that's happening.
- 5.3 Aside from the parietal and the DLPFC—excuse me, well, the DLPFC as well, the PFC overall, turning down, you have a neurochemistry that's really hard to pin down exactly. It's really hard to say exactly what's going on. There's something—and this is a point I want to make clear in this lecture as well. I can be reductionistic in my approach and say, "This part of the brain turns up, this turns down, this is active, this is less active." But in fact, the brain is complicated, and all these states are very hard to measure and pinpoint exactly, and say, "This is this, and this is that." I mean, there's just something about awe and spirituality and God that's hard to put in words. It's ineffable. It's hard to describe.
- 5.4 Let me give you an example. You know Dostoevsky? At some point in his life, they were about to kill him, execute him, in fact. He was in prison in Russia. He was taken to execution. Before execution, in fact, they had some drums rolling [makes the sound of a drum roll], and that was a signal that this was a mock execution. So he wasn't executed. In fact, we can tell because he survived to tell us the story.
- 5.5 But one of his partners, one of his prison mates, was so affected by the experience of almost dying that that person never recovered emotionally. They went insane. So that's one emotional reaction to almost death. But Dostoevsky, what happened to him was that he regained a new perspective on life. He gained a new appreciation for life, a new joy for life. Something deep happened, and he wrote this letter to his brother about how he saw life differently now, because he was almost deprived of life itself, this gift we have, all of us, that we are alive this very moment, which is highly implausible.
- 5.6 The fact that you are here and that I am here, this moment, this here and now of conscious experience is extremely unlikely. Think about it. Think about how unlikely life is. If your great, great, great-grandfather had coughed during Intimacy with your great, great, great-grandmother, merely coughing, you wouldn't be here right now.

Somebody else in that pool of sperm swimming would have been here. In fact, anybody in that chain of people, had they merely coughed during intimacy, you wouldn't be here.

- 5.7** Forget winning the lottery 10 times. The fact that you are here and alive is one in 400 trillion chance of you being here, Joe, and asking these complex questions. Extremely unlikely. Yet you are here, existing and attending my lecture, even though I'm sleep-deprived. You are here. Does that make sense?
- 5.8** What I'm trying to say here is that we use all these diagrams, and we use all these brain scanners, and we use all these machines, and these antennas, and these EEGs, and all these PET scans, and all these things, but they can't tap into the subjective I, the me, of consciousness. Does that make sense? In fact, if there was a Martian right here, and he was to hook up my brain and know all the pathways of my brain and know exactly what's happening in my brain, when I'm looking at my beloved over there and he tells me all these pathways are active right now, but I'll tell him, "I don't see love on that damn diagram. I don't see what I'm feeling. I don't see love. I don't see that beauty in her. I don't see it. Where is that on the diagram?"
- 5.9** And he'll tell me, "Well, I can see pathway A, B, C," but he can't capture my subjective internal experience. Does that make sense? That's what love is all about. I use papers, I use all these notes, but it can't capture what love is about, whether it be love of self, love of others, love of God, love of brother, mother. I can't capture that. That is a subjective qualia, an internal subjective experience that only you can extend, that you have access to. Does that make sense?
- 5.10** Now, when you do fall out of love, it's withdrawal. Dopamine centers were active. You were excited. She represented novelty in your life, new hope, but now she's gone. She went away. There's a crash in my dopamine centers. Now, I may seek that pleasure elsewhere. I may go out and find another partner initially to fill that gap. That's one approach. We have a homeostasis, we have a balance in our brain, and whenever we're deprived of some dopamine source, we seek it elsewhere. That's one reaction to heartbreak.
- 5.11** Obviously, there's an interesting syndrome, actually. It's called broken heart syndrome. Did you know that your heart can literally break, and you can get heart disease from having a broken heart in the metaphorical sense, in the social sense? That has to do with heightened cortisol and stress and noradrenaline, pumping through your veins and overwhelming your heart.
- 5.12** This has correlation, literally, with amygdala activation. The more the amygdala, the fear part of the brain, is active, the more likely you are to have this—it's correlated with this heart issue, this breaking of the heart or this damage to the actual heart, which can lead to heart attacks and things like that. It's quite fascinating. It's quite something.

- 5.13** On the whole heartbreak situation, first of all, you might ask: what is the trajectory for heartbreak? Let's say the girl, she vanishes. She's out of my life. She's gone. How does my brain react? Initially, there's a craving, week one, week two. Week three is about peak craving, a peak craving around week two or three. Then, eventually—obviously, this will depend on how long you've been together now. If it's been years, it's a different story. But generally, this is how dopamine will act in your brain: around week three, four, and five, it will start to go down, dopamine, and you'll start to have more of a general sadness instead of craving. Craving can continue. Your brain will adapt, habituate eventually. So that's a good thing with a survival mechanism.
- 5.14** But are there things we can do in order to help the situation, to make things easier? What do you think? Is there anything we can do to make the heartbreak less severe? You got it right there. Exactly, Belarus. You got it. It's my own point. What's that? Natalie. I know. I just like Belarus. Look, absolutely. Attention plus pleasure. Dopamine and acetylcholine. You got it. Exactly. That's part of it. Exactly.
- 5.15** So, before I get there, the first thing you want to do before you travel and do these things to encourage plasticity, the first, simplest thing you can do is avoid the cues that would trigger the dopaminergic urges. Does that make sense? If you're an alcoholic, don't go to the street where the alcohol is being sold. Or if you're a crackhead, don't go to the place where they're selling cocaine. Does that make sense? Because you have what? Urges. Basically, what you'll do is that the visual percepts, or the smells, or the sounds will activate the initial circuits.
- 5.16** You had the emotion circuits, the bonding circuits, in fact, for this person, and they will become activated. In practical terms, in terms of a romantic situation, it would mean if you guys would hang out in a certain park or in a certain place, avoid those places. Keep away from those places in order to keep your brain from triggering these circuits. Does that make sense?
- 5.17** Then, Natalie would say that you want to engage plasticity more. So, what you can do is definitely keep yourself preoccupied. You might say, Okay, look, the person keeps popping into my head all the time. Why? Because my serotonin is low and dopamine is high. But I will try to engage plasticity and attention and overcome my obsessiveness by focusing on something else. This is extremely difficult. This is where prefrontal inhibition plays in. Prefrontal strengths really come in. It's in these moments. Can you focus on another activity? Can you go and learn that new instrument? Can you learn that new language? Can you learn that new sport? And so forth.
- 5.18** This is very hard to do, but it literally will help rewire your brain. Why will it do that? Well, you're learning a new sport, tons of focused attention, acetylcholine, the nucleus basalis system in the midbrain, and then you have dopamine as well, reward from that new thing you're learning. Each time you learn a new step in that process—tennis, you're learning how to play tennis—you have a reward, and you lay down new circuits. So that's important.

- 5.19 Something else you can do is you can bond with other people, new people. Now, I'm not saying, Jonathan, I'm not saying you should go and find another girl a week later or two weeks or three weeks later. That's not what I'm saying at all. I'm saying bond with your family, bond with loved ones, bond with God. Go to a religious ceremony. Do things that will trigger oxytocin and melt away some of the old circuits.
- 5.20 And by melting away some of the old circuits, I know you will say, "I don't want the circuits to melt. I don't want to forget about that person. I still like them. I want to end up with them potentially. So why should I go melt them away?" But what you're doing here is that you're creating a new story around them. You're not necessarily just pressing the delete button on them, but you are creating a new story.
- 5.21 In fact, over time, what happens is that as we withdraw, initially, there's a lot of tension, anger, resentment. But as time passes and dopamine flattens a bit, serotonin and oxytocin pop up, you tend to be more empathetic, and you tend to see the entire picture. You can see, "Oh, maybe I was wrong. Oh, maybe I was too stern. Oh, maybe I was too rigid. Or maybe I shouldn't have shouted on that particular day. Maybe I was not being generous enough." So you do a cognitive reappraisal of the situation.
- 5.22 Your DLPF up here comes online, and you can really see the bigger picture. That's why often you find that during second marriages, people have better marriages because they can learn from all their mistakes. They have DLPFC activity much more. They can see the perspective and really help shape future relationships. Does that make sense? Good.
- 5.23 There's a fine line between liking a girl and then liking her too much, and then liking her to the point that when she's out of the picture, you can't let go, and you may do something stupid, like feeling an urge to go to her house and see if everything is okay or something. Do you know what I mean? You don't know what I mean. We are stepping into the stalking territory here.
- 5.24 And honestly, as a teenager, there was this girl that I liked in school. Oh, man, did I like her. I really liked her. She was special. She had depth, she was smart, she was empathetic. I wanted to marry her. And mind you, I'm an old-fashioned guy. So the way I wanted to do it, ideally—have you seen the Godfather? Michael Corleone goes to Apollonia, in the mountains, and says to her father, "I want to ask for your daughter's hand and marriage." Do you know that? That's my style. I like that old-fashioned now. That wouldn't work today, would it? No, I don't think it would.
- 5.25 Anyway, I liked that girl. She was something else. We had a bond. But then, as life would have it, I screwed it up again. My tendency to screw things up. I screwed it up. I always screw it up. I don't know. I shouldn't have taken her for granted. Sometimes you just don't take her tears into account. You become so selfish in your own sense of pride, and you feel on top somehow. You feel like now you are the one in power. You don't literally say that, but you feel it. And then, you take her fears for granted, her

feelings for granted. You're not empathetic enough, and you think her feelings will always last, which is never the case.

- 5.26** Years later, I was driving by her house with my friend. It was after my sister's wedding, so I was in that state of mind, and I felt like, "It's been like 5 or 10 years. What might she be up to right now? Should I go drive in the parking lot, sit there for a minute, and contemplate nostalgia about, "Damn, I did that." But I also asked myself: Am I stalking her right now or what? What do you think? A little bit? I'm not sure. I'm not sure. But maybe one day I'll tell you the whole story about that girl because it became really crazy.
- 5.27** I ran away from home, ran away to Italy. As a teenager, you can understand how impulsive I was. 25% absent rate. Remember I told you? I would run away from home. I went to Sicily, ended up living on the streets of Sicily, writing poetry, with my hat. I was a poet. I'm still a poet. I write poetry, but I wrote poetry, lived near theaters and stuff. I would watch people go to theaters and live in the houses there for a short while. I was madly in love with a girl. I was crazy. I was a crazy teenager, very impulsive. Anyway, one day, I'll tell you the whole story.
- 5.28** Now, when you are stalking, the stalking brain is a brain that's—so low serotonin is the love brain. Low prefrontal activation in the prefrontal cortex is the love brain. Heightened, OCD-like tendencies, especially in the ventral striatum and basal ganglia, overactivated, that's the love brain. But it's even more activated in a stalker. A stalker's circuit—the orbitofrontal and ventral striatum and basal ganglia—that whole loop is hyperactive. It's literally OCD. It's OCD at its worst.
- 5.29** But you also have things like the amygdala and the fear parts of the brain being hyperactive as well. So there's fear, there's anxiety, perhaps. There is high cortisol, so there's stress; they have a lot of stress in their system, these people. The point is: it's a brain that's very close to the OCD brain. It's an OCD brain, and we also already mentioned that the love brain is already an OCD-like brain.
- 5.30** But you know, one of the reasons why people actually do stalking, you know why that is? Because it works sometimes. You know that. It can work. There's only 15% chance of it working. So it's very low. But that's why we still have it as a trait in the population. It's because it can work. Now, 15% chance of working is better than a zero chance. So that's why people will do it.
- 5.31** People that stalk have this weird perception of—you know when serotonin goes down, dopamine goes up. We talked about that. These two balance it out. But if dopamine goes too much up, guess what happens? If dopamine is flooding the system too much—you have in the mesolimbic circuit, that circuit that goes from the limbic and dopamine-producing parts of the brain to the frontal lobes, producing dopamine. If it becomes overexcited and overflowed with dopamine, what happens is you can have something what's called magical ideation or magical thinking.

- 5.32** In fact, if I was to give you a pill right now that boosts dopamine, you might get delusions. Too much dopamine in the prefrontal lobes creates delusions. That's why somebody with OCD who has tons of dopamine in the circuit may say, "Oh, I feel like I touched a doorknob over there, and now my entire body needs to be sanitized," for example. There's an aspect of delusion in OCD. In Parkinson's, people with Parkinson's, when they take drugs that boost dopamine, they may develop delusions as well, which is very common. Having tons of dopamine, too much dopamine, in fact, can create delusions.
- 5.33** The stalker is a great example of that. Their brain, they are highly delusional. That means the girl smiled at me, or she, in fact, not even smiled at me, forget smiling at me, that'd be a home run. She just smiled at somebody over in the supermarket, and you go, "Oh, she noticed me. Oh, she was sending me a signal. She was unconsciously smiling to make me want to approach her." So they create—they have these delusions that will help them, propel them in their quest for getting her back, their stalking behavior. Does that make sense?
- 5.34** What is jealousy about then? Is jealousy a good thing? Is it a bad thing? Why do we have jealousy? Where does it emanate in the brain? Jealousy comes about in the orbitofrontal cortex. Orbito—here, just beneath the cortex. I mentioned that earlier, I believe, it's involved in—it's the bridge between emotion and brain, like the higher-order thinking in the brain. It helps translate emotions into cognition. I want you to go away with a full appreciation for the orbitofrontal cortex, not merely me saying it's involved in emotion and it's a bridge.
- 5.35** It's an amazing structure in the brain, for God's sake. It's amazing. You know why? It's amazing because when it's fully developed, it makes us human. What do I mean? A little child, a baby, for example, doesn't have a full sense of self. They can't relate to other people in the way that we can. So a child, for example, can a child feel jealousy? I'm talking about not jealousy in the—it's a tricky one. Romantic jealousy, that's what I'm talking about.
- 5.36** So jealousy can be felt—even dogs can experience jealousy. You know that, right? So, jealousy is a complex human emotion that I would say is a bit controversial because it does have a primal aspect to it. But generally, the orbitofrontal cortex and what makes it so amazing, and perhaps one of my favorite structures, is that it makes us human in the sense that it allows us to connect self with others.
- 5.37** So you have damage to the prefrontal lobes, and the orbitofrontal cortex is damaged. Now, you won't feel embarrassment anymore. You won't sense that. You might go out in public naked, for example, or you might do something completely strange. Why? Because the orbitofrontal cortex is involved in embedding the self in relation to others. That's the orbitofrontal cortex. So embarrassment, romantic love, and jealousy tend to be in that part of the brain. That's why it's so important.

- 5.38** Now, I want to say something else on the orbitofrontal. I guess that's just my point that the orbitofrontal cortex becomes hyperactive, and so you have heightened jealousy in romantic interactions.
- 5.39** I wanted to touch upon whether jealousy can be good. Can jealousy be good ever or is always bad? Like 50 years ago or 20 years ago, they would say that if you're jealous, you're immature, you're childish, you're no good. That's how the view was. But do you think that's true? Do you think so? What do you think?
- 5.40** I think jealousy can be good because it can protect.
- 5.41** It can protect. Honestly speaking, right? So if a man, if you liked a man, and there was another man in the picture, and that guy that you liked saw that and became jealous a little bit and tried to, in a respectful way, push that other guy away, but in a respectful way or even show his territorial behavior, how would you feel? How would I feel? I don't know.
- 5.42** You don't know.
- 5.43** Do I like the guy? He's being pushed away?
- 5.44** Yeah, you like the guy. Basically, he's showing you jealousy. He's showing signs of jealousy.
- 5.45** Yeah, I can go like, "Oh, he likes me, too." It's desirable. It's a good thing.
- 5.46** Exactly. It's a good thing. It can be a good thing.
- 5.47** He wants to protect me.
- 5.48** Yes, absolutely. There's pathological jealousy. That's the guy who... I had some of that pathological, I saw a girl once that I liked, and then there was this guy; she was sitting next to him. I became very angry. They were sitting a little bit too close for my liking. The guy was a bad boy. He was trying to do his moves. I became very jealous. I became angry. Mind you, I was 14, 13, 14. That's not good. You shouldn't become angry or become agitated. That's not the goal. But having some jealousy is good. In fact, the girl appreciated my jealousy. She's like, "Oh, this guy, this guy is a little bit—he has an edge to him. He's a bit manly." It's okay. It's fine. You show that you have protection for that girl. So it can be a good thing. It's not exclusively bad.
- 5.49** Have you heard about De Clérambault's syndrome? De Clérambault's syndrome. It's a curious disorder. These are folks who are convinced, usually, it's often women, for some reason. They are convinced that a male of high social status, like a mayor or a billionaire or whatever, a famous person, is in love with them. They have this delusion, and it's not actually the case.
- 5.50** It's curious because De Clérambault's syndrome is not well explained. We don't know why it exists, but it does exist, and it's a pretty robust phenomenon. Why do we think it might exist? What could be the rationale for that in the brain? Now, it can exist the

other way around. Trust me, my colleagues, I've seen them, my colleagues, have this delusion that a young girl is into them as well. So it can go the other way around as well. But why does this delusion occur? What's going on in the brain? What are the triggers?

- 5.51** I think that delusions like this are important for a sense of ego and sense of self in creating who we are. Let's say you feel insufficient, you don't have a lot of self-esteem, self-worth. I think that's when these delusions might pop up. But I think they are triggered by some dopaminergic imbalance as well. So there's got to be some delusional tendency in that person.
- 5.52** What you often find in delusional disorders is that they can shift depending on context, depending on what's considered socially, like what the person is craving. So this is a delusion. If having a socially high-ranking mate is something that can make you happy, then the delusion will manifest as this. Does this make sense?
- 5.53** Let's start here. Let's start here. Absolutely. Jealousy can be possessive. It depends on where the jealousy stems from. If you have seen Beauty and the Beast, we talked about that yesterday, and you are Gaston, and you want Belle to be yours as a trophy, ventral striatum, dopamine centers. You want her. When she doesn't want you, you feel rejected, you feel your sense of ego is bruised, and now you become jealous. You want to hurt the guy that's trying to pursue her because the trophy is gone. Your self-image is being diminished. Does that make sense? That's a bad form of jealousy.
- 5.54** A good form of jealousy is that I have empathy, but at the same time, I can't help but feel like she's my girl. I want to show that girl that I feel something. I want to show her that I have bonded with her, and that bond should persist and should not be broken. Does that make sense? That's really the key.
- 5.55** Now, the first one is what? Ventral striatum, dopamine centers, self-related, self-referential. The other one is what? ACC. It's empathy, it's bonding, it's oxytocin, vasopressin. In fact, vasopressin itself is a territorial hormone that makes you want to protect the woman that you love. Does that make sense? I would venture that the Beast himself would be jealous if Belle was to go out on a date with another guy, and that's healthy jealousy. Does that make sense?
- 5.56** It decreases it, absolutely. The more safety, the more bonding, the more you communicate with your partner and have a safe relationship, the better. 100%.
- 5.57** Then we had a question about, there was a question about long-term—
- 5.58** Neural synchrony.
- 5.59** Neural synchrony. Oh, yeah. So that's interesting. I've noticed something. The question is neural synchrony. Our brains, when we are facing our lover, our brains synchronize. You'll find that alpha activity is found in my brain, but also in my beloved's brain. We have neural synchrony. Now, the question is: what happens then when we are apart?

In fact, forget going to a different country. Just moving away in space and not thinking about her will remove that neural synchrony. So it's really particular to a very specific context. Does that make sense?

- 5.60** But it reminded me of something else, which is—and I've noticed this—when you are in a particular country, in a particular place, and a girl is within reach. A girl is within reach; you can see her every single day. She works in the lab that you're working in, and you can run into her every day. I've noticed a peculiar thing, which is that if you suddenly travel to a different country, there's a change in your emotional shift. Have you noticed this?
- 5.61** There's no name for this, but I do think there's a true phenomenon going on whereby whenever the woman is inaccessible and out of the picture, there's a change in my emotional state. What I think it is is that your brain computes, "There's no reason for my dopamine centers to be highly activated at this point because even if they are highly activated, I can't access her, and it is a failed computing power."
- 5.62** Think of your brain as this machine that's always in the business of saving its powers for more important things. In other words, it's not going to be wasteful. It's not a wasteful brain. You having all this dopamine and being agitated makes no sense when you are in Denmark, and she's in Arizona. Does that make sense? So there's almost a shift in the psychology. There's almost a shift in the psychology, and you calm down a bit, you see the bigger picture, and there's a perspective shift.
- 5.63** I think it's an important phenomenon. It should be described. Maybe you should write a paper on it. Literally, I think it's a true phenomenon. I've seen it tons of times. I've experienced it. It has to do with the fact that what motivates us to action is what's called—when resources are within reach, our brain changes its state.
- 5.64** For example, I may feel jealous of my neighbor who's like me, but I don't feel jealous of Bill Gates. I might admire him, but I don't feel jealous of him. Why is that the case? Well, in the first case, the neighbor, who's like me, lives in the same neighborhood and has the same resources as me, it makes sense for me to be jealous of him because my jealousy will motivate me to act and try to gain whatever resources that he has gained that I'm jealous of. Does that make sense?
- 5.65** But the guy that's a billionaire, me being jealous of him makes no sense because he's too far out of reach. A beggar is more jealous of his beggar friend than Elon Musk. That's for the same reason. So I think that has to do with whatever is within access, we have more jealousy towards it and it motivates us to action.
- 5.66** So I think there's a phenomenon here: when you are away from the beloved, you shift, and that can be an instant. Like, you take a private jet—I haven't been on this—but you take a private jet, go to a different country. Within a few hours, you completely shift your perspective.

- 5.67** I have an idea: what if you do a trick? What if there was a pill you could take, or somebody would lie to you and tell you that the woman has traveled to a different continent? How would you feel at that moment? Is there ways for you to trick your brain into the state in an artificial manner? Could it be conducive to treating romantic obsession? I wonder. Does this make sense? If there's a way to get around that. The fact that you can break down that obsession in this way. Anyway, this is me thinking aloud. Thanks for the question.
- 5.68** Natalie, you had a question?
- 5.69** I'd like to know about long-term relationships, and how come it's the case that whenever a hard break-up happens, the body starts to break down? Why is it not just a limited to the brain activity and our mental state, where somehow our body starts to break down? What's the connection?
- 5.70** I think that we know the brain and the body is hyperconnected. You sit in a sauna, you're melting away in the sauna. Your muscles are becoming relaxed. Your muscles that are now relaxed send signals to your amygdala saying, "Buddy, calm down. Don't be anxious. Don't be frightened." My muscles are relaxed, and that's a signal that there's no danger. Although it is an illusion, because there could be a lion here in front of you, but because the muscles are relaxed, the amygdala was tricked. Does that make sense?
- 5.71** The opposite would be, I tense your muscles, make them more tense. You send a signal to your amygdala, and suddenly you become what? Anxious. You become stressed out. You become agitated, irritable. So there's a strong body-brain connection. Strong brain-body connection.
- 5.72** Does it depend on perception?
- 5.73** A lot. Hugely on perception. But what I'm saying is that when you have stress in the system, perception, interpretation of the relationship—the person died, "My wife, after 50 years, died, and I was bad to her. I didn't treat her well. I was a bad husband." You blame yourself. The point is, you have a negative interpretation of her dying. You're not going, "Oh, she's fine. She's in heaven. I'm going to meet her soon." That's one interpretation. That's healthier because you were more positively seeing the situation.
- 5.74** But if you have this other view, I literally had a friend of mine—it was not a friend; it was actually a school teacher. This is a very sad story. He was my school teacher. This guy was a bit of a harsh guy. Actually, he almost beat me up when I was seven. Imagine this. One day, I was doing bad stuff, and he took me and he shook me. I was like, "What the heck?" It was in the ghetto. Teachers had to be rough, but beating a kid? It's too much.
- 5.75** Anyway, later on, this guy, he was from the Middle East. He had a Danish wife, and he didn't treat her well. He didn't treat her well, and she died from a heart attack, or a

stroke, or something. She died. And the guy went crazy, completely crazy. Every single day, he would call me. He would find me. "Help me. I'm missing my wife. I'm going crazy. I can't stop thinking about her. It's all my fault. I treated her badly." He did treat her badly, and he ended up in an insane asylum.

- 5.76** Think about this. He went insane just from his perception of him having been a bad husband. He probably also precipitated the stroke by stressing her out so much, perhaps. I'm not saying directly, but there could have been influence. High stress can also lead to strokes in some ways. But it shows you how these things can affect you. How these things can affect you.
- 5.77** I often think about how love can exist. So the wife is now gone. You're the man. You're maybe 70, you're 80. It was actually my neighbor. It happened to my neighbor. He had been married to this woman, and they had been together since they were 14. She died. This guy was such a beautiful person, always smiling, always greeting me. His name was Peter. He was always this kind person, very social, outgoing, and his wife died. Initially, he tried to date other people. He was 70, 80, but he still tried dating other people. He was telling me, "There's women everywhere. You don't worry about it."
- 5.78** He always told me, he told me, with the high school situation, he said, "Pursue love, never let love die. Go pursue it." It was very romantic. In fact, I would listen to him and I would succeed each time. But this guy, Peter, within a year or so—damn, this was sad, he hung himself. It was my friend. We used to hang out, just talk. He was 80, and I was like 15. We just talked about life, meaning, the world. The guy hung himself. Can you believe this? That's one way that—
- 5.79** I think what happens here is that, think about, you've been together for 80 years—not 80 years, 60 years, let's say—and you have built your hippocampus, your prefrontal lobes, your Wernicke, the meaning part of the brain has created a whole story around that other person. These circuits are crystallized. They are firmly anchored in the brain.
- 5.80** By the way, you have a certain circuit for, let's say, you have a circuit you create for riding bicycles. Let's say you like that; that's your thing. The longer it lasts, the more you ride the bike, you do this activity, temporarily, in time, the more it lasts, the stronger the circuit is, and the longer it takes to break down. In the case of Peter, think about this: he's had this woman for all these years, Sam. Think about this. He loved her so much, Lillian. He loved her so much. Those circuits just couldn't be reorganized. Plasticity couldn't do its trick. He tried—novelty.
- 5.81** He was a smooth talker. All these women at the local club or whatever you would go to, old people's homes and stuff. He would go and talk to them, and he was a charming guy. But yet, one date, two dates, he always saw Lillian in this one, saw Lillian in that one, and eventually, what happened? I used to pass by his door afterwards and feel very sad for his passing. But that's one way that plasticity can work against us if it's too crystallized, in fact.

- 5.82** You might say that first love can create that impression, too, because first love is so profound and the teenage brain is so hyperactivated in the dopamine centers. Testosterone is so high that the plasticity of that teenager, also the BDNF systems are not completely shut down yet, so the brain is hyperplastic. That impression lasts for so long.
- 5.83** That's me becoming the stalker after my sister's wedding and still going and seeing if that girl is still there, expecting her to come out with her scarf and brown hair and big deer eyes, coming out, and me looking at her and saying, "Hey, how are you? You're doing well?" Of course, it never happened. She was probably with her new boyfriend somewhere. That's what we're talking about: the first love. It stings, it's hard, but it lasts, and that has to do with plasticity as well. We seem to compare every other person to that initial person, that first love.
- 5.84** I do wonder about your question. And by the way, other questions? We can throw them in. I feel like your questions to me, especially in this lecture, help me weave in my own content in a better way because I know what you want to hear versus me giving you A, B, C in a very static way.
- 5.85** I do feel like the whole idea of widows and things like that—you mentioned widowing and all that being a widow—I do feel like I want to touch upon how we process love once, let's say, the wife is dead or the husband is dead. How can we process that in the best manner? What comes to mind, a movie that comes to mind is Ghost. Have you seen Ghost? You haven't seen Ghost. It's an interesting movie.
- 5.86** For those who haven't seen it, it's a classic. A guy, a man—actually, I love it because I also talk about it in my book, not to brag or anything. Because becoming a ghost is something that I'm very interested in, how the brain can create ghostly shadows and all that. Now, what's interesting is that Patrick Swayze is the character that plays the male protagonist. What happens is that he's with this girl. He loves her. He's very romantic with her. But then he's killed. He's killed. He's killed, and he becomes a ghost.
- 5.87** He becomes a ghost, and he walks about as a ghost in that city where they live. He becomes this ghost. He's stuck between these two worlds: the physical world and the ghostly, the spiritual world, let's say. And he's trying to warn her because she is potentially being killed. She's about to be killed by somebody who—the person that killed him is also wanting to kill her. That's context.
- 5.88** Why is it so important? I think it's important because I feel like what you see in movies like this, the way that Molly, the female character, processes the loss of Sam, is very characteristic. There's a bonding of the oxytocin and the vasopressin system that cannot be broken. It keeps going. It keeps lasting. It outlasts death. I think that truly, because the brain has this way of reusing its circuits, I think that if somebody like her was to truly find solace in a future romantic relationship, she would not create

completely new circuits. She would repurpose those circuits, those Sam circuits. She would reuse the Sam circuits. Does that make sense? That's one point.

- 5.89** Let's think about the movie differently. One way to think about it is that it's inaccessible to somebody else. Let's imagine for a minute that Sam in that movie didn't die, but was inaccessible to Molly. He just went into—he traveled to a different place. The question is: for how long would the love of Molly be there and be steady with his absence? That taps into absence. So he's not dead. He could be in jail. He could be in a different place.
- 5.90** Again, that's a big question. Taps back to dopamine and the brain and bonding circuits, and how easily they break down. What I think is, in that movie, have you seen the scene where Patrick Swayze, the Ghost, is able to persuade this lady who's in the spiritual realm, she's a person who can communicate with ghosts and stuff like that. She's able to go and communicate with Molly, letting her know that Sam is actually a ghost and wants to give her a message. Have you seen that part? You haven't seen it, Sarah? Okay. A different generation, maybe.
- 5.91** All right, Sarah, look, you know what he does? Sam says this to Molly—so there's a door. There's a door here, and Molly will not let the woman into the house who's saying that Sam is a ghost. Makes sense, right? It's a bizarre thought or a bizarre thing to say. Then, Sam says to the lady outside, he says, "Push this quarter in, this penny, beneath the door," and she does that. So she pushes the penny in. And now the penny is here, the penny is here, and Molly can see the penny.
- 5.92** Then, the ghost takes the penny and pushes it on the door, and the penny is flying like this, but it's still on the door. It's going up, up, up, up, up like this. So far, so good. Not that impressive because it could be a magnet. But then it flies in the air, and it flies all the way over to Molly. At this point, she cries. She realizes something supernatural is going on. What is my point of this? What's the punchline of all this? It's a nice scene, but is there any neuroscience to this?
- 5.93** I think that that very scene transforms her brain, giving her a signal, a cue, telling her that love can overcome even death itself. In a similar vein, if you are absent from a lover, you are far away, the love will wane and die, eventually, but small cues, a letter, somebody giving a message, a small signal can do the trick to revive the dopaminergic circuits. Does that make sense? Is the analogy clear, or is it a bit far-fetched? Is it clear?
- 5.94** I think that the way you think about dopamine and how it works is that it eventually goes down to baseline, but having some cue, any cue—it could be an email, could be a signal, could even be in our age, a "like" on Facebook or Instagram or whatever we use, TikTok. You might know this. People do this in relationships these days. They go and then they're maybe absent, but then they go like something. These cues can revive the circuitry of love. That's my point. A bit extreme example, but I hope it made clear.

- 5.95** Speaking to maybe infidelity and the impact on the betrayed, what is happening in the brain as far as like, betrayal trauma, especially in long-term marriages and relationships?
- 5.96** That's a good question. Cheating deteriorates trust. So safety circuits, those neurons that would produce oxytocin and other neuropeptides for safety and bonding, prolactin is also a little bit involved. I didn't mention that earlier, but it has some bonding role, get eroded. When somebody cheats, you literally create a schema in your brain that the world is dangerous, that you cannot trust. Your amygdala, the emotional core of the brain becomes hypersensitive to any similar situation where you feel like you might be cheated, somebody might cheat on you.
- 5.97** You develop—it's a recipe for an anxious, clingy attachment style, like having somebody cheat on you. You become paranoid. It is really that. It's a concept and schema you develop. Prefrontal cortex: meaning, self, other. "How do other people, how may they betray the safety bonds that I'm laying down? They might cut that bond. They might betray me." So literally, you will become paranoid and hypersensitive to that.
- 5.98** The memory is encoded very much in the hippocampus and the amygdala part of the brain, meaning—I'm not trying to be sophisticated on you here, but what I'm saying is that, bringing a neuroscience lesson, is that emotional memories are encoded in the very place where they arise. Did you know this? It's interesting.
- 5.99** If I feel a certain feeling of jealousy or somebody cheats on me and I feel horrible, that memory is encoded in the amygdala itself and the hippocampus itself, and those circuits going to the prefrontal will literally be laid down. The next time I have a similar situation, it turns out that those circuits, very same circuits, will become activated as if I'm experiencing that as the very first time. Does that make sense? That's my answer.
- 5.100** Do you have any recommendation as far as how to heal that or to overcome it?
- 5.101** It's a great—how do we overcome cheating? First of all, people that have been cheated on are more likely to be cheated on in the future. We know this. Why? Unclear. It could be that you have learned that that's a normal behavior, perhaps. It could be that maybe you just are the person that people are likely to cheat on. Maybe you don't have boundaries. Maybe you don't have self-respect, and that could be because your mother was absent, dad was maybe violent. It could be whatever trauma in your history that makes you more likely to accept things like that, maybe have a low self-esteem and all that. So I think that's a reason. The people that cheat are more likely to cheat again. So it runs. It has this arch to it where it's a continuation, really.
- 5.102** How can we prevent cheating? Was that the question? Or heal?
- 5.103** Heal.
- 5.104** Heal. Healing from cheating: it's a hard one. How do we heal from cheating? I think the way we can heal from cheating is by meeting partners that are safe, that have a safe

approach, and that show us that cheating is not something that's acceptable. I think that's the trick. The reason I say that is because we know in neuroscience that something called cognitive re-interpretation of events typically has a hard time convincing your behavioral pattern and your actual emotions. It can penetrate the emotional system, but often there's a block there.

- 5.105** That's why I can tell you over and over again that by being afraid of dogs, that is a ludicrous idea. You will know that is the case, but you still fear dogs, for example. Does that make sense? Now, if I actually bring a dog to you and you learn to pet it and play with it, your emotional system and your brain go, "Oh, I didn't die from that." Does that make sense?
- 5.106** Your brain, cognitively, can be weak in the sense that exposure, an actual behavioral exposure, is much more potent. That's why I'm saying, merely knowing that the world is safe, me giving you 100 lessons on neuroscience and all this and cheating and all that, is not going to help as much as you meeting a partner who teaches you, "Cheating is unacceptable. That's something we will never do."
- 5.107** What you can do is look for partners that look decent. By looking decent, I mean you can sense that these are ethical people. Like, literally, the girl in the ghetto when I was 14, cheating was nothing for her. She would, the next day, literally go find the next guy with the big muscles. I was crushed. It was very crushing for a 14-year-old boy. That always stayed with me, always. Crazy, how that works.

Lecture 6: Love's Shadow Side

At a glance — Love's shadow side and the dark triad.

- 6.1 There is this dark triad, personality trait. You have the narcissist, you have Machiavellianism, and psychopathy, which is the three branches of this dark triad. Narcissism, as you know, involves sense of self-entitlement, thinking that you're better than others, "I'm great." You have grandiose thoughts. Machiavellianism is a bit different here. It involves a sense of planning, scheming, tricking people, trying to con them and so forth for own purposes and gain.
- 6.2 Psychopathy is a completely different category, but again, you have traits of being cunning and wanting to cheat and all that, but it is more than that. It's really about having no empathy at all for other people. In fact, your nervous system is completely shut down in the sense that you don't get aroused by doing anything criminal. Your amygdala is the emotional core. So the amygdala, normally, if you go and rob a bank, or you get stopped by the police, your amygdala fires away. It sends signals to the hypothalamus, and the hypothalamus, the hormone part of the brain, sends signals to the pituitary to release cortisol. There's adrenaline and so forth in the system circulating. That's the normal person.
- 6.3 The psychopath, there's no such thing. Their amygdala activity is more quiet, and the medial prefrontal lobes are decreased. There's no sense of more decision-making. It's all the outer layers here, the DLPFC. It's cold cognition, cold thought, and outsmarting people. Does that make sense?
- 6.4 The Machiavellian type does have emotion. The Machiavellian person— and obviously, I'm relating this to a mating situation. You see these traits a lot in mating. That's why it's relevant, but I'll get into that in a second. Machiavellian people rely a lot on the DLPFC as well, but they do have emotions, and they do have feelings, and so their nervous system is not completely shut down. Does that make sense? So the amygdala, the medial prefrontal lobes are fine, but they rely a lot on the DLPFC, that part of the brain.
- 6.5 Then, we mentioned narcissism. The narcissistic brain is one that relies heavily on the medial prefrontal lobes in self-referential thinking. In fact, it can be overactivated because they're always thinking and assessing self.
- 6.6 Now, do women find these guys with these traits interesting? Yes or no?
- 6.7 Yes.
- 6.8 Really? Do you like men with these traits?
- 6.9 Yes.
- 6.10 No, you can say yes. You like them, really?

- 6.11 Yeah.
- 6.12 Why? Because they're confident? They're not—okay, look—
- 6.13 Charming?
- 6.14 Charming.
- 6.15 Charismatic.
- 6.16 Charismatic?
- 6.17 I think there's a mystery factor to some of them.
- 6.18 So would you like to end up with a psychopath is what you're saying? A narcissist?
- 6.19 It's too late.
- 6.20 It's a good question. So women do find these men attractive, but it tends to be for short-term mating. I assume that's what literature tells us, that you would prefer a partner who is kind, generous, and emotionally available, and not one trying to cheat you and rob you of your resources. They try to, they do that sometimes. You don't want that, do you? Presumably not, right? So we know that it's more of a short-term strategy.
- 6.21 But females, younger females often get tricked by these men. Younger females get tricked because they get persuaded by their risk-taking, their boldness, their charisma, flamboyant clothing: pink shirts and hats. So they have these characteristics. By the way, sometimes people think it might be narcissistic because of this clothing. They feel like, "What is going on? Why is he wearing this flamboyant—?" My mentor used to say, "Jalal, you look like a drug dealer," because I used to wear pink shirts and hats to the laboratory, too. That was my thing in California. So, definitely, it can—
- 6.22 But I think it serves you well. I think females do like males that dress slightly edgy—correct me if I'm wrong, this is my own thoughts on the neuroscience behind it—edgy clothing because it suggests a rule-breaking tendency. They're not afraid to stand out, to be different. The brain likes novelty and things that flip the script a bit, and it's a bit different. And so I think females can get turned on, but females at the same timelike confidence, but not overconfidence. Is that true? Can you tell me?
- 6.23 For example, I've had situations where I had my hands burned because I was too overconfident in the female situation. I came across as too overconfident. I wasn't overconfident, but in my behavior, in my language, the way I spoke, I came across as overconfident. The brain doesn't like that. It gets suspicious. In fact, the brain, when it comes to personality and assessing a personality and charisma, there's two branches.
- 6.24 Branch A is competence: how competent is this guy, for example? So by moving your hands, for example, when you give a lecture, or you're speaking, you come across as charismatic. You have a clear voice, the way you speak, the way you dress looks good.

Competence. You're a bit stoic, a bit don't smile too much. Competence. That's all competence cues. Women like that. So you want to have that overall.

- 6.25** But research shows that clearly, if warmth is missing from that equation, it all derails. You need warmth as well. Competence and warmth equal charisma, the one that females like. And it's both genders, by the way. But for males versus females, they want females to be slightly more warm versus competent. But let's take the male example.
- 6.26** So if I want to get with this girl over here and I'm interested in her, and I'm completely stoic, and I'm stern, and I'm trying to convey competence, maybe I'm telling her about my job— actually, I would never do that personally. You know why? If I met a girl, I would present myself as a person who's just cleaning the streets. Seriously. Not lie, but I would not reveal anything about myself that suggested that I might be slightly successful, or even have an education, or have a job. I wouldn't. I would just be completely nonchalant and kind and generous.
- 6.27** Then, once she figures it out herself, her dopamine spikes. "Oh, he actually has a job. He actually is doing something in his life. It's not just a bum who sits on the couch. Much better than revealing all your cards, much better. Dopamine likes that. Now, if you reveal your cards, what you're telling her is that, "That's all I am. I have nothing else in there." So you're telling her, "That's it. I'm insecure, and I'm telling you right now, this is what I got, versus you saying nothing, and her finding out. So that's a dopaminergic hijacking you can use. It's a way of making her—
- 6.28** It's also better for yourself. Why would you feel like your personality depends on what you're doing for life? Your personality should depend on who you are. You can lose your job tomorrow, you can lose your health tomorrow, you can lose your family tomorrow. Be sufficient in yourself. That's how you should program your medial prefrontal lobes to be self-sufficient, or God-deficient, if you want—God-sufficient: rely on God. That's another way to look at it, to have a spiritual angle if you want to go that route. But don't depend on your materialistic gains and achievements because that's not going to get you anywhere.
- 6.29** Anyway, warmth and competence. If you are too competent, showing too much lack of emotion and financial, that you are, maybe, you show that you are rich, indirectly, of course, she will lose trust. Without warmth cues, without smiles—authentic smiles, by the way, a real, authentic smile, it's called a Duchenne smile. It involves the eyes, right here. You have to have the eye muscles be activated. In fact, a real authentic smile is slightly tilted upward on the left, slightly. Why? Because you use the right hemisphere more. The right hemisphere is involved in emotion. So a fake smile is more symmetrical. A real smile is more left-tilted.
- 6.30** So these cues of warmth, of smiling, authentic cues, like the guy from Groundhog Day when he became more empathetic and his ACC came online, his TPJ, he became more empathetic. It shows. You can't hide those cues. It's automatic. That's why you can't

fake charisma. Charisma, you cannot fake, because charisma, you need to have an authenticity to you, a warmth, a real caring, a real love inside you. Otherwise, it's not going to work. She will figure that out instantly. You have to have warmth and competence. But for males, slight competence might be better than too much warmth because it suggests a bit of a bodyguard effect, as we talked about, the strength and so forth. Does that make sense?

6.31 Oh, by the way, I just wanted to add there on the neuroscience of love and all that, the right hemisphere, I believe, is more important for empathy, and not only empathy, I mean empathy in the cognitive sense and romantic sense here, that you need more of the right hemisphere. The right hemisphere is involved in decoding faces and especially decoding emotional expressions. So if I want to make sense of what Natalie is feeling and her expressions are, "Is she authentically happy right now, or is she bored right now?" That would require my right hemisphere. It's very important to know this, what we call lateralisation or functional asymmetry, using fancy words for meaning that the hemisphere is very important for being able to read emotional cues.

6.32 Now, females, as you'll know, are very good at this. They are very good at it. Their right hemisphere is extremely powerful in detecting subtle facial expressions. They just know. So many times I was in the vicinity of a girl, and she could just read my emotions, read them. Literally, she knew that I was angry or I've had some resentment. She just knew. I was like, "My God, is this telepathic? How in the world can she know that actually I am feeling a bit sad right now?" or whatever? They just have this radar.

6.33 Whereas us men, we're looking at her and we're like, "What is she feeling? Is she sad? Is she angry? What's going on?" We scratch our heads. That's because we don't have the same fine-tuning of the emotional circuitry. Well, why is that the case, Sarah? Well, it turns out that when we were living in the caveman days, women would sit and look at facial expressions, and those circuits became more developed and more sophisticated, whereas men were out hunting, and so their skills became much more, like having physical and motor skills.

6.34 Even spatial skills. It turns out that men are spatially— I'm going to just say it straight out—superior to women when it comes to spatial navigation in mazes. Why? They were out hunting all day. They need to know, "Where am I going right now? Where are the gazelles?" and all that. You scan their brain, and those areas of the brain light up like a pinball machine because they have more spatial awareness, whereas a woman might be lost in the maze quite quickly. It is what it is, but you have emotional powers, so different powers.

6.35 We talked about how the Gottmans, John and Julie Gottman, can predict divorce by 94% accuracy. So this is true. The four horsemen of divorce, there are four aspects. First, there is criticism. Basically, what I'm saying here is that the Gottmans or these folks in the lab would look at a female and a male sitting across each other and having a

chat. Then they would look at four components and say, "What does this represent? What does this mean?"

- 6.36** So first of all, they look at something called criticism and the level of criticism, finding faults in your partner, intense complaints and so forth. That's one. Defensiveness is the second one. How defensive are you? You know these couples, they chat, and then they start defending themselves really quickly. They're very fast to do that. So that's one. Stonewalling is the third one. Emotional withdrawal, they seem avoidant, they seem not there. They're not present; they are not really engaging. That's stonewalling. And then the most powerful predictor, which is contempt. Contempt. That's scorning, that's the scorn rolling of the eyes, sarcasm, subtle expressions of disgust, and so forth. So these are the four predictors, the fourth being the most powerful predictor.
- 6.37** Now, there's a neural substrate to these predictors. Criticism involves OCD circuits, the orbitofrontal cortex, which is involved in error detection. That's the error detection circuit of the brain. Defensiveness is more TPJ, lack of empathy—that's the up here. Stonewalling is suggesting low dopamine, low oxytocin. Then, of course, contempt is high amygdala, high cortisol, and so forth. Even high insula. The insula is the disgust part of the brain, so if you feel disgusted by a partner, we talked about how social disgust and actual disgust come about in the same region of the brain, the insula. So, if you have social disgust from your wife, that can trigger the insula. Does that make sense? So you have these components to it.
- 6.38** Let's talk about cheating a bit more because you had some really good questions over there. Can you remind what's your name again? Sorry.
- 6.39** Danielle.
- 6.40** Danielle. You had some really insightful questions about cheating, and so what makes you cheat? Let's think about this a little bit. What makes you cheat? Well, there's factors like, as we mentioned earlier, having poor boundaries and all that, that makes you cheat. But literally in your brain, what factors can lead to the cheating? There are different components to this.
- 6.41** One aspect is that in a relationship, eventually the dopaminergic circuits habituate, they dampen down. So that's obviously one part. You're no longer as excited when you see your partner. So there might be at some point, a craving to novelty. We know this. This is called the Coolidge effect. You know the Coolidge effect? It's an interesting effect, Ramachandran used to tell me this. We were thinking about how you can apply this to humans, how you can use this to help romantic relationships, if there are tricks to using this. Our conversations in our laboratory was often about how basic effects in neuroscience can be used to actually have practical applications. Let me tell you about the Coolidge effect.
- 6.42** When a hamster is—it may mate, it may engage in mating behavior. Then, at some point, it stops. It can't mate anymore. It runs out of mating energy. Now, it turns out if

you introduce a new mate, suddenly all its powers will come back, and it can mate again. This is called the Coolidge effect. That's why my mentor was saying, "Is there a way to reignite the relationship and the passion you have for your wife of 20 years or 30 years by tricking your brain?" That's called the Coolidge effect. The Coolidge effect is interesting. That might be one aspect of cheating. One reason that would push you towards cheating is the novelty. That's one.

6.43 Secondly, what really kills passion and dopaminergic activity is routines, everyday life being very—doing the dishes, picking up children, and there's no novelty at all in the relationship itself. So not only is the partner not novel, but even their life is in complete routines, and everything about you is revealed. There's nothing else to get out of you. Does that make sense? You've basically laid yourself out there, and there's no intrigue anymore. Right, Natalie? You've got to be creative. Now, there's ways around it. I'm not saying it's a dead end; I'm saying that's what leads to cheating.

6.44 And then that new female comes to your work, starts the new job, you see her every day. She's pretty, and your prefrontal cortex, which engages in time travel, will start making thoughts up. "How would it be to be with this girl? How would it be different?" Maybe she's showing some signs, maybe she's smiling a little, maybe she's being charming. So it's like, "Oh, maybe there's something between us," and the fantasies begin.

6.45 At this point, at home, everything is boring, and you start arguing with your wife day and night. That leads to amygdala activation, hyperamygdala activation, so the fear and anxiety parts of the brain are going through the roof, as cortisol is becoming elevated. In fact, this is becoming chronically elevated. Now you have chronic elevation of cortisol leading to what? What does chronic elevation of cortisol lead to? Atrophy of what part of the brain becomes thinner? The prefrontal cortex. What is the prefrontal cortex important for? Decision making, logical thinking and being restrained, being inhibited. Okay, perfect cocktail for madness.

6.46 Your prefrontal brakes are out of the picture. Perhaps you're not sleeping well because of all the fights. The prefrontal cortex is functionally completely dampened down, and your amygdala is screaming, "I need something to calm me down." And at the same time, you're not getting oxytocin and bonding from your wife, so your oxytocin centers are saying, "Goodness, I need somebody to bond with," because oxytocin, we know, is calming the amygdala. There's a link there. Whenever you secrete oxytocin, the amygdala goes down.

6.47 So now you have all these desires, all this need for new dopamine, all this need for oxytocin, all this need for dampening the bloody amygdala and your prefrontal cortex is under attack every single day, thinning and thinning, and thinning, and then what happens? Your impulses get the better of you. Maybe you ask, you start to talk with that girl, get her number, and then eventually end up cheating. That's the path to cheating.

- 6.48** Of course, it should never happen. Never should it. It should never happen. And in rational, wise people that knows how much this hurts a partner, they would never allow such a thing. They would stop at the very instant of finding themselves attracted to it. They would stop themselves knowing that really, cheating on somebody could be worse than killing them. I mean, it's horrendous. It's horrible. But this is the path. It has these triggers. Does this make sense?
- 6.49** Actually, there are a few—let's talk about this later. There are some things you can literally do to help yourself, stop yourself from cheating. I know you're interested in this, Danielle, but I think it has to do with making yourself more empathetic, seeing the other person's perspective more. Maybe if you have the desire to cheat, at that point, you stop, you empathize with her pain, see how she might feel about that, and so there are techniques like that that could actually prevent you from even going down that route.
- 6.50** Let's go over to another part of this lecture. Now I would like to talk about our genetics, our epigenetics. We have genes, and genes can be turned on and off. We talked about this earlier. If you have bad experiences, you may turn on genes for anxiety, depression, stress. If they turn on in you, they might stay there and affect your children. Remember we talked about that?
- 6.51** We talked about the rat experiment. Did we talk about the rat experiment? They smell something, a negative smell. We talked about that. Good. There's a similar study that's really interesting, where females, women who were pregnant on 9/11, a group of these women ended up with PTSD trauma. Does that make sense? They witnessed 9/11. They were present; they witnessed the horrible act. They were pregnant. They developed PTSD.
- 6.52** Now, their cortisol systems are all messed up because of this. They have chronic low cortisol. You can have chronic high cortisol, but also too low is not good either. There's a dysregulation in the system. It turns out that when their babies are born, they have the same dysregulation in cortisol. They have the same low cortisol. Does that make sense? This is an example of how epigenetics works. As we mentioned, within the DNA, you have methyl groups turning genes on and off. If the right genes are turned on or turned off, that can lead you to be predisposed to all kinds of disorders throughout your life.
- 6.53** When you think about children, a baby's brain is hyperconnected. There's a hyperconnection between neurons. There's about 50% more synapses in the baby's brain. If a normal brain, the connectivity, you have a certain connectivity, you have 50% more connections between neurons in the brain. We have axons, and we have the axon going to the dendrite. Then, these branches and the synapses will be 50% more amply available in children.

- 6.54** In fact, there are 15% more neurons in the baby's brain. This is important because there's a process called pruning. In the pruning process, the following occurs. The baby's brain is vastly plastic. It's very malleable. If you stress the brain too much, it will affect the plasticity of the brain and how it rewires. It needs to rewire. It needs to cut some of these connections. That's the whole idea. The idea is that you want to keep the good connections, sculpt new ones that are in a good direction, and remove excess connections that are not used. Is that clear? So there is that pruning process.
- 6.55** In the baby's brain, of course, we have excess BDNF and GDNF, Jonathan, neurofertilizers of the brain, making the brain more likely to change itself. Heightened plasticity in children. One way to assist the child in developing their brain and keeping the right circuits and removing the ones that are not very helpful is through stimulation. This could be cognitive stimulation, learning new things, and so forth.
- 6.56** But from an emotional perspective, that's when, during childhood, you want to make sure that you hug your children, that you show them affection, that you show them love, that you show them safety and trust, that you have a role model in the father, the father that shows what a father is like, and shows them that masculinity. So these are part of the sculpting of the baby's brain.
- 6.57** There's various techniques a mother can use. One is called the kangaroo—let me see, what is it called? It's like a kangaroo type of hugging. It's when you have your child close to you. You know what I mean? It's close to your chest. That's in particular, I don't know why, but all the studies are looking at that, where you have your child close to your chest physically. That seems to be very important for development of the child, intellectually, emotionally, and so forth. Those are some of the techniques to be aware of, a kangaroo type of carrying of the child.
- 6.58** There's tons of studies showing that rats, though they aren't exactly humans, but rats that are hyper affectionate, grooming their rat babies, all that will lead to epigenetic changes such that the rats, the rat babies, the offspring, will be much more resilient to stress later on, much more able to cope with stress, much less stress-sensitive, having a much less reactive amygdala, just by hugging, just by grooming and so forth.
- 6.59** It seems there's a similar pattern in humans. There's these findings from Romania that I found really interesting. They did these studies where they showed that kids growing up in these institutions, like orphanages and stuff, where they got very little love and affection and things like that, their IQ, their cognitive development, their brains were thinner in various places, in the prefrontal regions and so forth. So there was tons of damage just by being in these orphanages. Their brain would wither away.
- 6.60** Curiously, if you could get the child into a foster home by age eight, you could somehow circumvent some of these effects just by having them do that by age eight. Why? Because there's a critical window for plasticity. The critical window goes from anywhere between eight and twelve. That's why language, for example, we know that

language, when you learn a new language after eight or twelve, you will definitely develop an accent, like my accent. I learned English after eight or twelve, so there's a little bit of— that's a big accent, in fact. What's that? Yours, too? Absolutely.

- 6.61** The brain in childhood, though, is very plastic. We know this going all the way back to things like Hubel and Wiesel's study. I won't go too much into that. But if you have a kitten and you cover its eye from the first few weeks after it's born until week, I think it's week eight or something, from week three to eight, then it will literally become blind for life if it's covered. That's because the cortical, all the cortical tissue that's devoted to that eye, will go to the other eye. Does that make sense? But if it's done after that, you have no issues because you are beyond the critical window of plasticity.
- 6.62** Plasticity is because you have tons of BDNF and GDNF in the system. When you are a child, you're paying attention to everything, so the brain is hyperplastic. But as you reach maturity of puberty and the critical window ends, the nucleus basalis and the attention system dial down, and then you have no longer all this BDNF floating in the system, and your brain is less likely to change. Does this make sense? Good.
- 6.63** So yes, physical touch can leave a lasting impact on the DNA. And similar for men, there's so many studies showing that men who play with their infants, it changes that child and their prospect for better IQ, intelligence, and so forth. So it's not only maternal, but it's also paternal. It has the same effect.
- 6.64** Let's weave in a few questions as we finalize. I'd like to have your input on which direction to go.
- 6.65** Can you speak more to the teenage brain and what love looks like, and maybe piggybacking off of some of the developmental critical phases? Are there any in adolescents?
- 6.66** It's a very interesting question. So the teenage brain—a perfect question. What's going on in the teenage brain? You have low prefrontal cortex because the prefrontal cortex only matures around the age 25, so you're more impulsive now. The dopaminergic centers of the brain are hypersensitized to any reward sensation and so forth. So you have hyperdopaminergic sensitivity, one.
- 6.67** Next, you have hyper oxytocin as well. You are very likely to bond. So when you find that girl that you're interested in, now you have low inhibition and low prefrontal cortex. Heightened dopamine, so not only she's beautiful, but she's amazing, she's stunning, she's my life. You have heightened oxytocin, so you are more likely to bond, and you create these extreme bonds, in fact.
- 6.68** You have heightened testosterone, so you're a risk-taker. You are bolder. You are free doing things out of the ordinary. That can be fatal, that can be dangerous. It might lead you to run away from home, for example, if the relationship is turbulent because you

are in a testosterone-type risk-taking brain. You're stuck in that teenage brain. In effect, the teenage brain is a recipe for disaster when it comes to love.

6.69 That's why the first love experience can be so impactful because it's grounded in crazy neurobiology. Imagine all the love circuits being on steroids. Everything is just hyper and exciting, not only novelty, not only because your brain is dopaminergic, but it's also the first time, the first time anything is amazing. Imagine if your brain is already hypersensitized to dopamine and you already have this sensitivity to reward. So it has that component as well; it's novel. It's the first time. The first time of anything will have more of an imprint on the nervous system. So, a great question.

6.70 Before we go to the question, just keep the mic there, before we go there. I want to add, but then, as we age, there's a gradual decline in the sensitivity of the dopaminergic system. That's when we talk about things like, "I became jaded. I wasn't as excited." Often, I would meet my mentor, and in his late 60s, I would ask him, "I found you after this trip—" Basically, the short story is the following.

6.71 What would happen is that my mentor and I were very close friends, Ramachandran. We would often go for long walks on the beach, talk about science, poetry, love, Shakespeare, and all that. We had all these conversations. But I noticed, with age, each time I would visit him, his passion and enthusiasm—imagine this guy was 60 and was like Peter Pan in his persona. Very excited, very happy about life, but I found this decline year after year in his excitement.

6.72 And he would tell me. He had a lot of self-insight. He was, after all, one of the greatest neuroscientists. Not bragging, I'm just saying. He said, "I think I'm becoming jaded. I feel like my dopaminergic centers are less excited because of my age, or perhaps my Parkinsonian system is affected, too." It could be he had Parkinson's. So, there's definitely that effect.

6.73 But the positive: even though we become more jaded and our dopaminergic centers are less sensitized with age, there's a positive bias effect whereby we, preferentially, as we age, attend to positive stimuli in the world. We have this amygdala declining in activity, the fear parts of the brain, as we age. That's why old relationships, relationships in older age, can feel more calm, more grounded, and more steady. I just wanted to finish up that point, so there's a continuation in the lecture itself. Good. Now the question.

6.74 You talked about the Coolidge effect, and it actually made me think, what is the neuroscience perspective on open relationships? Or currently, if you go on dating apps, there are a lot of other weird things you'll see, like ethical non-monogamy. Do you think it can actually have some benefits for the relationship and the bond within the couple, or it does more harm?

6.75 Open relationships and non-monogamous relationships. It's a great question. I wanted to cover that too, because I wanted to talk about arranged marriages and situations

where things are a bit different. In an arranged marriage situation, and by the way, I wouldn't recommend it. Personally, I don't know. I find it a bit odd. The romantic in me doesn't like it. Maybe for some people, it does work. In fact, it often can work for people. But how in the world it works for people, I have no clue. I'm wired completely differently.

- 6.76** But it turns out, if you go into a marriage with a prefrontal cortex being the decision maker versus your emotions, sometimes you do learn to fall in love, and you learn to associate that female with something positive. But again, it depends on the level of arranged marriages. If you haven't seen her at all, I mean, I would personally completely think that would be not a positive thing because what if you're not attracted to her at all? Not talking to her either. I don't know. It seems like a recipe for disaster without having to have some context and so forth.
- 6.77** Now, your question, particularly on marriages including several partners: I think that our brains these days are not wired for that. I think it's difficult to have several partners because how do you treat them fairly? You all agree that I will have three wives. Three wives. First of all, love is very obsessional. It's very one-directional. So how can you have that towards three? That shows you it's not going to be the obsessional Hollywood-type love, meaning the obsessional infatuation-type love. It definitely won't be that. It's going to be some love that's more, I'd say, less obsessional and more maybe you have bonds with them.
- 6.78** From a neuroscience perspective, it could work, but I feel like it would require a lot of plasticity and a lot of agreement between these partners. What was the question exactly? Sorry, I'm trying to understand how to answer this. Are you asking me if it's an advantage? Or it's good, the brain wired for that? If you look purely in the brain, we didn't even have marriages. People in the savannah didn't obviously have partnerships in that way. They may have had some rituals and things like that. But the question is: could it work? Is that what you're saying?
- 6.79** Well, you talked about the Coolidge effect and how it leads to cheating, and cheating is very common.
- 6.80** Yes.
- 6.81** So, can actually an open relationship be somehow—
- 6.82** A way to get around that? Is that what you're saying?
- 6.83** Something like that. Or it always will do more harm, actually, to your relationship?
- 6.84** It's a great question. The short answer is: I don't know. I would feel like there would be tons of jealousy. There'd be tons of jealousy when there's multiple partners. Intuitively, I don't like it. Maybe that's my upbringing, that's where I come from. I don't like it when there's multiple partners in a relationship. But just like a male and a female. That's in my situation, obviously. It's two people.

- 6.85** But I suppose there could be an arrangement where it could work. So I don't think it's always bad, but somebody's always going to suffer a little bit. So, from a neuroscience perspective, I can't speak to it more than to say that it would need a lot of prefrontal regulation if you were in that partnership. You would have to be very well aware that you would be driven towards jealousy. You have to be extremely fair to them.
- 6.86** I think that the opposite scenario—I took a male having three wives, because that's the most culturally available example that's out there. In many cultures, you may see this. But the vice versa, a female having four husbands, you see that in some regions of Nepal and other places. I don't think any man could go for that. And that has to do with the wiring. It has to do with the male wiring. They might do that in some places of the world, but it has to do with the fact that men don't want to deploy their resources towards another man's child. That's the key thing.
- 6.87** That's why, even a female—and this study shows this—a female, if she has a child from a different marriage, some males may be put off by that because it feels like they have to take care of that child. It can potentially be a put-off. Some men are above that, and I respect that a lot. I think they can see the bigger picture of how beautiful it is to take care of another man's child.
- 6.88** In fact, if the woman is divorced. I think that's a beautiful thing. But I think males are too territorial for allowing a wife to have multiple partners. The other thing, the other scenario, has been seen throughout cultures, but I think it's a very harsh and difficult way to live. I couldn't personally live like that. That's my answer. Did I answer your question? Am I being politically correct here or not? I don't really care about politics, but anyway.
- 6.89** The high frequency, is it a positive thing for the brain?
- 6.90** Yes, in the gamma sense it is, because it's high frequency, not in the beta type. Beta is different. It's when you're stressed out, sympathetic arousal, thoughts all over the place. That's the beta state. Gamma is more far-flung regions, but coordination as well, a lot of coordination of the wave, so it creates more clarity, in fact. It's clarity of attention, and emotion, and thinking, and so forth. So it has all these components.
- 6.91** And I do want to talk about eye-gazing because you do find that eye-gazing and neural synchrony, getting to your point of brain waves, doesn't only work for partners, but it also works for infants. In fact, it works in a way that it can lead to increased volume of the brain by continuously having these eye-gazing sessions with your child.
- 6.92** It has this aspect of not only are you reflecting your own nervous system to them, if you have noradrenaline and you have high stress and so forth, that is being transmitted during the gazing session because the child will just pick it up and then the brain will go to alpha, like you are in an alpha state, a more calm and steady, but over time, you might see things like actual brain volume changes. Practically, that means you should definitely connect with your child, look your child in the eyes. This lecture teaches us

really the importance of—and I want you to go home with this. It's not only theoretical abstract stuff. Hug your child, kiss them, play with them, eye contact. This really helps. This really helps.

- 6.93** Does positive love impact us for the long run, making our brain change to become smarter?
- 6.94** Can love make us smarter, you mean?
- 6.95** Can it make us healthier? We know that loss of love deteriorates our health. So would strong positive love increase our health and make us smarter?
- 6.96** Very good. Yes, yes, yes. Actually, what I thought you were asking was: can dark love make us smarter? That's what I thought you were asking. Actually, going through a tumultuous relationship and getting out on the other side, can that make us smarter? So let me answer this question, actually, answer both.
- 6.97** So the first question is, yes, definitely love and being in a healthy relationship that's stable and that's based on safety: less strokes, less heart disease, less anxiety, less depression, less propensity towards trauma, better cortical thickness, all these indicators, chronic inflammation goes down. It's a buffer for life's uncertainties and the harshness of life.
- 6.98** But the other scenario I think is true as well, which is, I do think that even though love can be harsh and hard and turbulent, you still learn from that. There's a strength that comes out of you on the other end. There's a strength that gets into the circuitry wired in there. And I think that love can be positive in that regard in any way.
- 6.99** I think the worst thing a person can do is shy away from love. Fear of being hurt, fear of rejection, yes, it hurts. You can be hurt, but you can also not be hurt. It can also be a good relationship. It can also be a healthy relationship. And if you get hurt, so what? You get over it and learn from it. So we shouldn't be scared of having our emotions bruised a bit. So that's my thought on that.
- 6.100** I guess, why? What is your why with all of the neuroscience of love? Why does it happen? Is there a greater purpose for it?
- 6.101** I see what you're saying. I see what you're saying. Is there a greater purpose for it? It depends on who you ask. If you ask a reductionist scientist, he'll tell you that no, there's no greater purpose to it. It's all for survival. It's just what it is. If you ask me, as a religious person, spiritual person, I'll tell you that I believe it's God's design. Is that what you're asking, for example? Could it be something along those lines?
- 6.102** So the question is a real tricky one because scientists usually hide their religiosity or their belief in God. I live in that world. I know them. I know them, okay? They're my buddies. When we go around in the lab and talk about God and all that, they're not very fond of talking about God and beliefs and all that. But there's nothing in the religious sense and belief in God that goes against the brain and neuroscience. I would

put it that way. In fact, for me personally, the more I learn about the brain, the more I learn about the activation of circuitry, and the more I learn about the neural structures, I gain more belief and faith in God. That's just me personally.

6.103 Many scientists may think differently, and I frankly don't care. That's just my belief. I found that in my own life, some of the greatest scientists out there had a belief in God. They believed in God and faith and things like that, but often would keep it hidden from the world. But as I said, personally, I don't give a damn. I am who I am, and I think that's a key lesson, if we can end on this.

6.104 In life, whatever you do, be yourself. I came from the bottom. My parents were poor, ran away, came from Iraq, war, to Bulgaria. Refugee camp, I was in a refugee camp in the ghetto. I witnessed all kinds of things, but I was always proud. I met Danish people with blonde and blue hair, and here I am, this guy, this little brown guy. But I was always proud because I knew God created me in this way, and I am proud of who I am, and I will never hide my parents. I would have friends that would hide their parents. We would go to school and they would hide their parents. I would always kiss my parents in front of my friends. I don't give a damn who's looking. If the girl doesn't like my parents, I don't care. I'll take my dad to a lecture at Oxford, put him in the front row, and that's it. You don't like my parent? You don't like my father, who has grown old? He doesn't look maybe like Brad Pitt anymore.

6.105 Having an authentic self and not caring about what anybody thinks about who you are and your background, if there's something I can convey in this course, that is it. Just be yourself and don't care what other people think. That's the recipe for any success. I'm not saying I'm successful, but any success in life is that, that grounded and personally, belief and faith have driven me. Is that a good answer? It makes sense? All right. All right. Thank you.

Lecture 7: Sexual Minds

At a glance — Sexual minds and early attachment circuits.

- 7.1 In childhood, obviously, we are laying down all these circuits, these attachment circuits. "I'm a securely attached person." "I'm an ambivalently attached person." "I am an anxious and clingy attached person." We lay that down in childhood. But there's one notion that—we also lay down a lot of our sexuality in childhood. Now, this is very Freudian, but I think it's interesting anyway. And this is the notion that— have you heard during sex, for example—when we talk about sex, I may get a bit shy, so I may look down and all that.
- 7.2 But people have like dirty talk. "You're so dirty, baby." What do you say? What are the terms people use? The point is that, truly, we use these terms and scientists, some will say that this goes back to potty training and stuff like that. I know it sounds a bit— but that's one notion that we were exposed when we were potty trained, and these are the Freudians, of course. And because of that sensitivity, the circuits are being repurposed for that. So that's actually one view. You have to know this view, but it exists out there. And it could potentially explain why we have this going on. But of course, is it the whole story? I'm not sure. Definitely, the whole aggression part we talked about yesterday, and it's crucial, the fact that aggression and sexuality, for example, are in the same part of the brain. And does that explain fetishes?
- 7.3 Let's take a step back. A mother looks at her child. Mommy looks at little Joey. Sorry. Little Thomas. And then we scan his brain and see, or we scan the mother's brain and see what circuits are activated. We find that the prefrontal goes down, similar to a man and a woman being in love. Prefrontal goes down. TPJ may go down. ACC becomes activated—empathy. This pattern now is similar to the mother and child as well. So there's a similar pattern. I'm talking about the similarity.
- 7.4 You will find that in this activation of networks, there's a crucial difference between the activation when it comes to two parts. When a man looks at his lover, you will find the hypothalamus activating. Whereas when a mother is looking at her child, the hypothalamus is not activating. Is this clear? So this is the crucial difference. There's also an additional difference, which is that when a mom and child are gazing at each other, the face area tends to light up, especially in the mother. So, the fusiform face area, we have a region for processing faces that tends to be very active as well.
- 7.5 But my key point was the hypothalamus. This marble-like structure, we have two of them, and these are involved in things like sexuality, things like love as well. But they are the key difference between the love—like what we call maternal love or paternal love, and romantic love—the hypothalamus. My mentor always used to tell me that you can't really have romance and romantic love without a sexual overlay. I'm not sure. Is that always true? I think it's always true. You can't be romantically inclined towards

someone without that romantic tinge to it. That's really the hypothalamus driving it, the hypothalamus.

- 7.6** Now, as you were hinting at, the hypothalamus is also involved in aggression. That's why we have sexual rape scenarios and things like that. That's why you say, "F- you." Have you heard the term, "F- you"? Why do you say, "F- you," to somebody, aggressively? Why use that term at all? It's a very hypothalamic term, right? "F- you." It shows us where—it points to this fact that the hypothalamus processes both sex and aggression.
- 7.7** But can it explain fetishes, is what you're saying? When people do damage to their skin and all that, and they put maybe to their nipples, they put stuff in and all that— chains and all. Okay, well, that's a very interesting question. Very interesting question.
- 7.8** Well it turns out, one theory says that when you are a child, a lot of these people with these tendencies, when they are children, they tend to have chronic illness. We know that because we can go and look at their files. We can go look at their history and see, "Oh, this person who's putting all this chain around his neck and his genitalia and all that, he was at a hospital for a very long time." He had a chronic illness. So what's going on? What's the correlation? What's happening?
- 7.9** You know what one key could be? One key factor could be that they learn to fuse sex with pain in the brain. So now, think about this. You know what dystonia is? Dystonia is the following: I play the guitar, and I use these two fingers all the time. I play the guitar. After one month, two months, six months of me playing, what happens? I can no longer move these fingers separately. They get stuck. Why is that the case? Because in the motor strip of the brain, the fingers are placed right next to each other. They are adjacent in a motor strip. Meaning—
- 7.10** In the brain, there's a strip of cortical tissue on the outer layer of the brain. So the cortex is six layers deep. On the outer surface is something called the cortex. Right here is a motor strip controlling all your movements. Okay? Right controls the left side. Left controls the right side. You have it on both sides. Now, this motor strip has an entire human being mapped onto that strip. And that strip, the way the human being is mapped onto that strip, it's in a way that actually looks like your human physical body. So it has a leg and the foot would be next to the leg and then the stomach and the arm. It literally looks like a human being. Is that clear? And the fingers also, they're next to each other.
- 7.11** Now, Rachel may ask the following. She may say, "How come they are next to each other? Does that even make sense? What's the functional logic?" Nothing in the brain is a coincidence. Nothing in the brain is just sloppy wiring. There's always a purpose, and the idea is the following. Whenever you use a bodily limb or a finger—I tend to use this thumb often with this finger. And then with this finger coming next, when I grab a glass, I go like this [makes sequential sounds] and grab. Is that clear?

- 7.12** The brain says, "Oh, I should cooperate with this finger, this finger next, this finger next, this finger next. Oh, it might be a good idea to map that next to each other so I don't have to do a lot of work, so the circuitry is quick with axons. I go quickly here." Does that make sense? There's a lot of cross-wiring between these regions of the brain and the cortex because these have to cooperate. Is that clear so far?
- 7.13** Then in the motor strip, right behind it, there's a sensory strip, a sensory map of their entire body. The sensory map is also like a human being. We call it a homunculus. But it's kind of deformed. It's off a little bit, where, for example, you might find the lips being huge. You might find the feet being large. Genitalia are much larger than normal. The tongue is gigantic. Why do you think this is the case? More receptors. You've seen my previous course, right? Good.
- 7.14** There are more receptors in these regions, so they need to—they have to have more cortical representation in the brain because these are more sensitive. In fact, we often use these regions in sexual scenarios: the tongue, obviously, genitalia, of course, and then—but not the feet, do we?
- 7.15** Now, here's a question on the feet. One theory suggests that—so it's interesting, even though that shape, it's shaped like a human being, there's one exception in that shape. You know what that is? The feet are right next to the genitalia in that map. Did you know that? It's a human being. Everything is shaped like human form: big lips, big tongue sticking out and all that. But then suddenly, instead of the feet being down there, they're up here next to the genitalia. There's a deformity in the map.
- 7.16** We don't know exactly why that is the case, but some have argued that this could explain why foot massages, massages are so erotic. And that's why you would never go for your wife getting a foot massage by your friend or something. You never go for that. Okay, so that's actually one theory that could explain that.
- 7.17** Anyway, that was a tangent. When you go back to the whole children's in the hospital — your brain tends to link things together in the world and then map them on so they become fused. Now, imagine if you are in a hospital and you're constantly being probed. Doctors are coming and sticking metal things into you, probing you, looking at you naked. You are in pain, physical pain. You may have cancer treatment. You have all that. Your amygdala goes through the roof. Cortisol is chronically activated.
- 7.18** How do you deal with that? One way could be to engage in things like masturbation or other forms of self-pleasure, in order to escape that pain as a 12-year-old boy, for example. You engage in these sexual fantasies and masturbation and so forth, which then leads to a reduction in the amygdala. It leads to endorphins being released in the brain. You're releasing all these endorphins, which is dampening down the amygdala. It leads to temporarily a reduction in cortisol, and you feel good. It's a temporary escape from the pain.

- 7.19** But at the same time, now you've learned to associate pain with pleasure, sexual pleasure, in fact. Is that clear? Now there's a link in your brain. Each time you're being probed with that metal thing, you are doing some sexual act or fantasy to escape it, but it becomes fused, it becomes linked. Your brain associates the two. And for that reason, that could explain why a lot of people go on and do these kinds of acts later on. It becomes something that's wired in the brain, and they feel like whenever they do put metal into their bodies and so forth, they have sexual fantasies and sexual desires.
- 7.20** I think that kind of plasticity also explains things like why some people become sexually attracted to animals or even objects. Very bizarre, but they do, right? It has to do with some pairing in childhood. Early on, when the brain is very plastic, they associate these objects with pleasure.
- 7.21** In fact, pornography is one of the culturally very relevant examples of plasticity and how plasticity can change your brain. People might start with something extremely soft, then that's not good enough. They keep going and keep going, and escalating, and the dopamine system keeps wanting more and more because it leads to habituation over time until they eventually— This could actually lead to [where] people go into some hardcore stuff as well because they feel like the softer now can't titillate their dopamine centers, and they need more rough stuff in order to get them going. So that's one way it can escalate as well.
- 7.22** So the Coolidge effect works potentially through pheromones, which are chemicals that can be picked up in the olfactory bulb. Now, are pheromones a real thing? It's this chemical that scientists will say—and this is why I mentioned yesterday, whenever you want to meet somebody, you're interested in somebody, don't do too many emails and DMs. You have to meet them in person because smell is a huge part of it; it's a huge part of it.
- 7.23** In fact, men will smell the T-shirts of women pre-ovulation and will say, "Oh, this woman is more attractive or this smell is more attractive." And vice versa. So women, when they are in pre-ovulation, they will find men's smell more attractive. So there's something about smell and our hormones that needs to be taken into account at all times. Is that clear?
- 7.24** And so some people will talk about pheromones being a factor. These are chemicals released by the hypothalamus and then picked up by the olfactory system. Olfaction, of course: sense of smell. There's huge controversy whether this is actually the case, whether there is such a thing in human beings. In other words, we don't know for sure whether we have this, but it's speculated that we do have these pheromones. In other words, it's speculated that we react to them and we can shape our desires based on them. The Coolidge effect is very much a pheromone phenomenon.
- 7.25** Now, the Coolidge effect is interesting. So you have a hamster, it's mating all day. But then it gets tired. You bring in a novel mate, and it will mate again. Now, there's a

circuit in the brain of this hamster. It goes from the amygdala, the stria terminalis of the bed nucleus of the amygdala—a fancy name for a specific structure in the amygdala—it goes to the hypothalamus part of the brain. You know the hypothalamus by now, right? This structure, if I were to zap this structure, guess what happens? Even if I don't bring a new mate, it will still mate with the old mate. We've just found what? We hit on a sex circuit in the brain. So this has been shown. Think about this. And if we sever this circuit, guess what happens? It will not mate at all. It will stop mating.

- 7.26** So the amygdala itself and the stria terminalis of the bed nucleus—it's a fancy name, but this structure is important. See, once you get into the nitty-gritty of neuroscience and really want to get deep, go to the amygdala, you will find—within the amygdala, there's like 30 parts of the amygdala. There's this particular part. And even things like anxiety, for example, we talk about anxiety being the amygdala and fear. No, actually this part of the brain, the stria terminalis of the bed nucleus, is where anxiety is in the amygdala itself. You know what it is? It's like when you get to know somebody, the more you get to know them, you get into the details of that thing. And so that's how the brain works as well. Amygdala, then you go more deeper into it. And that goes to the hypothalamus. And then you have things like you have a sexual circuit.
- 7.27** Now, this is in mice and rats. The question is, does it work in humans? Could it work in humans? Perhaps. And if it does work in humans, it could be cool because then—we talked about how we can make people more sexually active into their 80s and 90s, right? How can we sustain sexual activity in people? If we somehow can get into that circuit in humans and selectively stimulate that, perhaps that could be it. Now, we don't know for sure [whether] that circuit also is the circuit for humans as well. But if it is, it would be pretty amazing, wouldn't it?
- 7.28** Yeah, sexuality is a complex one. It's shaped by a lot of factors and also a lot of unknowns. There's a lot of things we do not know about sexuality. For example, why people even have same-sex preferences. This is a huge, huge controversy. A lot of scientists will shy away from this completely, including myself, because it's dangerous territory. If you go there, you might destroy your career. It's risky stuff. You might say the wrong thing. Anyway, luckily, it's not my area of research.
- 7.29** But there was a study in the 1990s that came out in Science. A big paper, a big splash. It made a big splash. The hypothalamus of the brain, the anterior part of the hypothalamus was found to be smaller in gay men versus non-gay men. It was found to be smaller, actually resembling the hypothalamus of females, much more. Now, this made a huge splash in scientific literature. "Oh, we found the brain circuit and the brain hub for homosexuality." It became a huge thing.
- 7.30** A couple of issues with this paper. First of all, it was only 40 men, and many of them had AIDS. They died of AIDS. So it wasn't clear whether it was the AIDS, the disease causing this shrinking of the hypothalamus, or whether it was their sexuality. So it's very correlational. Small sample. To my knowledge, it hasn't been replicated. And as far

as I know, the scientist who made the study, he was gay himself, which could create a bias. I think he was. I'm not sure, but I believe that was an issue. And also, I don't think it's been replicated. It was one study, and I think it was dead people. They were looking at postmortem and things like that of brains. But you should know of this study, of course. It's an interesting study. One should be aware of it.

- 7.31** But really with homosexuality, I should mention there are various ideas and theories, but it's not really clear. The picture is quite fuzzy. It's messy. We don't know. Is testosterone lower in homosexual males? Or excuse me, in homosexuals versus heterosexuals, for example? The answer is we don't know. It seems like testosterone is actually the same. So it's not low testosterone that could cause homosexuality or contribute to it. The short answer is: we don't know much about this area. It's still open to investigation, but there's no real clarity in terms of— there's a lot of confounds, genetics, epigenetics, plasticity that plays in, that can shape that as well. Is that clear?
- 7.32** There's a region of the brain called the OFC. We know the OFC by now, right? The orbitofrontal cortex. It's a what? A bridge between emotion and intellect. It's evolved in OCD. It's a kind of a break for the brain. There was this guy, Joe. A brief summary, I've talked about this in length in previous courses, but a brief summary. He suddenly, overnight, developed these sexual urges, went to brothels, tried to pick up women left and right—and he was married—he became a sexual monster overnight, pretty much. As time passed, he even made sexual advances towards his teenage stepdaughter.
- 7.33** At this point, his wife stepped in, and he was captured by authorities. And as he was being potentially imprisoned, he was to see a judge and all that, he had a severe headache, and he took himself to the hospital. In fact, he let them know that he had the severe headache, and he was examined. And they looked inside his brain, and inside his brain, they found a large tumor on his orbitofrontal cortex. They removed the tumor, and his sexual urges disappeared overnight. So as I always ask, "Did the tumor reveal his actual sexuality or—what do you think?"
- 7.34** What we do know that is the orbitofrontal cortex really is the brakes in the brain. It's the brakes, including the sexual brakes, on sexuality and sexual impulses. We all have desires and things that we do inhibit. We do inhibit. They flare up in our dreams, and we have sexual dreams. But are we those people? Are we the person that is driven by—the impulse-driven person, is that really the real you or the person that's a balanced you with the brakes? If that makes sense. I don't have the answer. I don't know. It's really hard to say.
- 7.35** There's another interesting case. This is Oliver Sacks. He talks about this case. It was a 90-year-old woman; quite a cute case, actually. She was 90 years of age, and suddenly she had all these sexual desires. She said, "Doctor, doctor, I feel all sexual. I feel like I want to be with all these men, and I want to flirt, and I want to just have fun." Oliver Sacks is examining this lady, and he's like, "What are you talking about? What's going on?" And things like that.

- 7.36** It turned out this lady had syphilis, sexually transmitted disease, early on, and it had penetrated her brain. And what it can do is it inhibits the prefrontal and the orbitofrontal cortex. It's like the tumor. It removes the sexual brakes. Now, she had this when she was young, and then it reemerged. The virus can sometimes be latent and reemerge when she was 90. As a 90-year-old woman, she had those sexual—it's called Cupid's disease. That's what it's called in the book. It's quite interesting.
- 7.37** Do you feel like testosterone has fallen over the years in men? Are men less masculine? Testosterone is falling, sperm quality is falling. And the question is, why? Why is it falling? Why this decline to begin with? We don't know for sure. It could be environmental exposures, it could be food, it could be chemicals. A lot of things going on. It could be sitting on social media all day and not being physically active. It could be sleep, poor sleep. These are the factors that play in. But there seems to be a decline for sure.
- 7.38** Now, this needs to be a practical course as well. So we have to look at how we can improve our testosterone. Now, testosterone is not only good for men, but also for females. It goes both ways. It's not only—of course, testosterone is in the male, what we associate with male and male strength and all that, and masculinity, but really in both sexes, we need it. And estrogen as well for males.
- 7.39** So what can we do to naturally boost our testosterone? Sleep, first of all. Deep sleep. You have to get a lot of deep sleep. Now, when you go to deep sleep, you cycle through the different stages of sleep, you hit deep sleep after a few, maybe that could be in the first 90 minutes of sleep, towards the end, you will hit deep sleep. Get a lot of that. You have a lot of that in the first part of the night. Meaning if in the night you sleep eight hours, the first four hours, you have a lot of deep sleep, a lot of testosterone there. But also in REM sleep, you'll also get that. So you can monitor these sleep stages using some watch or something. Men who sleep five hours at night have about 15% reduction in testosterone. Can you believe this? It's crazy. We can't have that.
- 7.40** Exercise is important, but not all kinds of exercise. We talked about how aerobic exercise, like biking and running and all that where heart rate is up, that's good for BDNF and brain plasticity and endorphins. But if you want to have muscles and muscle strength and testosterone, you need resistance training. So you need to go to the gym, maybe do some deadlifts, do some bench presses, do something so you can build some muscle mass. Resistance exercise. High intensity can work as well. So HIIT: you're on the spin bike and you're going quickly, and then you pause. One minute quickly, and then you pause and keep going.
- 7.41** A crucial point: don't overtrain. Don't overtrain. Overtraining is bad for testosterone. Meaning, when I used to go to the gym, and I wanted to impress a girl. I used to go for 2 hours a day because I was hoping to run into her, so I would exercise for 2 hours a day. And eventually, my testosterone was just down because I was going on all kinds of machines. And then I'd take more to impress her a little bit. And then after a few weeks

of that, I had no more testosterone in the system, and I was looking all flat and all weak.

- 7.42** Testosterone works exactly in a seesaw balance with cortisol. Testosterone and cortisol are balanced out. They wrestle it out all the time. So the HPA axis is hypothalamus, pituitary, and adrenals; cortisol and high cortisol. Testosterone, on the other hand, is the hypothalamus-pituitary and then the gonads. But these two forces are always in opposition. If you have chronic high cortisol, you will have chronic low testosterone. Is that clear? You cannot have both.
- 7.43** And so if you're always stressed and you're always running about and agitated, what happens is that your testosterone will go down. And then you will be like me and try to hide it— perhaps if you're overtrained, you try to play cool and like you're smooth and all that. But you can't because you can take a picture of your face, it's like a photo of your face, and show it to a scientist, and he will tell you, "This is a high-cortisol face."
- 7.44** Can you believe this? Just by looking at a face, he'll tell if this is chronic high cortisol. It will look different. Literally, it becomes different. It looks different. If it's high testosterone, you can see it. It will just look different. So there's no way of being an actor and playing it cool. It just won't work. You can't get away with it. So you have to actually have low cortisol. Does that make sense? Marathons and all that, don't do them all the time, they will kill your testosterone.
- 7.45** Muscles, you need protein for muscles. Healthy fats, in fact—we think fats are bad for you, but healthy fats can be good, like olive oil, nuts, avocados, fatty fish. And then supplements, take good supplements—zinc, vitamin D, make sure you get vitamin D. There's a huge thing about fasting these days, and I do this often. I fast a lot, so I may do intermittent fasting and things like that. But in fact, too much fasting can also kill your testosterone. So don't do too much of that. Obviously, stress manage. Manage your stress, meditate, breathwork, avoid alcohol and toxins. This is the recipe. After this, you will have high testosterone.
- 7.46** Are there any supplements we can take that actually have been clinically shown to have an effect? There are. There are. So vitamin D, I mentioned, if you're deficient, especially. Zinc. Something called ashwagandha, has some trials supporting it. There's a herbal remedy called tongkat ali that is shown in clinical studies to have some effect. And then something called DHEA, a precursor hormone. So these are the main chemicals or compounds that could help. Anyway, we shall look into these.
- 7.47** Oh, yeah, man, there's a few things that can kill sexuality and sex drive overall. Having low testosterone overall is one of them. Being highly stressed is one of them. Being anxious is one of them. Depression, mental illness overall. Because mental illness is really all about the amygdala being hyper-activated and affecting this HPA axis in a way where you have chronic anxiety, you literally will always see low testosterone—like in OCD, low testosterone; depression, potentially; anxiety disorders, and so forth.

- 7.48** But there is actually a disorder called hyposexuality, where you don't have sexual drive all the time. You're just low on sexual drive all the time. About 10% of females have this, and 15% of men. This is an interesting disorder. These folks just are not excited about sex. There's no sex drive at all. And then you might ask, what about if you look inside their brain is there any signs of their brains being different than the rest of us? What do you think? Probably, right?
- 7.49** So it turns out there are differences. Let's think about some of the differences in the brain. In this lecture, we talked about one crucial brain structure for sexuality. The brakes for sexuality. What are the brakes for sexual lust? What is the main brake? OFC, right? OFC, orbitofrontal cortex, correct? It turns out when you scan the brains of these folks and they look at sexual images, when they look at sexual images, their OFC is all wrong. It's inhibiting their sexual drives. It's putting a restraint on their sexual drives. In a healthy person with a normal sexual drive, he's looking at these photos and his OFC shuts down as it should. The prefrontal and the OFC goes down. In these folks, you don't have that. The OFC is inhibited. There's a pure inhibition, a clear-cut inhibition.
- 7.50** And then in regions like we talked about the premotor area yesterday. What's the premotor area? Planning of action. So if you want to aid somebody and you're meditating and you want to go to help somebody over there, your premotor area might light up. It turns out when a normal healthy person looks at a sexual image, his premotor area also becomes active. It also dances with activity, becomes hyper—it flares up. In these folks, it's dead. There's no activity in the premotor area. And also in the somatosensory regions, there's no activity.
- 7.51** So, Natalie, what I'm getting at here is that there's—what's that? [inaudible student] No, what I'm saying— I'm not saying you have this. I'm saying what I'm getting at here is that there's a clear-cut brain basis for this. It's not some mumbo-jumbo. They're not just saying this. There's actually a brain basis. So low sexuality, obviously, is a bad thing. Having no sexual drive is a bad thing.
- 7.52** But what about having too much sexual drive, a hypersexuality? Is that a good thing? It's not a good thing; it can become an addiction. And so there's also a condition called hypersexuality, where you are hyper-involved in compulsive sexual acts. So these folks will consume pornography. They will pick up women left and right. They will have to engage in sex compulsively all the time.
- 7.53** In their brains, again, there's a clear signature that would explain this. It has to do with the ventral striatum and the nucleus accumbens, mostly. What's the ventral striatum and nucleus accumbens doing? We talked about this. You know this. I know you know this. Release of dopamine, right? Dopamine, ventral striatum, nucleus accumbens. Remember these structures: nucleus accumbens—dopamine. These regions of the brain, they are the hubs for dopamine and pleasure and anticipation of pleasure.

- 7.54** Now, if I give you a drug like L-DOPA or anything that can boost dopamine in the brain, what happens? You might artificially create hypersexuality. It can also happen artificially by taking a pill, by popping a pill. It really speaks to the dopamine circuit and how the dopamine circuit works and how if it's overactive, because dopamine is not only—think of dopamine as not only the anticipation of pleasure hub in the brain, but dopamine is also the—think of it as the movement part of the brain when it comes to motivational movement part of the brain. It has to do with a lot of movement as well. And so my point is it can lead to sexual compulsive acts. A lot of compulsivity that has a reward component to it.
- 7.55** Let's say somebody has a desire to do something compulsively, like engage in a sexual act. The dopamine will not only work in the ventral striatum, but also spill over into the basal ganglia and allied regions. So this is a bit of a technical point, but what I'm trying to convey is that dopamine works in the structure for anticipation of basic reward, in the frontal lobes and so forth, but also in the movement and the compulsive act, what you might call a behavioral addiction. So that's the role of dopamine and how hyperdopamine can lead to this bizarre illness.
- 7.56** Can we treat these illnesses? What do you think? We can treat them. How might one go about treating these illnesses? Balance the chemistry of their brain. How? "What drugs, doctor, would we use?" Now, L-DOPA might make them even more sexual. We don't want L-DOPA. We don't want to add fuel to the fire, you see. Dopamine is causing it, right? Remember, dopamine is causing it. How can we balance out dopamine? Serotonin. So SSRIs could work. They might balance out—
- 7.57** In fact, the problem is then you might end up with hyposexuality and low sex drive because that's one of the features of SSRIs. So it's always tricky. In that case, the doctor might say, "Oh, 20 milligrams of this, fluoxetine 50 milligrams. This 70," and then, "Oh, it's too much. Now, you have no sex drive. Let's go a little bit down." So that's why tapering and all that happens. But absolutely, SSRIs.
- 7.58** BDNF. Could we use plasticity? You're saying plasticity? Yes, plasticity. I would say, generally, exercise would be a good way to get some of that desire out and then rebalance the circuit towards serotonin. You have tons of release of endorphins and other feel-good chemicals when you exercise. I would say that's one. But really they have all kinds of tricks like desensitization and CBT exposure therapy techniques. Now, luckily, I'm not a therapist, so we won't go into all those techniques that have to do with sex tools and all that. That's not my thing. All right, so I'd like to take some questions.
- 7.59** I've seen a few studies that are saying that Gen Z is having less sex than previous generations. Is there any neuroscience behind that?
- 7.60** They're having less sex? I think so. I think they're having less sex. I haven't seen evidence or data, but it sounds very plausible. I'll tell you why. You know why? I think it

has to do with the fact that people are getting their dopaminergic urges covered from social media and the internet. We only have so much dopamine available. We talked about how dopamine always needs replenishment. There's always this arc where if dopamine goes up, it has to go down beneath the baseline. I think that definitely makes sense.

- 7.61** If you're getting all these images thrown at you every single day, then that can—but even beyond that, I think when you look at these females, I'm taking male as an example, males are more visual. If you're seeing all these picture-perfect females, pornography or whatever sites people are using, and they're like, "Everything is perfect about her!" She's been through cosmetic surgery a trillion times, and she looks like an AI doll. Now, when you look at your wife, you may not get turned on because your dopaminergic centers were just biased towards that.
- 7.62** We have something called acquired taste. An acquired taste is the fact that you may not be born with a liking for coffee. You don't like coffee, but then you drink it and you get some dopamine spike from that and you just feel like coffee tastes good. That's an acquired taste, for example. Sexuality is very much driven by an acquired taste. You have this woman, you're sexually with her, you develop a desire for her body schema and her body type, for example. It's an acquired taste. Now, if I get my acquired taste shaped by this perfect doll, then anything that is less perfect than that will not create arousal. I think that's a huge barrier. I think that can lead to low sexuality between couples, for sure. But yeah, it's an important question.
- 7.63** Does physical touch have an impact on DNA in romantic relationships?
- 7.64** In romantic relationships? Absolutely does. Yesterday, we talked about how physical touch in childhood, so if you touch your child a lot, that's going to change the DNA and the methylation within the DNA, so you switch genes on and off. Now, the question is, if you apply that to a romantic scenario, a sexual scenario, for example, and touch between two lovers, will that turn genes on and off? The answer is yes. So this is a lifelong thing. This is a lifelong thing.
- 7.65** However, it's just worth keeping in mind that epigenetics and brain plasticity overall is much more ample in children. It's just much more easy to change the brain of a child. So that's worth keeping in mind. But epigenetics is lifelong. So it's a good question. Lovers touching and all that will change your DNA, which is a good thing.

Lecture 8: Modern Love

At a glance — Modern love.

- 8.1** Now, in this lecture, we'll talk about two aspects of love that are very important. A is AI, so love and AI. Then secondly, we'll go into courtship, courtship itself. How do we master courtship? How do we become Romeo? How do we become Casanova? How do we woo a woman? How does it all work? What's going on in the brain? So we'll cover those two.
- 8.2** I also want to make sure that in this final lecture, all your questions get answered. I don't want you to go home and feel like Baland didn't answer your questions. He was just rambling on the brain and the OFC, and the orbito and the ACC and all that, and he didn't answer your questions. So we'll take some questions now, and we'll weave them into the lecture. And at the very end, we'll also have some more questions. You can keep them AI-related and courtship-related, but also generally, because this will be the final one, and we want to cover, we can talk about anything you really want to talk about.
- 8.3** I would like to know what causes people's brains to become stuck. Whether it's with LTD, not moving on past grief with PTSD, there's so many different ways that our brains become stuck, while other people are able to move on with life. Then others are, again, I feel like, just the word stuck, they can't move on. We understand neuroplasticity, that anybody's brain can change and grow. Yet some people seem to not be able to. I really want to understand what causes certain brains to be stuck in that state of darkness, PTSD, LTD.
- 8.4** Or just grief, right? They can't move on past this.
- 8.5** Even if they're working through it. They're in therapy, they're healthy, they're exercising, they're getting sleep, they're doing all the things that are recommended to do. Yet here they are stuck. Why?
- 8.6** I want to try to answer your question with a story. This is one of my favorite stories. It's the story of Joseph. You heard the story of Joseph in the Bible and the Quran as well? It's very classic. I love the story. So Joseph—I'm going to just narrate the story overall. Joseph, as a kid, his brothers take him. They're jealous of him. So their ventral striatum is firing away. They don't like the fact that he's getting all the attention from the father, Jacob. They throw him into a well. And then in the well, he's eventually picked up and brought to Egypt. And in Egypt, he's sold as a slave.
- 8.7** Eventually, as he's sold, this is interesting, he's bought by, there's a minister or a king of Egypt that actually buys him. So he's bought by a very high-ranking man. He becomes a slave. Eventually, a woman tries to seduce him. So we have seduction, clear-cut seduction. He's trying to run, and she pulls his shirt. His shirt is pulled from the back. So

we have seduction in that story. He's put in prison. In prison, he eventually becomes a dream interpreter. Then, he becomes a ruler himself, and he is reunited eventually with his father.

- 8.8** Why I love this story is because it's perfect for this course. Everything in this course is in that story. Jacob loves Joseph, literally loves Joseph, and he cannot get over the loss of Joseph. He cannot get over that loss. It is killing him. In some narration, he even goes blind out of grief. So he's stuck in some plastic state of brain plasticity, and the loss of Joseph just doesn't register.
- 8.9** I think there's two components to that loss. I think one component is the following: how loss and grief can stick. Component A is closure. So in a case like Jacob, who loses his son, Joseph, because he's never really convinced that he's eaten by a wolf, there's what we call an open loop of questions. His brain is always questioning: Is he alive? Would I reunite with him someday? And that closure prevents him from ever getting over the grief.
- 8.10** Anybody who is losing somebody they love, anybody they love in their life, and that person leaves, they may end up in a situation where they cannot get over that person because there's a lack of closure, especially if that person leaves suddenly. If they go to a different country, and they disappear somehow. You know what I mean? You know the difference between a relationship where you sit down, and you break up, versus somebody who just leaves? That's very important. Closure is very important in that whole thing.
- 8.11** Secondly, I think that beyond that, it really depends on how our nervous system is wired. Some people, in the way they form bonds, they may form bonds in a way where the circuitry is laid down—think about circuits in the brain as, initially, you lay them down. So you lay a circuit for this bonding with this person, Father Jacob to Son Joseph. There's a bond. But that bond, depending on your nervous system and how strongly you love, meaning how strong your nervous system is, meaning how sensitized your nervous system is—
- 8.12** Dopamine, for example, as a teenager, first love, a perfect example. Perverse love, as a teenager, we talked about yesterday, how the dopaminergic system is hypersensitized in a teenager. For that reason, because of that hypersensitized dopaminergic system and a hyperactive oxytocin and vasopressin system, their bonding is extreme. Their bonding becomes extreme, and the circuits are laid down in a very crystallized manner.
- 8.13** Now, I think in terms of loss, what can happen is—this is factor A. Let's say generally, you are a person with a strong, sensitized dopaminergic system, and your bonding hormones are also higher than the average person. I think that's factor A.
- 8.14** Factor B, I think that if you are a person who keeps reinforcing the love story in your head, you keep thinking about it like the open-loop scenario, "Oh, one day I'll be reunited with this person. The person is not completely gone. The person will reappear

somehow." I think the circuits keep getting reinforced. It keeps thinking about it. Literally, the attentional resources of the brain is what shapes circuitry and keeps them alive. If I was to give you a pill now and you pop it and you forget all about the person that you love, literally, other circuits will take over. And because it's a use-it-or-lose-it brain, the other circuits would shape your brain in a way whereby that love circuit, that Joseph-Jacob circuit, would melt away. Is that clear?

8.15 I think it has to do with our dopaminergic system, how sensitive it is, and then our bonding system, how amplified it is, plus how we reinterpret the story. And especially, I think in the first few years after the breakup, when the brain is very malleable, and the story is at its peak in terms of it hasn't really waned in the system. It's still fairly novel a couple of years after. That's really when you want to try to reinterpret and try to meet new people and things like that. I think that would be my answer. It's a great question. Thank you.

8.16 My question has to do with AI. My question about AI is, obviously, it comes in different mediums. How is it affecting relationships, personal and with other people? Because as we know with AI, it gives us an immediate response. People can, like you said, fall in love with that response. How is it affecting our relationships with other people, human beings as well?

8.17 Basically, yes, this is a huge problem. Think about it in this way. I've lately been using ChatGPT a lot. I use it for my work. It could be any other search engine. It doesn't have to be ChatGPT, Grok, whatever. So I use this—I'm just saying that because we don't want to advertise. That's what I'm saying. We don't want to advertise for a particular one. But let's say I'm using this engine, this AI.

8.18 I use it a lot now for my work, but then I meet a girl, and I start using it as well. I use it because it's effective. Have you used it for those purposes? Be honest. For romantic? And what is it? It's very calculated, very Machiavellian, isn't it? Okay, good. I'm not the only one. That's good to know. You ask it all kinds of questions, and it's very humanlike in its responses. It seems very human. It seems like it knows all about you and the situation, and it's very smart.

8.19 And then you ask it about various things, and then it says, "Oh, don't worry about this," and, "Oh, don't think about that," or, "Oh, don't do this." Its responses are very humanlike, and our brains were never really wired to distinguish between an AI and a human being in the sense that if the feedback seems humanlike, it has those cues of caring and affection and attending to our needs, we feel it's human.

8.20 So eventually, what can happen is attachment to it. Not only do you become intellectually attached to it as something that can help you in various pursuits, including romantic ones, but you become attached to the AI itself. Perhaps you may create a bond to it. Your oxytocin system and your bonding systems become activated. This is a huge problem because some people actually do develop psychosis. We're seeing this

where their brain cannot distinguish between what's real and not real because these machines are so lifelike.

- 8.21** There are other AIs that will have faces, and they will have eyes, they will have smiles, so they are even more directly tricking you. This is a huge problem. I think as we move along, we must be careful because, really, ChatGPT is a psychopath. It's psychopathic. It is, no doubt, because it has no medial prefrontal cortex. It cannot consult its limbic structures. It literally gave me the worst advice. Any fool, following this advice, would break a girl's heart right away: withdraw, become more mysterious, women like novelty, hot and cold. What are you talking about? It's a human being. She has feelings.
- 8.22** You can sense that ChatGPT wasn't there in the room with the girl when I saw her emotions. I can sense she's highly sensitive. I can sense that she gets hurt when I withdraw too much. I can sense that she feels easily rejected. I can just feel it. She has some insecurities, like all of us. And for God's sake, I don't want to hurt that girl by going away for a month. That's insane. But yet it says, "By all means, stay away. Don't go near her yet. It's too early. You should withdraw. It's much better. Dopaminergic systems work on reward prediction error. So you should not go the next day because then it means you are too safe, and that will categorize you as a safe person, as a friend zone potential." It's nonsense. You have to be reliable in relationships, too. So don't follow these machines for dating advice. Is that clear? I want to make that completely clear.
- 8.23** As human beings, we consult not only our limbic centers, but our emotional states. Our insula is always mapping our gut feelings, our heart rate, our sweating. It's all being monitored. So our human responses and our conclusions are based on all this combined. It's not just based on some DLPFC calculation up here.
- 8.24** It becomes even more tricky with dating apps. The problem with dating apps is the following. There are several problems, but I think the key problem is the fact that it keeps us stuck in the dopaminergic stage of romance. You can think of romance as a dopaminergic state initially. It's intoxicating, it's novel. I'm talking to this girl, and then everything is exhilarating. It's fun. But at some point, you want to go to the bonding stage. You want to create bonding.
- 8.25** But the trouble is, the dating apps keep reinforcing what you might think of as the dopaminergic stage. It keeps reinforcing, "Oh, look for this image or that image," or, "Did she like this?" This goes for social media as well. It doesn't allow for bonding. In fact, one of the key problems with these dating apps is that you cannot show traits that would signal a good, long-term mate. One of them being, I mentioned kindness and generosity and all that, but even things like emotional stability.
- 8.26** The key predictor, perhaps, that a female likes in a man is emotional stability. How emotionally stable is this guy? But can you show emotional stability on a dating app? No, you can't. It's very hard to show any deeper trait. What you show is the flashiness,

the muscles, the resource acquisition potential. It could be fake as well. People fake these things. So really, that is an issue with the dating apps.

- 8.27** Then, of course, most of all, most importantly of all, when we get to know a new person, how do we evaluate them? I've mentioned smell so many times in this course. Smell, how they move, their elegance, their poise, how they smile at other people, do they seem socially kind? Social proof. Social proof is a huge concept.
- 8.28** You've seen *Coming to America*. I've mentioned this movie before. Have you seen that scene where Akeem goes to the bathroom, and these two African men from the same country see him? Have you seen this scene? They go, "Oh, my God." He's a prince in his own country. So they want to take a picture with him, and they hug him, and they actually go down on their knees and like, "Oh, Akeem, this is the greatest experience in my life," and he goes down on his knees and things like that. They're very passionate in that part of the world.
- 8.29** The woman, Lisa, sees this and immediately goes, "This guy, there's something about him that's different." It's an example of social proof. Social proof. In that movie as well, Lisa will tell Akeem, the prince— this is what I like. He actually portrays himself as somebody who takes care of goats and sheep and things like that. That's his job, he says, his actual job. But he's a prince, as you know.
- 8.30** Now, Lisa says, "Akeem, there's something about the way you carry yourself, the way you walk, the way you talk, the way you express yourself. It's almost regal." I think these are the cues that a dating app will never pick up on. How do you carry yourself? Do you seem confident? How's your body composition? How are you composed, self-composed? Are you calm? All of these things will not show on a dating app at all. So this is the problem with that.
- 8.31** Is it possible to stick an antenna on your head, EEG, and you look at flashing images of women on a dating app, and you're seeing these images. Could a dating app, could an EEG, pick up on that and tell you which women you find attractive? Is that possible? It might be, right? It turns out the answer is yes, it's actually possible. Just by monitoring the activity and the waves of the brain, this machine will tell you whether you find lady A, B, C, or D attractive.
- 8.32** It primarily works for people that have a strong attraction or strong rejection tendency. So they're like, "Oh, I definitely like this girl, or I definitely don't like this one." These are the guys it works for. But I think it brings up something crucial, which is that in the future, the technology could be used to really decipher what types you like and just give you 20 people that you generally like in a very specialized and personalized way. So it could be very powerful in the future.
- 8.33** So we talked about hybrid vigor and genetics. Gabriella, remember I mentioned, if I have a certain set of immune system genes, and this woman over here has a different type of immune system genes, I tend to find her more attractive because we create

what? A potential offspring with a hybrid—it's called a hybrid vigor—and a better immune system. Could you use this in a commercial setting? Get genetic testing and tell you which folks have different immune system genes. That's one way to use that knowledge for that.

- 8.34** Or we talked about the different subtypes. Natalie, what did we say? There were the dopamine types, the excited types that always ran off and used all their money on fancy trips and things. Remember the dopamine types for couples? What was the other one? Serotonin types, the boring ones sitting at home and playing all day. And then, there was testosterone types. These are the ones who want to guide and be strong and take care of their partners. And then we had the estrogen type, the more nurturing, the more arguably feminine and, to some extent, a little bit more passive as well. We got these four types.
- 8.35** If you could genetically tap into what category you fell into or use some measure, then what? How could that be used? Matching, right? Genetics, for example, might tell you how— obviously, it's very hard to measure the level of proportion of these chemicals in your brain, but it might be possible genetically to know how you're predisposed and then match you in that direction. Some companies will talk about that. So I think the future of dating and dating apps— there are tons of these things.
- 8.36** But if you ask me, it sounds pretty simplistic. I think so far, this is way too calculated. It doesn't capture the essence of love, genetic testing and even apps. You're seeing all these women. For me, it never works. Never works. To this day, I see girls on social media, sometimes in my DMs. It does nothing to me. I see a picture, and I don't see anything else. Then I go down to my supermarket, and I see that girl with brown hair and the way she walks. I smell the perfume and everything, and it does something to me that's different. And that's the one I want to pursue. Not the one with the long CV and DM. It does nothing to me.
- 8.37** I think Shakespeare was right about love and what love is. We should never lose that spark. That's my own personal insight, what I wanted to share. Some people might disagree, but that's my own. I think love is so much more. I think you could even be so romantic to say there's an element of destiny. When I say destiny, you could interpret that in the spiritual sense, or you can interpret that in the deterministic scientific sense. That's up to you. But I think there's something there, too. That's something where you're just supposed to meet somebody at the right time, and when you do, it just clicks. That's my own experience. I can't put words on it and explain that further, but that's my own romanticized view of the world.
- 8.38** Before we venture completely into courtship and dating and tricks and tips and all that, there's one component that I don't feel like I've covered in this course, but I think it's so crucial to what love really means because it often gets overlooked somehow. I think it's crucial for that romantic bond, and we could easily be fooled to forget this crucial aspect, which is forgiveness.

- 8.39** I have, at times, had interactions with women, and maybe they have done something towards me. Looking at it now, I could totally see how it's just human nature, a human reaction to something. It could be me. Let's say—I'm going to think of something. I text her something, and I may seem desperate. I seem desperate. Or I may overtext, or I may seem overexcited, and she pulls back. Now, in this moment, if you are self-righteous and you can't forgive and you can't see the big picture, you're in trouble. Or she does something that hurts you. You have a certain date, and she doesn't show up. Or she shows up late, or whatever it could be. My point is we have to have the ability to forgive. We need to have forgiveness as well in our repertoire of tricks, if you could put it that way, in our toolbox.
- 8.40** We need to be acutely aware of the human condition and how human nature, how the brain, how psychology works, and not judge too harshly, even when people make mistakes. I think being lenient, being flexible, being forgiving is key for not only gaining new relationships, but also maintaining them and not going from one relationship to another. I think it's really key. I think it's really key.
- 8.41** Courtship. How do you win over somebody's heart? What are the ingredients you need? As I said before, I don't think this can be formalized in a theory or a recipe, but I think there are certain things that we can think about. Eye contact is helpful. Mirroring is a huge thing. Our brains can detect other people and how they mirror us. Facial expressions—our brain, especially the right hemisphere, is hypersensitive to facial expressions. It turns out my visual cortex, the vision part of the brain, has a super-fast highway that goes from the eye and vision to emotion, even before facial recognition.
- 8.42** That's why when I'm in that, let's say, in a social scenario in that supermarket, and I can sense that girl is angry, I can sense that in a split second, even before I have recognized that that is the girl herself, believe it or not. You know what I mean? It's super fast. It goes beyond visual recognition. We can hyper-visually detect facial expressions of emotion. We have a machinery for that. Therefore, when you are in social interactions, and you engage with a female or male or whatever, be aware of this, be aware of this mirroring, mirroring her emotions.
- 8.43** As a practical tip, though, I would say, and this is something that I got burned myself, doing this, but I think when you go and see somebody, try to be happy when you go see them. Now, if you go and you haven't slept—me—you've been stressed all day—me working—and then you've had four people call you and say this and that about work, and then you're completely riled up and agitated and slightly annoyed, and then you go see her. How do you think that's going to work out? Not very well.
- 8.44** So my suggestion would be, especially if you have a demanding life and there are stressful factors, maybe you have had a couple of weeks of bad sleep, try to at least get some rest. Maybe meditate, go exercise a bit before you go see the girl, because she will know; she can recognize your facial expressions. However, she can't necessarily know that your irritability is not about her.

- 8.45** This was how I got burned because I was stressed. I had a thousand things. And then suddenly, maybe there's a girl coming into your life. You're trying to seem not stressed, but you still are stressed. Everything is going on, and then she sees you. We know that cortisol and stress hypersensitize the amygdala. So when I go to that, let's say, a gym scenario, and you see her. You see her, and then she does slight things to annoy you; you become doubly annoyed now because you're already stressed. She can sense that. It becomes this negative feedback loop where everything is just negative. She sees you, and she pulls away more, and it becomes a disaster. So my point would be, if you just relax at home, maybe watch a Disney flick, *Beauty and the Beast*, maybe you'd be better off. That's what I'm trying to get at.
- 8.46** But mystery is good. Having some mystery is good because it allows for the dopaminergic centers to search for that concealed aspect of you. The dopaminergic system is always titillated by searching and looking, and making things out. That's why if I have a woman here over here who's naked, excuse me, but she's naked, and you have one that's partially naked, it turns out I might find the one partially naked, the concealed one, more attractive. You know why? Perceptual problem solving. My brain is trying to solve, "What does this girl look like? What are her curves exactly like?" So it just titillates these dopaminergic neurons. It is a feature of the brain. Perceptual problem solving.
- 8.47** Another allied concept is conceptual problem solving. Similar thing. In any romantic situation, don't reveal everything about yourself all at once. That's not being somebody who's Machiavellian or trying to outsmart her. You're just tapping into human nature. On the first date, don't tell her about all your dreams, your childhood, and everything about you, because that will just kill the dopamine. Think human nature. Again, think human nature. Gradually, let her know. Gradually, let the person know about your situation, who you are, but keep that mystery.
- 8.48** But mystery, taken too far, will activate the amygdala and create anxiety. Basically, you have this battle going on initially between the mystery and dopamine; you want to reveal yourself slowly and gradually. But too much of that creates amygdala overactivation. The person becomes unsafe. You don't really know where you have them. Who are they? This is what happened to me as well. I became a little bit too mysterious. I followed ChatGPT too closely. This got me in trouble.
- 8.49** The other side is the friend zone, meaning the oxytocin system. You need that for romantic bonding. You're creating a bond, you have oxytocin, and you want to let her know that you are safe, that you're not psychopathic, that you are reliable, that you're a potential long-term mate. These are the things that the bonding circuit will help you convey. By showing up, let's say, again, let's take the gym scenario, you show up consistently. Not completely consistently, but fairly consistently. That can help. You have some consistency to you.

- 8.50** The trick is the balance. Too much safety, too predictable early on, and she categorizes you perhaps as more of a friend. Why? Because the mystery keeps things alive early on. So some mystery is good, some mystery. But that's the balance. The balance.
- 8.51** Mix things. Use novelty in a creative way. In terms of dress code, try to have different clothing styles. On day A, wear a suit. On day B, wear a different suit. And then, on day three, come with a leather jacket. Completely flip the script because now you're doing a reward-prediction error. You're showing, "Oh, this guy has many sides to him." I literally mean that you want to use novelty as much as possible. That can actually help spark relationships, to have some novelty in dress code and things like that.
- 8.52** Novelty can also be going to different places, not doing the same things over and over. When I was in my young days, women would compliment me for this, so I take this as a good thing, but I think it has to do with my erratic nature, especially at that time. Now I'm a bit more stable, but in my teens, I was extremely erratic. So what I could do is, one minute, I could be walking around in the ghetto with a girl, and then all of a sudden, I would have a thought, "Oh, I'm going to take her to a restaurant one hour away," and then get a taxi and just go. I would do that sometimes. I would walk there, and then go take a taxi and go somewhere else.
- 8.53** The women loved that. They found, "Oh, this guy is extremely spontaneous." I mean, one minute, you are in a boring place, a ghetto. There's nothing, just graffiti on the wall. There's nothing interesting. And then suddenly we are in this fancy restaurant, and I'm using all my salary on her. Not very smart, by the way. It was stupid. It was stupid as hell. I got in real trouble for it. It was my erratic nature. But it worked wonders. Now, had I been more creative, I could have done something less expensive. The idea is sometimes you just want to pull the rug underneath her, and that can be done by doing something extremely unexpected.
- 8.54** Now, there's various ways to do unexpected things. Let's take the withdrawing situation for a minute. Early on, you are getting to know each other. For some reason, maybe she sees you're insecure, or there's something going on, and she withdraws. She withdraws from you, and you withdraw as well. Do you know this? Have you known this rule, the "no-contact" rule and all that? In that dating world, there's a lot of words. In the last few weeks, I've been exploring that aspect of neuroscience.
- 8.55** So, apparently, there's something called the no-contact rule, where you go your separate ways, and there's a psychological warfare of sorts: who's contacting whom first. I know you've experienced that before, right? You've had situations both going in that separate camp. The trick here, from my neuroscience reading, is: what you want to do to break that vicious cycle is to introduce warmth, some warmth, but in a surprising way when she doesn't expect it. That can break that cycle.
- 8.56** So let's see, she withdraws, what you can do then— again, neuroscience, you have to be careful. Don't go home and do this right away. But I'm saying you would first

withdraw even more initially and then introduce warmth, with sending flowers, for example. From a neuroscience perspective, if this is applied correctly and not interpreted in an evil, malicious way, it could actually work.

- 8.57** She withdraws. You don't want to reinforce that behavior by going and chasing. Why? Because whenever somebody does something that could break the ties of that relationship, don't ever reinforce that in a positive manner. She withdraws. Don't ever chase her because then she learns what? What rule did she just learn? Me withdrawing means he's going to chase me more. So she's going to go away even more. Automatically, this is the rule of human nature. So if she withdraws, you withdraw more.
- 8.58** But then she thinks, "He's gone." Her amygdala goes off the roof. But then, you introduce warmth. That's where the flowers come in, the message, the text, something, and that can reignite the whole—that no-contact rule where things are in a closed situation. Does that make sense? I feel like for this course, I became more diabolical somehow. I got some super powers that I didn't have before.
- 8.59** We talked about nonverbal signaling. Scarcity and availability—this is the huge paradox that people always talk about. "Am I too available?" I think sometimes people, like females, the ones that I've known, they overdo it. They do overdo it. They think they have to be unavailable all the time, "Oh, I need to disappear, otherwise he won't like me. I need to completely vanish. I need to be completely mysterious and elusive and all that." It works, but you can't be that scarce. You have to have some signals of safety.
- 8.60** In terms of neuroscience, of course, it's the orbitofrontal cortex that's involved in this, that really gets activated for scarcity and for anything in the world that's scarce. Anything scarce, it's the orbitofrontal cortex. And also the social proof component that we talked about. Showing a girl that you have people respecting you, that's also the orbitofrontal cortex. That's also the social relational part of the world. It always works wonders. Always, always. It always worked for me. It could be the thing that really tipped things in my favor. If a girl liked me and she saw other guys coming up to me and greeting me and being social and showing respect, it always worked. She would always find me more attractive.
- 8.61** Now, generally, let's say I'm in a high school situation. This happened to me. I was in high school, the girl, I know she's over there in the cantina, she's eating, drinking. And then, I'm going to walk in, and then, ideally, the high-status guys, the cool guys, in other words, come over to me. I'm not going to go over to them, but they're going to come over to me. That's even better. Somehow, she feels that's really—she becomes attracted. Why? Why does her orbitofrontal cortex—
- 8.62** You seem scarce?
- 8.63** Not scarcity, but it shows what? Social proof. But beyond that, why would she like that social proof? What is it? By the way, there's no clear answer, so let's think open box

here. What could be the answer? You know what I think it is? I think it could be status, but more than that. Because I've actually walked with a girl. The girl was next to me. We were walking. I knew she liked me. There was nothing going on, but I knew she liked me.

- 8.64** We were walking in school. And then, literally, some buff guy, muscles, came over to me, and the guy, because the girl was really pretty—she was actually very pretty. She was really pretty—that guy became even more friendly than he usually was because now I became high status in his eyes. So, "Oh, how are you, Baland? How are you doing?" All the hand gestures and all that. I was like, my God, her smile, you should have seen her smile. It went all the way up because something innate in her became activated.
- 8.65** I think it has to do with the fact that it shows somehow that I'm part of a tribe that will protect her. You know what I mean? The brain must have some algorithm to detect which males are part of the social fabric and can protect her not only as an individual, but as a whole tribe. I think it taps into that. What do you think? That's my guess. There was something more in her smile than just, "Oh, that's a cool guy." It was more like, "Oh, that's really nice." She just smiled, "Who's that guy?" in a positive way. I've had that occur many times.
- 8.66** So this is something to think about. If you are in an environment, let's say, a school or university, and you know that the girl is going to see you, then before she arrives, or during the time that she's not there, try to prepare the environment, like Groundhog Day. Try to interact with people, try to make friends, try to be social. In that way, then, when she sees you, you're optimally ready to titillate her orbitofrontal cortex. Does that make sense? These are tricks we can use, but these are healthy, good tricks, not diabolical tricks.
- 8.67** I think that humor and playfulness works. I found that always works, humor. I wonder why, because too much humor is not good. If I was the guy in the gym and the girl saw me just going around laughing and joking with everyone, would that work? Would you find that guy interesting? No. Ask yourself, why? I want to know why. How come you don't find him that interesting when he's too goofy, and he's laughing and joking all the time? Why is that not attractive to you? I mean, after all, he's being social. He's showing some social proof, right?
- 8.68** I would ask, why can't you be serious or take things seriously?
- 8.69** That's one part. I think that could be true. Yeah, I think so. I think that's a good point, Sarah. Anything else?
- 8.70** Maybe you're not having control. I think that someone who is humorous and knows when to interject and make a funny comment or whatever, there's control involved.

- 8.71** Control, yeah. So seriousness, he can't be serious. Control, yes. I'm going to answer that. Those are good. What else? Anything else? You know what I think it is? I think you're right. I think Sarah is right. I think it has to do with, this guy doesn't have the ability to be serious in the sense that if he was a leader of the group or he had responsibility in a social hierarchy, he would be more serious. Surely, somebody laughing all day doesn't reflect a man who has any responsibility in his life. So I think that has to do with that.
- 8.72** And I think frontal control, when you are laughing and joking and being playful exclusively, that signals what? It signals low prefrontal control. It could potentially be an indicator of you not having prefrontal control and inhibition over your limbic structures. So I think that plays a role.
- 8.73** But I also think it is a signal of not being a protector as well, in the sense that a male has to protect the female. He has to protect her. A woman sees him as laughing and joking too much signals he doesn't have a dangerous edge to him. Think Beauty and the Beast. The Beast had the edge. He had the edge. He had the strength. When Gaston came at the end. And when Belle came back, initially, when Gaston came, he literally almost killed the Beast. You've seen that, right? He [almost] killed the Beast. The Beast had no strength to fight back. No strength.
- 8.74** But then, as Belle came back, his energy came back. He became energized, and he was able to fight off Gaston. Gaston literally said, "Beast, are you too kind and generous to fight back? You don't have any strength in you?" But then when he saw Belle, he did get his strength. His bestiality came out. I think some bestiality is important, is attractive for a male to have. And I do think that what is most attractive, the proportion would be being serious most of the time and then having some elements of joking and laughter. I think that women find that most attractive. What do you think, women? Does that sound true? That's my own assessment.
- 8.75** So with a woman, you don't act like Baland, just joke and have fun. You can't do that all the time. You have to be mostly serious, but then have the dopaminergic centers firing more optimally once in a while. Show some laughter, show some fun. Is that a pretty good algorithm? What do you think? Okay, good. As we are exploring this course, we are also creating new knowledge. I'm not sure how this has been written up in the literature, so it's very interesting to talk about.
- 8.76** I think laughter and joking is crucial as well. Having humor, in my own experience, can be very powerful. I think the reason it's powerful is because I think a joke is a window into how intelligent you are. I think it is a window into how creative you are. I think it's a window into how cognitively flexible you are, how spontaneous you are, how out-of-the-box of a thinker you are. And so if you do not laugh at all, you indicate rigidity, inflexibility, over-seriousness, and that's just off-putting. Women don't like that. They want somebody who is cognitively flexible, creative, smart, and intelligent because that

is the best scenario for resource acquisition. A man who knows how to pull off a good joke or has good sense of humor.

- 8.77** In fact, in my own life, the best creatives, the best scientists, the top scientists in the world, in fact, were hyper-humorous and hyper-joking all the time. They had great sense of humor. Ramachandran being the most brilliant example of that: so warm, so much humor, so much fun. A great sense of humor.
- 8.78** Another example would be, I talked about in my previous course, what's his name? My brain is blocking it right now. The dream researcher, Allan Hobson, one of the greatest dream researchers. His humor was out of this world. Extremely creative, extremely fun. I think that is a sign of creativity and intelligence, having that.
- 8.79** Some of the other scientists—there are the top scientists up here. They have a creative streak and also usually have a bit of a spiritual, mystical streak to them as well. And then, you have the one that is down here, the successful one, the top one, when you say, "This is a good scientist. He's a tenured professor at a great university." But he isn't great, great. He is just successful. There is no greater view of the world, and creativity and humor is usually lacking as well. It's interesting. So I think that's what women like.
- 8.80** Posture, standing back with your shoulders back. I sometimes forget this. I sometimes do like this. It's my thing. I always write on the computer and all that. I think that's what does it. But if you can have posture that's good and relaxed, people pick up an anxiety really quickly. It's off-putting because anxiety is contagious, so we are naturally wired to not like anxious people. We don't like them, at least when they're anxious. So shoulders back, seem composed, have slow movements overall. Don't be too rushed in your speech and when you're talking.
- 8.81** I think overall, what this is signaling is the fact that you are more in a low cortisol state, because all these microbehaviours signal high testosterone, being calm, being relaxed, not speaking in a fast manner and being rushed and all that. I think that's what can really work. So having that, being mindful of it.
- 8.82** You might say, "Hold on, I actually am stressed. I also, like Baland, have all these things I have to do, these daily tasks, and maybe my sleep was off, so I am stressed. What the heck do I do now?" Here, I would say, I would merrily say the following. I would say, as Jonathan reminded us, there's tons of interactions between the body and the brain. Somehow, if I can relax myself before going and meeting that girl, maybe I can do a sauna session, maybe I can do some stretches, a little bit of exercise might be a good idea. Anything that can calm me down, meditation. And then, actually, when I do meet her, what I can do is I can act like I am more calm and more relaxed.
- 8.83** Now, I told you it's very hard to act out these things because she might pick it up. But still, you can do some acting, and eventually, if you actually do relax your body a bit and do remain calm, there's feedback going back from your body telling your brain, "He's

calm because things are safe and okay. It's fine for me to relax a bit." You can trick your nervous system in this way.

- 8.84** Did I tell you about the studies that we wanted to do? There's studies showing that if you read cartoons and you have a pencil in your mouth like this, you rate them as more funny. If it's like this, less funny. More funny, like a smile. Less funny, like this. We know this is the case. My mentor and I wanted to do studies where people with depression had smiling faces on their faces to see if we could trick the nervous system by creating a false muscle.
- 8.85** What I'm saying here is that there's tons of interaction between muscles, body, composure, voice. When you try to remain calm and act in a calm manner, relax your muscles, and breathe calmly, to some extent, you can trick your nervous system. To some extent. That's much better than being anxious and stressed and then not working on your body composure at all, and your voice, and all that. Does that make sense? Good.
- 8.86** I like the whole social [aspect of] how we are driven by social concepts and social proof. We have our own individual opinions and brains. I have my brain, Natalie has her brain, Joey has his brain. We have all our individual brains, but our brains are in many ways hooked up in the way we judge others. It turns out, if I show you a picture of this female, or a male, for example. I'll show you a picture of a male, and you say, "I find this male attractive. This guy is a 10 out of 10." He is Brad Pitt in his prime level.
- 8.87** What turns out now, you give him this rating. You then hear about all the other women's ratings, and you will change your opinion about him. Can you believe this? Literally, the orbitofrontal cortex and its error detection will go off and say, "Oh, my God, I made a mistake. This guy is not a 10 out of 10. He's like a 3 out of 10," just like the group. So it really shows you how, if you can weave yourself into the social fabric and get along with the other girls, too, so they like you. Even if it's a friend, it can be friends because now it's different. You're not going for them. But if the other girls like you, then you're in a better state. Does that make sense? Really, there's a key component of how our brain works in this way.
- 8.88** As we are ending, as we're coming closer to the end, I think something that needs to be emphasized, if I haven't emphasized it enough, is the whole balance between macho, confident, self-assured type, the strong type, the protector type, the bodyguard, Frank Farmer type, versus being more humble, being more kind, being more generous, more light-hearted, maybe a little bit clumsy. How is the right balance?
- 8.89** There's got to be a balance there because surely if you do scenario A, where you're just completely macho and you do what you think might be the right thing, surely it is not the right thing. It is not the right thing, I'm telling you. So there's a balance there. I'm trying to get at the balance. The balance is also, don't forget humility. Being humble, to

a girl, can be extremely attractive. You're disagreeing with me, I can sense. No? You agree? Okay, good, good.

- 8.90** Maybe I expected you to disagree because there's a big caveat here. Being humble doesn't mean running over and buying her tons of flowers. It's being humble in your own person. Humility means you are not seeking validation externally. You're okay with yourself, and you're humble towards the world. You are grateful. Really, humility is about gratitude. That's what I think. It's being grateful for who you are, the blessings you've been given, and your situation in life, and going, "My God, I feel like I'm the most blessed person in the whole world."
- 8.91** Not to pinpoint myself, but I do feel very blessed. I'm not saying I'm humble because saying you're humble exactly is when you're not humble. "I'm the most humble person on Earth." It doesn't work. I do feel grateful, though. I do feel tons of gratitude. So even though I may experience a rejection here or this or that, I do feel extremely grateful for my situation, for life. Extremely grateful.
- 8.92** I think having that humility about life itself is very charming, and I think the girl can pick up on that. Because macho-ness, the guys on social media, they're very macho, and they talk down on women, and they speak, "Women are this—" They have to do this to women and that. These types—I'm not going to mention names, but you know who I'm talking about. These types, they are insecure. They are insecure, really. That's what I think. I think inside themselves, they're lacking something: respect and dignity. That woman is like your own mother. It's like your own sister.
- 8.93** One of my friends, I told him about a girl that I liked. He kept talking about sexual stuff, "Did you do this to her? You want to do this and that?" You know what happened? My heart broke. How can you talk about a woman in this way? Seriously. This is potentially my future wife here, and you're talking about her in this way. He's no longer my friend, let me put it that way. He wasn't a close friend before, but after that, it broke my heart. Really, it broke my heart. Respect. Respect her. This is like your sister in the sense that she's a human being, and she could, like your sister—some guy, one day, also, might want to date your sister, and your mother and all that. And so I think that's hugely important.
- 8.94** So humility is important, and having a sense of dignity and respect. As long as you do that with boundaries and understanding human nature and how courtship works, and the aspects of courtship that are needed, work on yourself, build some muscle, have some testosterone, which makes you more assertive and strong when you need to be, I think you'll be in good shape. I think you'll be in good shape. On that note, let's go over to the questions. Let's go over to the questions.
- 8.95** My question is more of a general type. I guess I would like to draw a parallel between a human brain and that of an animal, and primarily maybe our pets. Our human brain is a sophisticated machine, and it makes us who we are, makes us human. But at the same

time, it caused debilitating conditions due to, for example, extreme negative emotions or abnormalities in your brain. When we talk about the animal brain, it's a lot simpler, but yet animals seem to be like bundles of pure positive energy, and it's so easy for them to love.

- 8.96** Sure, especially dogs.
- 8.97** Dogs or cats. So then maybe there's a way for our brain to learn or to stay perpetually in a state of goodness and love, like people in romantic relationships experience and maybe learn how to perceive in love versus out of love?
- 8.98** It's a good question. It's a great question. I think, first of all, actually, I doubt whether cats are very loyal creatures. They're not very loyal. They don't have love in that sense, I think, but they're very cute and beautiful. I think dogs are extremely loyal. They can become jealous of their owners, even. We know this. But the problem with a dog is that the common dog has been bred. It's not the real dog. The real dog is the wolf and the wolf-like dogs. Why? Because we live in a dangerous world. That's the problem.
- 8.99** If we all became lovey-dovey too much, like the dog, and always in love and never had a thickness to our skin and a strongness to us, we would perish. I think it's the balance again. The balance. So yes, we can learn from the dog in how playful it is. It has innocence that's not been changed. It's still pure in its innocence. So if we can regain that and not let our past disappointments push us away from that, we'd be in a great state.
- 8.100** Can we learn that? I think we can learn it, but I think it's less about neural circuitry. It could be about neural circuitry. We could absolutely zap your brain on the left side of the brain with a high frequency, and then low frequency in the right. I'm being very straight with you: that would create less anxiety and stress and more happiness. It would. It would just be a good mood. And then, at that point, you might want to maybe engage in some talk therapy and expose yourself. More than talk therapy— expose yourself and then push yourself to be more loving. But forget the brain scanning or the brain TMS. You could do that in extreme cases, but I think just being bold enough to expose yourself to loving situations, I think, is the key. Just say, "Am I going to gradually reveal more and more."
- 8.101** You've seen the Good Will Hunting? You haven't seen Good Will Hunting. Good Will Hunting is this genius. He sits in the office with a therapist. He cannot love. He's too perfectionistic. And then the shrink says, "You don't want to reveal yourself to this girl because you think you're too perfect and you don't want that perfect image to erode in her eyes." Perfectionism sometimes can stand in the way of authenticity and truth. So gradually, but willingly, expose yourself because only in the imperfection, you will find love, true love. It's in the imperfection. It's all the flaws. It's all the things that are messy and that are not perfect that love will come. So you have to gradually go there. Does that make sense? Thank you.

